Let's briefly go through some of the learning outcomes, the expected learning outcomes of this course. I'm going to talk to you about some things that you should know about the time you finish the course. First of all, we want you to be able to explain the main concepts in HCI and why they're relevant both in a theoretical and a practical manner. We'll do that through describing a set of design approaches that embody the theory and the practice. We'll look at some practical examples. Think about how we can employ different techniques in different situations and try to contextualize some of our design decisions in place and think about what the implications that there are. We're also going to expect you to be able to use the appropriate design approaches and interaction prototyping methods, develop new interfaces and technologies.

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This isn't just about this process of thinking about how things have been done in the past. We're looking at matrix and tools and techniques that we've used in the past but it's not being fooled thinking and saying, "Well, what happens if you don't have something new? What happens if you have a new design challenge or a new piece of technology comes out that changes the work that people work or the way that they interact?" We want to provide you with a robust enough set of strategies to be able to overcome those challenges and to be able to design useful tools and techniques that, so it's interesting problems. We also want you to be able to make use of usability and interaction design concepts in critiquing, using and building interactive solutions for a range of applications. We want you to understand where the basics of this come from. This research was done in the early '90s and formalizing some of these design concepts.

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But the work is changed every time and the manifestation of these activities is changed. We understand a lot more now than we did in the past and we'll continue into learning more and more about how people interact with technology, as technology becomes more pervasive, more ubiquitous, it's all around us, everywhere in our lives, we have this interesting challenge that we have to be able to facilitate different types of design and directions. We're going to look at traditional matrix as a way to critique things. But we're also going to move forward and think about strategies to build technologies for new and interesting applications through solving interesting types of challenges and looking at design concepts and how they fit at given context. Again, this is much about conceptualization. We also want to be able to use appropriate evaluations techniques to constructively criticize interface designs and explain how they can be improved. When we think about design as being this intuitive sets of processes, I will produce a design, evaluate it, change it, re-evaluate it until we get to an intended outcome.

Play video starting at :2:39 and follow transcript2:39

In this case, we want you to be able to have process of skills and now that you do that, to constructively criticize, make positive suggestions for improvements and then implement those improvements in a useful way. We also want you to be able to explain new application areas, not the advanced technologies and thinking about what the implications for a design perspective. Thinking about what this means for people as we move into an era where we have things like self-driving cars or autonomous systems that will automatically do things for us. What are the implications on people's lives and their interruptions? How is that going to affect people of a given set of situations or circumstances? How is it going to change it whether they work, whether they live, whether they engage with entertainment technologies? All these things are

considered in the context and it's very important that you understand these looking forward, not just looking backwards.
Play video starting at :3:30 and follow transcript3:30
Thinking about what's to come and how you're going to be able to be a robust enough software engineer to be able to solve these new and interesting challenges. How we will learn? How are we going to learn on this stuff? It's a lot to cover, [inaudible] agree? But ultimately, were looking at a mix of theoretical and practical tasks, so lots of this will be around discussions, quiz questions, activities, reviews and of course, these video lectures to look into it. This is a chance to learn and critique and work together as teams but also to go away and just think about things and reflect, to do some reading. Occasion, lots of different types of activities to support your learning and to encourage you to learn different ones.

So in this lecture, we're going to talk about design and we're going to talk about what design is now. This is a word that you probably heard before, and it's probably been used in a specific context. So people often refer to design as being these kind of traditional practices of people designing in a very creative way. It might be things like painting or producing portraits or producing physical manifestations of things. So things like sculptures, that design can also manifest in a number of other ways. So we can think about design as a software engineering practices. We can think about it as an engineering task.
Play video starting at ::41 and follow transcript0:41
So we might design a railroad or a house or car. So manufacturing in these processes also in part designed strategies and what we do in this field is essentially we take all of those things together and we look at drawing from other industries to try to understand how we can better facilitate design. So it could be thinking about how things look and feel. So design might be all about those aesthetic properties of things. It might be about looking at the properties in terms of the look and feel how things are to experience as a set of experience design constructs. So how does this phone feel in my pocket? How is it to use on a day to day basis?
Play video starting at :1:18 and follow transcript1:18
Does the screen illuminate enough and given context? Is it working in a dark setting? Is it working in a light setting? It's the color contrast ratio, good and all these other aspects that may not focus around functional things. So it may not be about completing a set of tasks, but it's oriented towards these self constructs that are really important in our day to day life. So does it fit my workflow? Does it fit my dynamic as a human being and I interact with this in a way which is meaningful to me and allows me to do things without getting in the way.
Play video starting at :1:48 and follow transcript1:48
So this could also to do with things like structure. So we might be thinking about the architecture of our system, we might be thinking about the layout of objects specifically on a page, so it might be about the orientation of these objects. Do they fit together in a nice way? Are they constructed in a way in which I can understand things? So this whole fields that pop up around this, things like information architecture. Do you think about things like the navigational layouts where objects fit in a page? These ideas of ontology ease and categories of things?
Play video starting at :2:17 and follow transcript2:17

How do things grouped together? There's lots of different ways that you can structure things. And architecture and layout are another aspect of design that we're going to consider. And then finally we might think about formalizing specifications. So you might want to define things in a clear linear way. So we could have a diagram which might just be representing things like X and Y. Or we could think about things like UML diagrams, which is all about modeling certain structures of systems.

Play video starting at :2:42 and follow transcript2:42

Or it could even be a prototype. So it might be a physical manifestation, we could do some paper prototyping or it could be something digital, it could be a directive, it could be static. There's a whole manifestation of design activities that can take place to facilitate these sorts of actions that allow you to build systems that work in place. So let's have a look at some examples of things that we might see. So let's first of all consider the context of design in terms of when we do things. So when do we do design now? In most cases you probably experienced designers happening at the very start of the process with designer house and we build the house.

Play video starting at :3:17 and follow transcript3:17

You go and live in a house. That makes perfect sense as a set of contract. But actually design doesn't have to happen just at the start. It can happen all the way through and we can redesign and we can change, we extend our houses, we paint in different colors. So it's not to say that once the house has been built, we can't make changes to its structure, to this architecture, to the dynamics to look and feel. And so in this sense, design isn't something that just happens once. So all the time things are redesigned, this idea of it being iterative, formative, informing constructs over time and those constructs changing based on feedback, based on needs, based on once, based on desires, based on all these interesting enrich requirements, solicitation processes.

Play video starting at :3:59 and follow transcript3:59

And of course design does not happen in isolation. So we need to design with all these things in mind. What's the utility of the application? What's the aesthetic value and what are these trade offs, other things that we have to lose out on if we want to gain more in some other area? And how do we make those decisions? How do we decide to paint our house green to provide practicality so that we can produce interesting visuals. We have to think about these things and we have to think about what is the impact of our design decisions on human beings and on other parts of our working processes.

Play video starting at :4:32 and follow transcript4:32

How is a design decision in one place going to affect something that we do later on? If I start to uncertain architecture or a piece of technology or a certain design constructs or things that I want to do in terms of the way in which the system is built from a technical perspective and then dependent on those things to do other things. So I have to think about those interdependent relationships and how things affect one another. The manifestation of this is really important to consider. So what design lessons can we learn? What can we learn about design and how can it in part certain changes in the way in which we work? Well, first of all, we can learn a great deal from other people and we can borrow from other fields and we can talk about how does this own working properties of ergonomics?

Play video starting at :5:15 and follow transcript5:15

How do people hold things? How do human factors relate to the way in which people deal with technology? But we can also learn lessons from places like psychology, traditional designs such as artistic designs, creative designs, and we can also borrow from other places. So we can look at user centered design, which we've already covered in some of the other courses. We can think about things like service design, participatory design where we take individuals and we engage in a practice where they actually design the systems that they're going to use. So they are involved in the design process, not just as a stakeholder, but as a designer, somebody who makes suggestions as somebody who can design and build the system with us to ensure that we build something that's useful and usable, has that utility. And we can also think about socio technical system design, thinking about those larger constructs of what does it mean when people come together with technology?
Play video starting at :6:5 and follow transcript6:05
How does that manifest what the applications of that? What the implications of that? So we're thinking both in words, in terms of our own industry, our own ideas, our own concepts are only learned ways of working, but we're looking externally, we're saying, well what are other people doing? How can we use those practices? How can we engage in those tools and techniques to enable us to do interesting and useful things? So this is a good question. What is good design?
Play video starting at :6:29 and follow transcript6:29
How do we know that we're doing good design? Well, I'd like you to come up with these ideas. I'd like you to think about this yourself. I could tell you about what I think good design is, but I think this is a more open ended activity. So what I'd like you to do is to go on the discussion forums and I'd like you to list five generalizable features of good design. So I want five properties that will tell me good design in all possible contexts, any context you can imagine building and designing a system in five things that you'd like to have in your design. Five features of the design, five components of good design that are common amongst all of these different practices in different generalizable situations.
Play video starting at :7:9 and follow transcript7:09
Then I'd like you to take those five suggestions that you come up with, and I'd like you to prioritize them in terms of importance. Which of the five is most important when you're thinking about design, which is least important. And can you rank them in order? And then I'd like you to do another exercise following that, and I'd like you to prioritize them in terms of relevance to our field, to what we do in computer science. In software engineering these five things are important in this particular set of contexts. So you prioritize them in terms of general importance, and then you prioritize them in terms of how important they are in the software engineering field. Hopefully, that makes sense to have a go at the activity and let's have a chat about what things you think are common in good design practices.

So in this lecture we're going to talk about design, we're going to talk about what design is. Now, this is a word that you've probably heard before and it's probably been used in a specific context so people often refer to design as being these kind of traditional practices of people designing in a very creative way, it might be things like painting or producing portraits or producing physical manifestations of things, so things like sculptures, but design can also manifest in a number of other ways. So we can think about design as a set of software engineering practices, we can

think about it as an engineering task, so we might design a railroad or a house or a car. So manufacturing and these processes also impart design strategies and what we do in this field is essentially we take all of those things together and we look at drawing from other industries to try to understand how we can better facilitate design. So we could be thinking about how things look and feel, so design might be all about those aesthetic properties of things, it might be about looking at the properties in terms of the look and feel, how things are to experience as a set of experience design constructs. So how does this phone feel in my pocket, how is it to use on a day-to-day basis, does the screen illuminate enough in given contexts, is it working in a dark setting, is it working in a light setting, is the colour contrast ratio good and all these other aspects that may not focus around functional things, so it may not be about completing a set of tasks, but it's oriented towards these set of constructs that are really important in our day-to-day lives, so does it fit my workflow, does it fit my dynamic as a human being, can I interact with this in a way in which it's meaningful to me and allows me to do things without getting in the way. So this could also be to do with things like structure, so we might be thinking about the architecture of our system, we might be thinking about the layout of objects specifically on a page, so it might be about the orientation of these objects, do they fit together in a nice way, are they constructed in a way in which I can understand things, so there's whole fields that pop up around this, things like information architecture, which think about things like your navigational layouts, where objects fit in a page, these ideas of ontologies and categories of things, how do things group together, there's lots of different ways that you can structure things, and architecture and layout are another aspect of design that we're going to consider.
Play video starting at :2:26 and follow transcript2:26
And then finally we might think about formalising specifications, so we might want to define things in a clear, linear way, so we could have a diagram, which might just be representing things like x and y, or we could think about things like UML diagrams, which is all about modelling certain structures of systems, or it could even be a prototype, so it might be a physical manifestation, we could do some paper prototyping, or it could be something digital, it could be interactive, it could be static, there's a whole manifestation of design activities that can take place to facilitate these sorts of actions and allow you to build systems that work in place. So let's have a look at some examples of things that we might see. So let's first of all consider the context of design in terms of when we do things, so when do we do design? Now in most cases you probably experience design as happening at the very start of a process, we design a house, then we build the house, we go and live in the house, that makes perfect sense as a set of constructs, but actually design doesn't have to happen just at the start, it can happen all the way through, and we can redesign and we can change, you know we extend our houses, we paint them different colours, so it's not to say that once a house has been built we can't make changes to its structure, to its architecture, to the dynamics, to the look and feel, and so in this sense design isn't something that just happens once, it happens all the time, things are redesigned, this idea of it being iterative, formative, it forming constructs over time and those constructs changing based on feedback, based on needs, based on wants, based on desires, based on all these interesting rich requirements, elicitation processes. And of course design does not happen in isolation, so we need to design with all these things in mind, what's the utility of the application, what's the aesthetic value, and what are these trade-offs, are there things that we have to lose out on if we want to gain more in some other area, and how do we make those

decisions, how do we decide to paint our house green, to provide practicality so that we can produce interesting visuals, we have to think about these things and we have to think about what is the impact of our design decisions on human beings and on other parts of our working processes, how is a design decision in one place going to affect something that we do later on, if I settle on a certain architecture or a piece of technology or a certain design constructs or things that I want to do in terms of the way in which the system is built from a technical perspective and then dependent on those things to do other things, so I have to think about those interdependent relationships and how things affect one another, the manifestation of this is really important to consider. So what design lessons can we learn, what can we learn about design and how can it impart certain changes in the way in which we work? Well, first of all we can learn a great deal from other people and we can borrow from other fields and we can talk about how does design work in the properties of ergonomics, how do people hold things, how do human factors relate to the way in which people deal with technology, but we can also learn lessons from places like psychology, traditional designs such as artistic designs, creative designs, and we can also borrow from other places so we can look at user-centred design which we've already covered in some of the other courses, we can think about things like service design, participatory design where we take individuals and we engage in a practice where they actually design the systems that they're going to use, so they're involved in that design process, not just as a stakeholder but as a designer, as somebody who makes suggestions, as somebody who can design and build the system with us to ensure that we build something that's useful and usable, has that utility, and we can also think about socio-technical system design and thinking about those larger constructs of what does it mean when people come together with technology, how does that manifest, what are the applications of that, what are the implications of that, so we're thinking both inwards in terms of our own industry, our own ideas, our own concepts, our own learned ways of working, but we're looking externally and we're saying well what are other people doing, how can we use those practices, how can we engage in those tools and techniques to enable us to do interesting and useful things. So this is a good question, what is good design? How do we know that we're doing good design? Well I'd like you to come up with these ideas, I'd like you to think about this yourself, I could tell you about what I think good design is, but I think this is a more open-ended activity. So what I'd like you to do is to go into the discussion forums and I'd like you to list five generalisable features of good design. So I want five properties that will tell me good design in all possible contexts, any context you can imagine building and designing a system in, five things you'd like to have in your design, five features of the design, five components of good design that are common amongst all of these different practices and different generalisable situations.
Play video starting at :7:9 and follow transcript7:09
Then I'd like you to take those five suggestions that you've come up with and I'd like you to prioritise them in terms of importance. Which of the five is most important when you're thinking about design, which is least important and can you rank them in order? And then I'd like you to do another exercise following that and I'd like you to prioritise them in terms of relevance to our field, to what we do in computer science, in software engineering. These five things are important in this particular set of contexts, so you prioritise them in terms of general importance and then you prioritise them in terms of how important they are in the software engineering field.

Hopefully that makes sense, have a go at the activity and let's have a chat about what things you think are common in good design practices.

So I'd like to talk to you briefly about what interaction design is not. What we talked about hundreds of different contexts and situations where interaction design borrows from fuels like informatics, information architecture from the media, from content creation around graphics generation and things like that. All the way through to fields that I'm interested in like audio engineering, adaptive sound context and thinking about designing instruments. In the same sense, interaction design isn't just about building applications, it's not just about thinking about, I'm going to do with some technology using a framework and I'm going to use react or some other javascript framework to build something. It's about thinking about the context of use and the implications for design. What does it mean to build a system for a specific context? And how is that going to impact people in terms of how they use it?
Play video starting at ::59 and follow transcript0:59
So we think about these often as a set of universal design practices or universal design constructs. Where we think about what the implications to smaller groups of users and how can that general is up to a larger group of users. What are the implications for instance, for designing a system in purpose and in place? How is that going to affect people? If we were designing a road system for instance, we'd be thinking about all modalities of travel, would be thinking about cars and buses, and we'll be thinking about how they might travel across those different journeys. What might they use a journey look like? The long journey and car can be an uncomfortable experience.
Play video starting at :1:35 and follow transcript1:35
And so we spend a lot of time thinking about manufacturing a vehicle was comfortable to being. In the same sense, a long journey would also be something that would be a consideration in terms of energy use. So we probably want to conserve energy over that long journey and think about things like fuel efficiency or whether electricity will be a better modality for that and how far we can actually travel using those different mechanisms. Whether or not we have to stop of course in a race that might be really interesting thing and it might be really useful to think about that. We want to stop as few times as possible to ensure we have the fastest race time. But if we're talking about allegedly journey where we're doing a long drive, we might have pauses built into that. So again, we have to think about a different set of design constructs with those different types of experiences.
Play video starting at :2:22 and follow transcript2:22
And what we want to do from an interaction design perspective because you want to validate those assumptions and we want to apply different context of use to our system to allow people who are experts and non experts and everybody in between to use the system in a way in which they can achieve the things they want to achieve with the system. The ultimate goal is utility, how we achieve that through a number of constructs. And we also have to think about those other values that might not immediately create a value proposition that can be measured. So we might see something that's more pleasant, but that might be more difficult to measure. And so we have to take that into account of the designing and building systems. And we have to build in metrics that are not just about performance, but thinking about those wide set of constructs and

how we can validate our design assumptions. How we can test, how we can evaluate informative way and how we can ensure the building a system is going to fit a given context.

So I'd like to talk to you very briefly about what happens if we design wrong and what happens if we don't have interaction designers thinking about all possibilities and all permutations. So I have with me a very, very lightweight laptop. It's very portable, it's very easy to use, it's very powerful, it's a lovely system, really well designed, really nice, the aesthetics are really good, beautiful, it has a really nice keyboard on it, a really nice trackpad, thoroughly enjoy using it, even the display is beautiful. The problem that I have is this. I have so many of these for this system because it doesn't have USB-A ports, so we have some USB-C ports on the side of it, which I have adapters for, and then I have to use adapters plugged into adapters, so I have this adapter here to give me a series of USB-A inputs, and then I have to plug another adapter into this system if I want Ethernet connectivity, so I've got adapters everywhere. Now this is a problem for me because my portable system is now really small, has suddenly become very large, and suddenly, instead of just taking a small bag with me, I have to take a number of cables, a number of cables that connect to other cables, and then I need power sources to drive those, and so on and so forth. So this is what we call a compounding problem in interaction design. It's a problem where when we create a design construct that isn't comfortable for people to use, they have to subvert that in some way, and so we have to add extensions on, we have to change the way that people work, and that modality of doing things is not necessarily comfortable for everybody. It affects the experience in terms of making things less accessible. If I forget my cable, I can't plug devices into my system. It makes my life more difficult.
Play video starting at :1:54 and follow transcript1:54
So the design construct around making the device slightly smaller, by the context of a few millimetres, has actually created some real issues for me. It makes it very, very difficult for me to now use my system the way I intend, and so there might be contexts of use where people aren't interested in plugging things in, where they have lots of Bluetooth technology, but for me, that doesn't fit my workflow, and so when we get this wrong, there can be serious implications, and it can make people's lives much more difficult and much more tricky, where they have to go out and buy lots of expensive adapters, and they have to adapt the way that they work to fit the system. Now what we want to do as interaction designers is we want to model a system that allows them to do all the things that they need to do, without imposing unnecessary constraints or restrictions on people. What we want is a system that can be used in lots of different ways, and allow lots of different individuals to meet their needs and wants in a desirable fashion, to use a system that's pleasurable to use, but also allows me to achieve my intended outcomes. Now that's quite difficult to do as we've seen from the foes of designs, and this is something I want you to think about as you go through this course. Is there a context where a design might work really well, or in other cases, it might cause problems for people, or it might cause things like accessibility issues, or it might not be usable for a given group of users, and this is something you should keep in mind all the time, and this is one of the reasons that we test different groups of users with different levels of experience and understanding. We want a broad demographic of people to be able to use our systems, and we want to enable them to do the things that they want and need to do.

So we're going to continue down this journey and we're going to think about some ways in which we can conceptualize systems from an interaction design perspective. So we can think about are intended outcome and then work backwards to think about the manifestation of the activities that would allow somebody to complete set of actions. Now the actions that they want to complete may vary from user to user, and you may find that your designing different types of systems where you have to facilitate different types of activities. So this isn't just about designing in one particular way, it's about thinking about a robust set of design strategies and then contextualized in there for each case. So every time you look at a new case, a new system that you're working on, a new piece of software, a new piece of hardware, you have to really think about your stakeholders and all the different moving parts, then decide how you're going to build a system that's going to meet people's needs. So let's think about what people's needs might manifest of as a general set of deliverables. So the first thing we might start out with, and the easiest thing to conceptualize is a goal.

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We have an intention, we have something that we want to achieve from the system. It's typically just one thing we want the system to do a thing for us. So it might be something like automation, it might be that I want a keyboard that can type or alpha numeric characters. That's a fairly easy goal. We can conceptualize that, we can manage it, it's tangible, we can measure it, we can measure progress against different types of metrics. So we know we have a certain number of characters in our character space. We know how close we are to achieving that thing, and we can design a system that allows us to do that.

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We could go across the ask your character space or you can think of another type of mapping that would allow us to input all of those objects and allow us to output different types of things as well. So what we're really thinking about with a goal oriented system is, can I achieve this thing? And can I do it well? Now those of you who are familiar with usability goals, we'll look at this as a set of usability constructs. As effectiveness, does it do its job? And efficiency, does it do its job well? Can we do it with limited resources?

Play video starting at :2:3 and follow transcript2:03

Can we do it in a relatively timely manner? Are we relying on lots of things, lots of moving parts to get it done or is it a simple process? So you can think about our goal in this manner. It is fairly straight forward and fairly simple. So let's move forward and let's think about more concrete examples of things that might not fit this manifestation. So let's think about scenario based design. Now this is a bit more complex than just a static set of goals.

Play video starting at :2:27 and follow transcript2:27

So in this case we're thinking about achieving an intended outcome. So we want to achieve something in the same way, all three of these things that I'll show you are the same manifestation of once, so intended outcome. But we have to think in scenario based design about the steps that are involved in successful failure. Now, a system could often fail user in many different ways. It's not just about designing a system that does task number X and that task can be achieved with one set of functions. And in fact, those of you who've got lots of programming experience would probably know that you have to build up lots of functions, you have to have quite a big repository of code to get it to do relatively simple things. And so systems design is very similar.

Play video starting at :3:7 and follow transcript3:07
It's about thinking about there are a set of logical steps involved in the process, which allows somebody to achieve something. Now those steps might not be linear. So a journey might go in lots of different directions before the user gets to the end of what they're trying to do. So it might be the case of reading a book for this course, for instance, you might decide, okay, I'm going to read the book cover to cover, I'm going to read from chapter one right through to the last chapter, and then I'm going to internalize that. Some people might read a chapter and then make notes. Other people might skip to the chapters that they want to read. And so there's different ways in which you can absorb that content.
Play video starting at :3:43 and follow transcript3:43
It's not just about, I want to absorb all the information from the book. So I have to read it cover to cover. There's lots of different ways to achieve your intended outcome. And scenario based design is really all about this. It's about thinking about all the different moving parts, different journeys that people might take for the system, looking at getting to that end point, trying to achieve something with our system. Now, if we take a step forward, we might find ourselves in a situation where we're doing something called story oriented design. Now this is a bit more difficult to map.
Play video starting at :4:13 and follow transcript4:13
And in this case you often see these maps is things like customer journeys or user journeys and some of those maybe scenario based, but others might not. So even between these two elements as distinctions in how you look at these things, but with a story, what we're really interested in and what makes it different to the scenario based design exercise, is that we might not have one single goal. So the system might not necessarily be about achieving X, Y and Z. It might be about an experience, it might be about enjoying yourself, and that's something that might be difficult to measure. So the time that you spend using the system, it may not be a linear construct. We might not just be able to say, I'm going to spend 30 minutes using this system, I'm going to spend an hour. If you think about a game or a movie, you don't usually make the choices of what you're going to engage with based upon how long it's going to take to do it.
Play video starting at :5:1 and follow transcript5:01
You actually think about certain aesthetic aspects and design aspects, thinking about maybe who the developers of these things are, or who the publishers are, or which film studios involved in producing this? Are you interested in the characters? Are you interested in the narrative? Have you read similar books by the same author or by other authors who write in the same genre? So this is much more thinking about that idea of a story, a narrative. It flows, it goes in different ways, in different directions, as different characters, you may have main characters, you may have sub characters subplots, there's a lot going on with the story. And it's much more broad, it's a much more broad way to think about designing a system.
Play video starting at :5:38 and follow transcript5:38
But that can be helpfully thinking about all the different elements and all the different individuals and how they interact. Now, the scope of the story is that it's broad, but it's not unlimited. So we're always going to have constraints, were always going to be limited, but there are perhaps less defined constraints in a story than if we talk about a goal oriented system. And so these are the three mapping of systems that we might see. We can think of goal oriented systems where it's just about some functionality and you can produce these very easily as forms of technical

specifications. We might have a scenario based design exercise which much more about a robust set of interactions that allows somebody to achieve an intended outcome through a series of steps. And then you can think of stories as more broad exercises that cover a broad spectrum of interactive components, all working together possibly multiple people, possibly lots of different levels of interactivity.
Play video starting at :6:29 and follow transcript6:29
But ultimately all three of these types of design exercises can be useful depending on the type of system in the building. As always, with this course, context is king.

So let's have a little think about the sorts of terminologies that we might see and what they actually mean in the context of interaction design. So like most fields in computer science and software engineering, a lot of the terms can be used in different ways depending on your specialism. So we'll just clarify some of these terminologies for you and just talk about what they actually mean in the context of interaction design. So we'll start out with words. Now words are really important to us. Language allows us to communicate ideas and concepts in a clear and concise way. But actually, if we think about the sentence structures of most sentences, they're somewhat convoluted and they don't necessarily communicate information in a really efficient way.
Play video starting at ::47 and follow transcript0:47
So our job as interaction designers is to use words in a way in which they can convey as much meaning as possible without confusing people about what they actually do. So when we think about words, some good examples of words that might be useful in the context of designing systems are things like stop. We understand stop. That means something significant, that something bad is happening or that there's some degree of danger. Words like go, those sorts of words are really useful in the context of interaction design. They allow us to express something that's very clear and concise. And this is something that you should get in the practice of as an interaction designer.
Play video starting at :1:22 and follow transcript1:22
In thinking about things that can be simply understood by all your users. So you want to be as concise as possible, be clear in what you're trying to express and use words in a meaningful way. In many cases, you don't actually have to use words at all. And that's the benefit of being able to use different types of tools and techniques in this space. Now the second dimension of interaction design and the terminology that I wanted to cover is around visual representations. Now there's quite a lot in the space that we work in. We deal with visual things all the time and they're quite useful for us because we can convey lots of information very quickly through visual structures.
Play video starting at :1:58 and follow transcript1:58
So that might be something like a picture of it or an image of something showing how it's to be used. It might be some icons. So we've all got icons in our windows that we're familiar with. Things like minimize, maximize, close. As people who use technology on a day-to-day basis, we're intimately familiar with these sorts of operations. And we're not going to be confused when we see the minimize icon up in the top right hand corner. We automatically assume we know what that means.

Play video starting at :2:23 and follow transcript2:23
If you change the behavior of that, then that might change the user's perception of how usable your system is. So if it suddenly does something different to other systems that are already out there, that might create a problem for people and you don't want to be reinventing the wheel in that way and thinking, well, actually, you know, I think that this should be designed this way. So if you're used to a certain way of doing things that can be really confusing for people. So it's about not only drawing from those schemas that exist, but trying to rely on things like the user's intuition, their understanding of the world around them and relying on things like metaphors, things that they can see in the real world can be modeled in a system in a visual way, using clues like logos, information around iconography, even things like typefaces that they might be familiar with might provide them a sense of comfort. So again, make sure that your visual structures give some added meaning to something and don't confuse people. Now the third dimension of interaction design, which is one which is perhaps really, really important and one that's maybe underutilized to some degree is the idea of physical objects or space. So this is thinking about things in relation to other things.
Play video starting at :3:31 and follow transcript3:31
So if we imagine ourselves in a room, we're placed in that room with a number of things around us that we can interact with. And that room gives us a sense of where we are and the sorts of actions that we can take place in. So we've got things like affordances that we understand, and this is another really important terminology. Now affordances are things that when we go to something, we have a certain expectation. So when we look at a door handle, we would assume that that door handle can be pressed and the door can be pushed open. Now that's a sense of an affordance, a light switch can be switched on and off. We all know these things.
Play video starting at :4:2 and follow transcript4:02
We're familiar with these things. It's an affordance of an object. Now often when we think about objects, we don't necessarily think about them in relation to other things. So when we design, we tend to design in isolation. And so it can be quite difficult for a user coming to an experience where all the individual elements on a page or on an app or on a piece of hardware or software are designed individually. So we can't just think about those things. We have to think about those things in relation to one another.
Play video starting at :4:30 and follow transcript4:30
Where do they sit? Are they at the front of the screen? Are they at the top left-hand corner where we might expect to start reading? Are they hidden somewhere in a submenu? So thinking about how we group things together and where things are physically placed in an interface is a really important thing. And this is the third dimension of physical objects or the space and proximity and orientation of these objects in relation to one another. And this is particularly challenging because now we work with lots of different devices.
Play video starting at :4:58 and follow transcript4:58
So we've got small mobile phone screens, we've got big tablet screens, even bigger desktop screens maybe. And we as interaction designers have the challenge of having to design experiences for all of those people at once. Now we've got two other dimensions that I want to talk about before the end of this lecture. So the fourth dimension I'd like to talk about is time. And this is the idea that essentially when we deal with media or interactive media, as I should say, we are expecting things to change over time. Now this will impact users in certain types of

ways. So we might find that the user navigating around a product may not understand or appreciate what that modality of work looks like.
Play video starting at :5:40 and follow transcript5:40
So they may not understand how things are happening over time. So when we're dealing with things like motion, particularly around things like animation, that can be a bit confusing, but also looking at things like sound. So sound is something that we tend to represent over time. We don't represent it as a kind of physical object. We think of sound as changing and manifesting in different ways over time. So the amplitude or the pitch of a sound may change as we work through a set of exercises. And so it can be another area that's quite tricky to design for, but it can have lots of value in your products or services that you're building.
Play video starting at :6:15 and follow transcript6:15
Now again, you need to use this sparingly and think about things like animation and how much value and utility do they add versus are they just confusing? Are they just creating unpleasant experiences for the user where things are moving around the page and that's confusing in itself? And then thinking about that mental map that we have, where do things live? Where was that object? It was here a moment ago. Now it's changed. Now the sound's changed.
Play video starting at :6:39 and follow transcript6:39
I don't know where that sound's coming from and so on and so forth. So the final dimension that I'm going to talk about in this lecture is about behavior. Now this is, for an interaction designer, easily the most important of these five, but it can vary in terms of how much value you get out of things. And so the design constructs around behavior can be quite tricky to build because people are relatively unpredictable. So what we're really thinking about here is an amalgamation of all the other things that we've talked about. We're thinking about people's behavior in terms of what we might be able to predict what people are going to do based on all these other elements. So we can think about increasing space on the page to make it clearer that objects live in certain areas.
Play video starting at :7:20 and follow transcript7:20
We can think about grouping objects together, creating icons, changing typefaces, making fonts bigger. There's lots of things that we can do that might actually impact the behavior. And in some sense, the behavior should really be impacting the way in which we design things. So if we find that people are doing things a lot, that might tell us that our design's not quite right. And so we need to change things. We need to change certain elements. And of course, the knock-on effect is still there.
Play video starting at :7:45 and follow transcript7:45
It doesn't go away. So when we make a change, we have to make sure that we're not just testing for that change, but we're looking at the broad spectrum of activities that exist within that scope of things like our goals, our scenarios and our stories and trying to consider that manifestation of actions. So does a change in your physical space or your proximity of objects to one another have an effect on the user's behavior? Does it affect things like other typefaces or other visual clues? Do they no longer make sense? And so on and so forth. So we're considering things in relation to one another.
Play video starting at :8:17 and follow transcript8:17
And when I talk about things being of holistic practices in this particular field in interaction design, what I'm really describing to you is the fact that people are unpredictable and there's lots

going on on a screen, even if we don't always think about it. Just have a look at your mobile phone and scroll through the number of icons you have in the text. Try out an app and just look and list everything that's in the app. And it will give you a sense of the fact that there is a whole magnitude of things that we can do. Lots of hidden features that we might not even recognize, things that we don't use in the app we might be completely not aware of. But in this context, it's important to recognize that new users coming to this system may have a hard time in dealing with this. Take some time to think and reflect on what I've talked about.

So in this lecture I want to talk to you briefly about the history of interaction design and where it all comes from and where we're essentially going with this as well and give you a sense of some of the things that you want to be aware of as we go forward, some of the technologies that might change the way that we think about things. So this is something that's been researched for a number of years by lots of different individuals in different fields. And in the early 80s people really started thinking about how systems could be designed to meet the needs of individuals who are now using these systems. It was essentially a process wherein as technology grew and technology usage grew, we had the opportunity to engage with these systems where we didn't before. And so there was a time period where programmers and developers were the only people who used these systems. They were the only people who necessarily had a need to use these systems. And it was only people who were really technically involved who were really interested in the power that computation could bring that were actually using those sorts of systems.
Play video starting at :1:7 and follow transcript1:07
Now over time we've thought about the manifestations of activities in systems and we've tried to design systems that do different things. So you can think about this in terms of your own experience of systems. A mobile phone system is very different to what you might see on a desktop or a laptop. And of course there's curated experiences as well. So you can think about standardized tools like the toolkits for VR, for instance, where that's a very different experience that you're going to get in that to what you might do when you're on your mobile phone browsing a website versus interacting with the virtual world. Now in terms of the history of this, it all dates back to a really important set of architectures that were really changing the way that we think about systems. So for many, many years, systems were very similar to what we see in terms of command line systems, or if you've used the terminal system before in Unix based systems, they were very similar to that.
Play video starting at :2: and follow transcript2:00
It was a series of inputs and outputs, and it was all about writing long commands to get things done. So some of you might be used to this way of working. And for me, I still find it quite a useful way of working if I'm trying to improve my efficiency. So if I'm doing lots of tasks repeatedly, lots of clicking and stuff probably won't make much sense to me. I don't want to spend hours and hours and hours clicking things. And so for a terminal or a command line interface for me, it makes sense. But for the activities that most people do, where they might want to click on a link if they're web browsing, or they might want to type some information into a document to be presented and read by somebody else, that modality of working isn't particularly

efficient. You could try doing something like that in Vim and find that you have some interesting challenges in dealing with that.
Play video starting at :2:44 and follow transcript2:44
And some of you might be really experienced in using those systems. So something like Vim or Emacs might be workable for you. But for most people, there's quite a steep learning curve there. And so in a sense, we've tried to simplify systems by building things that people understand, basing things around architectures that we know. So this is essentially where the desktop metaphor came from. Most people are familiar with the idea of having a physical desk. Most people go to work and work at a desk, or at least they did before the COVID pandemic took over. And this idea of working with a desk was fairly familiar.
Play video starting at :3:16 and follow transcript3:16
And so we took this metaphor and we applied it to systems. We said, well, this is your desktop. This is where you work. This is the environment that you work in. And everything that you'd expect to see on your desk is on your desktop, with a couple of exceptions. Now, most people don't have a bin on their desk. So the idea of the recycle bin might be pushing that metaphor to a little bit of an extreme set of circumstances.
Play video starting at :3:37 and follow transcript3:37
But for the most part, it makes sense for people. We think about having icons close to us, things that we need to access quickly. So shortcuts. We've got a main menu where we can access things that are kept deep within our desktop. So that might be a filing system or a set of drawers or some other component way of working with your physical environment. And that's a manifestation of that in a digital way. But all this dates back to an acronym from WIMP.
Play video starting at :4:4 and follow transcript4:04
So WIMP stands for Windows Icons, Menus and Pointers. And it's essentially a style of graphical user interfaces or GUIs, as we know them, that uses the idea of widgets. So it's got this idea that we can see in most of our systems that we see today. In fact, we haven't moved beyond this metaphor of having things like Windows. So we're all familiar with Windows. We've got multiple windows open most of the time. And some of us have multiple tabs within multiple windows open if we're working in lots of documents and lots of things at the same time.
Play video starting at :4:35 and follow transcript4:35
But we're familiar with most of these things now. So they're not foreign concepts, things like icons. Most of us can see icons when I look in front of me. I can see icons all around me. I can see icons in my window menu. I can see even icons on my hardware devices that will tell me that something's going on. I can see menus all over the place now, menus, hundreds of menus.
Play video starting at :4:55 and follow transcript4:55
In fact, menus have become much more complex and much more apparent in modern systems because, in a sense, we have more to do with systems. So we've got a larger number of operations that we want to achieve with our system. And having more menus and more sub menus and more items within those menus is a good example of how systems have changed over time. And any of you who do graphics editing or video editing will know that some of those software applications have really, really complex menu systems where you have thousands or maybe even tens of thousands of options. And, of course, pointers, which sometimes this acronym refers to pull down menus. But for the most part, you often see it used with pointers.

So we have a mouse, we have a pointer on the screen, which essentially tells us where the mouse is in relation to the proximity of the screen.
Play video starting at :5:42 and follow transcript5:42
So we can think about the mouse as being physically on the desktop or if you're using a trackpad, it might be embedded in your device. But the mouse itself, we can think about the manifestation of that on the system as a visual structure. We can see it as a little icon, a little graphical representation that says, this is me, this is where I live. And if you want to click on something, you have to move me in a certain direction. So we get that instant feedback. We can see as we move our mouse around physically on the desk or on a mouse pad or on a trackpad, we can see it moving on our screen. And so that makes sense to most people.
Play video starting at :6:16 and follow transcript6:16
Now, this style originally became popularized with the Apple Macintosh. And in fact, now most systems use it. So if you're using Linux or Windows or Mac or pretty much most other types of operating systems, there are very few that don't embed this architecture in the way that they've designed their systems. But this is essentially just a way of thinking about interaction. Now, some people disagree with this and they say that it's not an interaction style. I believe that it is an interaction style and I believe it's a learned interaction style because it's something that we impose on people. So when we talk about the idea, looking back to what I mentioned about affordances, about a door handle or a switch, there's many things in our systems now that maybe don't have natural affordances.
Play video starting at :6:58 and follow transcript6:58
So the idea of a recycled bin on a desktop is a classic example of a system that doesn't look right in place. But there are lots of things that we wouldn't even be able to represent in a physical way. There's lots of things that we do digitally that might be really, really difficult to represent in a physical way or impossible. So maybe you want to think of some of those examples and present them in discussion forums and talk about how you might manifest those things. So what things can you find in your system that you can do as part of a technical set of processes? It might be input, it might be output, it might be things going on in the background. But find something that would be really, really difficult to represent in the real world and give us an example of that and tell us why that example is so difficult to represent and maybe even have a go at representing it if you think that you can.
Play video starting at :7:43 and follow transcript7:43
Good luck.

So I'm going to talk to you briefly about a field that I'm really interested in. A field known as usability, which is essentially a subset of interaction design. We're really thinking about working processes and improving the work that people do in terms of making things quicker i.e more efficient. Or making them less error prone, i.e making them more effective or it might be even about things like safety and accessibility. So let's talk about some general things that we might do to enable our applications and our software and hardware and all the things that we're designing to facilitate these sorts of actions. In fact, a good way to do this is to actually look at bad examples of things and look at the opportunities that we have to fail users. So if we think

about road systems as an example, if we have traffic lights that don't work, that can cause problems, it can cause accidents.
Play video starting at ::59 and follow transcript0:59
But there's a certain amount of trust in that, isn't that? We assume that when we come up to a traffic night, that it will be working. And if you're a driver who drives in a country where roundabouts have lots of traffic lights at them, it's always interesting to see how people will react when the traffic lights are on. Even though lots of roundabouts in this country don't have traffic lights, when people come to a roundabout where a set of traffic lights is broken, they automatically stop. It creates confusion because even though most people know how to deal with roundabouts, there's an unexpected circumstances. So the first thing you want to think about is not putting things in your users way, don't create things that are going to cause confusion or make it harder to get to the thing that they want to get to. If you imagine having to navigate through a maze to get to something that's really important or something that you want to use on a regular basis, you can see how that would cause real issues for you as an individual.
Play video starting at :1:50 and follow transcript1:50
And a classic example of how that's done is looking at supermarkets and how they tend to put really, really popular products towards the back of the supermarket to encourage you to browse through. So as you walk through the supermarket, you look around you, you see other types of products and it creates the opportunity for impulse purchases. And actually all you want is a loaf of bread or a pint of milk, but that's hidden right at the back of the shop to encourage you to walk through and have a look around you and make those impulse purchases. Now, in a sense, it's valuable to a business to do that for interaction designers. That can be really dangerous. And if you look at website design, you often see that there's metrics in place that will show that if people spend too long trying to find something, they'll just jump off the website and go somewhere else, to try and find that information. Because they recognize that in the time they could spend looking for it on your website, they know that they can find it more quickly and more efficiently and more easily on a different website.
Play video starting at :2:44 and follow transcript2:44
So it makes sense to actually design experiences that don't give your competitive advantages. So this is not just about improving your own sets of processes, but also thinking about not losing customers, not creating experiences where people will move away from the services that you're providing and using an alternative. Yeah, now another classic example of things that companies do wrong is providing too much utility, providing too much functionality in systems. And this is something that I come across on a daily basis in my research. So I spend a lot of time looking at digital audio workstations which are essentially just a manifestation of the old mixing desks. There a chance to engage with a set of systems that allow you to process audio and allow you to use audio in meaningful ways. So you can do things like mixing and mastering.
Play video starting at :3:32 and follow transcript3:32
You can create tracks using these things from physical instruments, from digital instruments, and it all comes together through this unified interface where you manage this set of processes. Now, a lot of these systems have a huge number of features. And again, those of you who worked in graphics design or have experience of using these types of software packages might find that they have a similar set of constructs that there's lots of menus, lots of icons, lots going

on on the screen. And this can be intimidating for users. And so you often see with these software packages, they provide different views or perspectives. You can go into things and look at design views. You can go into things and look at architects view, for instance, with a lot of software packages where you're building or engineering software, there's often a gooey based view where you can represent things like widgets and then there's a code based view where you can write those out in code.
Play video starting at :4:24 and follow transcript4:24
So it's about providing the right experience for the users at the right time. And if you can do that, you can avoid all these issues of giving people the wrong type of interface or giving people the wrong type of information. And this is often done through things like customization. So one of the ways in which we all see this done is through advertising. So we have customized adverts in what we do. We tracked across the web now by many different websites and you'll often find that adverts are tailored towards you. So again, it's about making the experience of getting you to purchase things much more quick, much more likely to happen as well.
Play video starting at :4:57 and follow transcript4:57
But we can take these lessons that we learn from outside in the world and import them into our designs and think about things that we want to do. But also think about things that we definitely don't want to do, and we don't want to create experiences for people that are unpleasant or in some cases even unsafe. So we can look at dark design practices and we'll talk about this a bit more later on in the course. But we can look at ways in which system designers can encourage people to do things that they might not want to do. They can encourage negative types of behavior, and they can create, in some cases what we call uncomfortable interactions or interactions that are unpleasant. So we'll talk about this a bit more and we'll look at some reading into this activity as well.

So I'd like to do an activity with you where we look at some examples of things that might be unpleasant and we create situations where we might want to imagine a set of circumstances where we would actually want to create an unpleasant experience. So I'm going to give you a couple of examples and then hopefully you can come up with some more based around systems that you might want to design or build. And then we'll look at some reading around some things that people have already built and we'll look at some of the things that my previous or former colleagues worked on as a research article, looking at the idea of uncomfortable interactions. So let's think of a few. Now, I wonder a lot of people are familiar with is going to theme parks and visiting theme parks to have what you would consider normally to be an uncomfortable interaction. So that might be going on a ride where it travels very quickly or there's a plummeting feeling. If you've got like a vertical drop on a ride that might be something that in most cases will be unpleasant.
Play video starting at :1:8 and follow transcript1:08
But in this situation is desirable. And again, water rides where you get splashed with water for most people, if you're walking to work and somebody drove past you and splashed you from a puddle, you probably wouldn't be very happy about that experience. But in the context of a theme park where that experiences curated, it's something you go to for a thrill. You go there with the idea that you're going to experience something unpleasant. But there's a factor relating to that in terms of your own personal preference around actually wanting that experience and

enjoying the experience. It could be a social experience. It can be something you do with your friends or your family and it's nice to experience those things together.
Play video starting at :1:45 and follow transcript1:45
So that's one example of an uncomfortable interaction. So what I'd like you to do is to come up with some other examples and think about how you might design a system that would facilitate that. Now I'm going to have to put a caveat here and I'm going to have to say, I don't want you to design anything that could be considered unethical. So it's fine to give people these experiences where they come to things and where they have this experience where it might be unpleasant in some way, but it could also be desirable. Now there's a fine balance here between designing something that's going to be fun and a bit silly and a bit challenging of the norms versus designing something that's intentionally there to harm people. So it's not sufficient to design a system for creating intentional harm, but to create an experience where somebody might be uncomfortable, not her or in danger or in any real sense of danger. But things like the perception of danger might be something you want to design into your example.
Play video starting at :2:44 and follow transcript2:44
So I'll give you one more example before we go. And this is an experience that most people have had in their lives an experience that some people either really enjoy or really seem to hate, depending on who you ask. But when you think about films and games in the horror genre, you can think about those as being quite pleasant experiences for some people. So I recently got the chance to play resident evil seven in VR and it's an absolutely terrifying experience and I thoroughly enjoyed it. But there is no way whatsoever that I would be able to convince my wife to do that. So, those are two people who are interested in curating different types of experiences and the idea of uncomfortable interaction in this case, in my case would be something that would be desirable. But in other people's cases will be something that they would challenge and say, I don't want to experience that at all under any circumstances.
Play video starting at :3:32 and follow transcript3:32
So your job is to come up with some ideas and some designs of an uncomfortable interaction and think about how that might manifest in the system. And it would also be good if you could think about some of the safety features that you could implement to make sure that that system doesn't create unnecessary harm to people in the real world. So of course we want to create thrills and we want to create situations where people can get those desirable experiences, but we don't want to hurt people. So how could we make sure that we can do that without hurting people? Have a little think about this. And then we'll look at some examples in some research papers about how other people have done this sort of thing, and some of the careful considerations that took place to allow these things to happen.

Welcome! In this video, we will explore a fundamental approach to interaction design, user-centered design or UCD. UCD ensures we design not just for users, but with them, putting real human needs at the heart of our design process. So what exactly is UCD? User-centered design, UCD, is a broad term to describe design processes in which end-users influence how a design takes shape. UCD is more than just a method, it's a mindset. It is both a broad philosophy and a variety of methods.
Play video starting at ::49 and follow transcript0:49

There is a spectrum of ways in which users are involved in UCD, but the important concept is that users are involved in one way or another. The term user-centered design originated in Donald Norman's research laboratory at the University of California, San Diego in the 1980s and became widely used after the publication of co-authored book called User-Centered Design System, New Perspectives on Human-Computer Interaction, published in 1986. User-centered design starts with users, not features. It is a multidisciplinary design approach that actively involves users at every stage, so we can better understand their goals, behaviors, and environments. As Don Norman reminds us, it's not the user's job to adapt to the product, but the product's job to adapt to the user. That means we don't just test at the end, we include users early and often. One of the most common pitfalls in design is focusing on features too early in the process.

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But UCD challenges us to flip that approach. We begin with users, their goals, frustrations, tasks, and build outward. Instead of asking, what can this technology do? We ask, what does the user need to do? And how can we help them do it? User-centered design is grounded in several key principles. Early and active involvement of the user during the design process.

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Design decisions are based on what the user needs and wants. Clarification of user and task requirements, identifying these requirements is critical, and business goals should be aligned with clearly defined user needs. User feedback is incorporated into the product's lifecycle. Design teams collect and analyze user feedback and perform regular usability testing. The product is improved using an iterative design process. Developers make changes gradually as they test features and gain a deeper understanding of their target audience. This process isn't linear, it's iterative, flexible, and always user-led.

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We will have a closer look at the UCD process next. Let's walk through the UCD process. User-centered design is an iterative process that focuses on an understanding of the users and their context in all stages of design and development. Generally, each iteration of the UCD approach involves four distinct phases. First, understand context of use. Who are your users? What are their goals, environments, and limitations?

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As designers working in teams, we try to understand the context in which users may use a system. Second, specify user requirements. What do users need the system to do? Here, we identify and specify the users' requirements. Third, design solutions. Create wireframes, prototypes, or mockups based on user input. This is the design phase in which the design team develops solutions.

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Fourth, evaluate against requirements. Test those designs with users. Are their needs being met? This is an evaluation phase. Here, you assess the outcomes of the evaluation against the users' context and requirements to check how well a design is performing. More specifically, you see how close it is to a level that matches the users' specific context and satisfies all of their relevant needs. After this, the team makes further iterations of these four phases, and you continue until the evaluation results are satisfactory, which means you loop.

Play video starting at :5:26 and follow transcript5:26

And loop again. UCD is inherently iterative. Iteration is the beating heart of UCD. Rather than designing once and launching, we build prototypes, gather feedback, and refine. Even rough sketches can help us identify major usability issues early, and each round of testing reveals new insights until the solution fits user needs with clarity and ease. Next, let's have a look at some common methods used in the user-centered design process. We have already discussed the four core phases.

Play video starting at :6:11 and follow transcript6:11

Now we are zooming in to see how each phase is typically carried out, highlighting techniques typically used in practice. To understand users and context, we gather insights through interviews, surveys, field observations, and diary studies. These help us create personas that represent user types and needs. To define requirements, we make sense of that data using affinity diagrams, thematic analysis, and journey mapping. We also use user stories and task analysis to define what the system should do. In the design phase, we generate ideas through brainstorming, sketching, and ideation workshops and express them with wireframes and prototypes, from low to high fidelity. Finally, to evaluate with users, we run usability tests, A-B tests, and use think-aloud protocols or heuristic evaluations.

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Feedback forms and surveys also help capture users' responses. Together, these methods help ensure that we are designing not just for users, but with them at every step. We will explore many of these methods in more detail later in the course, and you will have the chance to try some of them in your own projects. Now that we have seen how user-centered design works in practice, let's briefly reflect on its advantages and disadvantages. On the plus side, UCD leads to products that are more effective, efficient, and safe. It helps manage user expectations and increase satisfaction. Involving users early gives them a sense of ownership.

Play video starting at :8:12 and follow transcript8:12

Products often need less redesign and fit more smoothly into real-world use. And the collaborative process often leads to more creative solutions. But there are trade-offs. UCD can be costly and time-consuming. It may require specialized team members and broader stakeholder involvement. Some user insights can be hard to turn into concrete design decisions. And designs may end up too tailored, limiting reuse for other projects.

Play video starting at :8:48 and follow transcript8:48

So while UCD offers strong benefits, it's important to weight them against practical constraints. Still, the long-term payoff, more effective, usable systems is worth it. To wrap up, UCD places users at the center of every design decision. It's iterative, evidence-driven, and multidisciplinary. And when done well, it leads to products that are not just functional, but better and empowering to use. As a final thought, reflect on how UCD supports each stage of the design process. From understanding users and defining problems to designing and evaluating solutions.

Play video starting at :9:35 and follow transcript9:35

It's not something you apply at the end. It's something that shapes how you design from the very beginning. In this course, you will learn different approaches that build on these principles, including design thinking and the double diamond, both of which follow and expand on the UCD process. So as you move forward, keep asking, who are we designing for? What do they need? And how can we make their experience better?

So in this exercise, we're going to look at an opportunity to do design in a refined way and thinking about the process of design and what it actually means to design meaningful solutions to problems. Now, this is essentially what interaction design is all about. It's about understanding the space, making sense of the things that are going on around us and allowing us to conceptualize these things in meaningful ways. So how do we do that? That's a really, really difficult thing to do, isn't it? It sounds really tricky, but actually, we can simplify it. So we're going to look at a strategy that allows us to simplify this process and add meaning and structure to the work that we do. So we're going to look at a specific process, and some of you may have seen this process before if you've done design thinking work. It's also prevalent in a number of different industries and different manifestations. So the way that people think about this as a conceptual idea is different depending on which environment you're in. But the techniques and tools are essentially the same. It's about thinking about this overarching process that brings value to a set of interaction design possibilities. So let's have a look at what we're doing. So we'll be exploring the double diamond concept, and we'll be thinking about how it helps us to come up with good solutions, how it helps us to form ideas, to refine those ideas, and then create something meaningful from that process. So we'll start out with the first diamond.
Play video starting at :1:31 and follow transcript1:31
Now, the first diamond is perhaps the most important, because if we get this part of the process wrong, then we're going to struggle with the next bit. Now, this process is really starting off with an idea or a concept or a question. We're thinking about, well, what are we doing? Where do we go from here? What is this thing that we're trying to solve? We start out with a lot of unknowns in this space, and we say, well, actually, I don't know where we are. I don't know what we're doing. You might start out with a brief. So it might be if you're working in industry or working for another company, you might get a brief come in that says, we'd like to build this thing. We'd like you to come along and help us to optimize our working processes, to make us more efficient, help us to deliver on some given goals. But we want you as a systems designer to come in and think about that whole process. And so as a systems designer, you're capable of doing this. You can come in, you can look at this as a question and say, well, actually, okay, you've got a good question and that is a good question. That's viable. We can actually come along and build that thing for you. We can test that thing for you. We can evaluate that thing for you, but we're going to take a certain process. We're going to follow a certain set of steps that allows us to deliver on this thing. And we want to make sure that we're successful.
Play video starting at :2:38 and follow transcript2:38
Of course we do. We want to do the best possible job that we can. So we're going to follow this process. We're going to follow the double diamond design process. Now that's a mouthful. So we start off with an idea and most of us can probably figure out where we go from an idea or a question or a concept or a brief or a challenge. What most of us will probably do is jump on Google and just go, okay, I'm going to look up this thing. How have other people done it? What's out there? How can I make sense of this space? How can I solve this problem? And that's a reasonable assumption. So we start out with a process that allows us to understand the space, understand what's going on around us, understand the rich context of this particular situation, this particular thing that we're trying to solve. So we follow this process of what we call divergence. So we follow along a particular axis. And in this case, it's this axis. So we'll follow along this axis in this process of divergence.

Play video starting at :3:34 and follow transcript3:34
So what this means is we're opening up ideas and concepts. We are broadening the scope of our thinking process to allow us to consider a range of possible permutations and options and considerations. We're taking lots of different types of things into account to allow us to encapsulate all these things and add meaning. So when we do our research, if we do a literature review, for example, we'll go away and we'll read lots of things and we might follow a path. So we might look at an interesting paper in the field, look at all the references, look at papers that are being referenced by and papers that are referencing from that and follow that trajectory and try to understand the dimensionality of the research. Now, design is no different to that. We're really thinking about this process of divergence and opening up in this first stage of the first diamond. And we're thinking about, well, as we move along this trajectory, how can we make sure that we do good quality research to allow us to come about with interesting and valuable solutions? And eventually we'll get to a point where we're happy, where we're satisfied that we've done enough research to say, OK, I understand the problem space. I've got some primary research. I've gone away and maybe I've asked people some questions. I've followed some focus groups. I might have even done some ethnographic research. So I've done some field-based research where I've watched people in what they're doing and I understand the domain. I understand the system.
Play video starting at :4:55 and follow transcript4:55
I understand the subtle connotations of the working environment and how people are doing things now. I may have also looked at lots of secondary research in this domain. So I might have a good understanding about what other people have said about the space. So what we're doing here is essentially we're at this point now where we have lots of unstructured research. And we've followed this divergent path right the way through to this point. So we've got here in this middle point, we've followed this divergent path where we've been quite open. We've opened our eyes to the world around us. We've looked at everything that we possibly can. And now we're satisfied. Now we've got all this unstructured research and we need to make sense of it. So what would be the sensible way to do this? Well, the sensible way to do this will probably be to go in the other direction, to go back down this diamond and follow it down the other path. Now this is a process called convergence.
Play video starting at :5:45 and follow transcript5:45
And what we're essentially doing here is given that we've got some research already, so we've formalized our research. We've got some good understanding about what we're doing at the top of this diamond. We're happy. This research is really important to us. It's really valuable. It gives us a broad scope of the space. We understand everything that's going on in this general domain. We want to refine that now. We want to go through this process of definition.
Play video starting at :6:7 and follow transcript6:07
We want to say, well, actually, what does this mean? We need to define this research in a particular way. We want to specify. And in some cases, this is a process of synthesis. So it's taking that research and picking out the good bits, picking out the bits that are interesting and valuable to us. So what we're really trying to do is to draw out meaningful insights from this process. So we're thinking about, well, what things are happening? How can we categorize our research into meaningful categories? How can we ensure that we're thinking about the sorts of things that we can do to deliver on this premise, to deliver on these ideas and concepts? There

is a gap in the market and we're going to fill that gap. And this is how we could go about doing it. This is how we might take this research that we've done and add meaning to that. And we get to a point here at the end of the first diamond where we want to think about now moving on to the next diamond. But this is far ahead of this process now. Once we've got to this point of convergence, we've actually reached the maximum point of convergence we can for this particular diamond. So we've basically said at this point, OK, we've got that initial brief, we've got that initial question, and we have some good understanding about it now. And we have to go forward and we have to add some meaning to this in terms of actually delivering on some of these concepts. So it's all well and good that we've understood the domain and we've understood the space. But now as software engineers, we have to go away and build something. As hardware developers, we have to go away and make something. As engineers, as people working in a business, whoever you are involved in this process, you have to deliver on these concepts. And we're going to talk about this in the next lecture. In the second part of this, we'll talk about the second diamond, the double diamond in this process. I look forward to speaking to you soon.

So welcome back. So in our last lecture, we looked at the first part of the double diamond design process. We looked at this process where we went and got some research and we tried to conceptualize that research in a meaningful manner. And if you remember now, we ended up about here somewhere, I believe, where we were essentially converging on the last set of synthesis around our research. So we've got some good understanding of the concepts and we're now actually thinking about this next step, this next process of delivering on this diamond. So I'll keep my first diamond, but actually we're moving on to the second diamond now, aren't we? Because we follow this process for the first time and then we understand what we've done here. But we now need a second diamond to allow us to bring this to fruition. So let's jump ahead and let's start on this second diamond. So we're now on diamond number two, return of the diamond. And diamond number two is very similar in terms of structure. So it looks the same as diamond number one. All diamonds look the same, to be fair. But the process itself is quite different. So what we're thinking about here is we're thinking about this process of actually delivering on the things that we want to deliver on. So our first process is really about understanding research and the dynamics of the domain. This process is about, well, now we have a refined research question. We've got some good understanding. We think we can build this thing. And so we're starting from this point now, from the end of the first diamond, moving straight on into the second diamond. So we're here. And let's go ahead and think about what that looks like.
Play video starting at :1:38 and follow transcript1:38
So again, we've got this same process. We've got this same way of working. We've got this divergent process up this axis, a divergent process. So we're opening our eyes up. We're looking at the world around us. We're saying, OK, we're going to take all possibilities and permutations into consideration. We're going to have an open world view about how to deliver on this thing. And then we also have this process of convergence. So in our second diamond, we follow the same process as we did in the first. We follow this process of opening up and then closing back down, divergence and convergence. But in this case, it's actually about delivering the thing. So in the last case, we looked at the research, the conceptual frameworks around

this. How can we ensure that we're in the right domain, in the right space, that our understanding is reasonable? We've tested our assumptions.
Play video starting at :2:24 and follow transcript2:24
Now we're actually going to go away and think about building this thing. So this first part, this divergent set of processes in the second diamond, is really about the process of development. And it's about development of ideas. So we've got a good understanding of the research space, but we don't necessarily have a good understanding about how to bring this to fruition. It's a bit like making a recipe. Over time, you follow this process of you refine the recipe ever so slightly. And over time, hopefully that recipe will improve. So you're achieving all the things that you want, and you've got a desirable dinner at the end of it. So in this process, what we're really thinking about in the first half of this double diamond is this ideation process.
Play video starting at :3:4 and follow transcript3:04
How do we conceptualize ideas and evaluate the success or failure of those ideas? So we take an open mind and we think about this divergent process is really about understanding our ideas in context. Are some of these things going to work? Probably some of them will. If we have a hundred ideas, if we conceptualize that from our research and say our research has shown us there's a hundred different ways that we can do this, we might have a synthesis process, which is about taking groups of 20 of ideas and picking out the good ideas from each set of 20. And from each one of those, you might have one good idea. So at this point in time, we now have five ideas that could all be viable. So we're opening up our process.
Play video starting at :3:44 and follow transcript3:44
We're thinking about how do we take these ideas and make sense of them? So at this point in time, when we get to the end of this divergent process, we have our ideas. We have all of our ideas that we could potentially use, and we have to now go away and think about delivery. So we've not left ourselves much time really, have we? This seems a little strange that we've spent all this time thinking, and then this last kind of small section of one half of the last diamond really. So it's only a quarter of our entire process. We're actually spending delivering on things. But by this point, we already have a really good understanding of both the research space and the conceptualization of our ideas. So we've ideated, we've come up with some ideas, we've evaluated those ideas. So we've actually followed an internal divergence and convergence process in here. We could have another diamond in there that talks about that, that process of divergence and convergence. And then we've got to this middle point here where we're actually halfway across this second diamond. We're nearly done now. We're 75% of the way there.
Play video starting at :4:46 and follow transcript4:46
So what do we do in this last bit? Well, this bit's the easiest bit. This is all about implementation. So we've conceptualized everything. We've got a good understanding of the space. All we have to do really is deliver. Sounds simple. So how do we go about delivering on all these ideas and concepts in relation to all the things that we've already talked about?
Play video starting at :5:5 and follow transcript5:05
Well, really, there's only three steps to this. We build, we test, and we iterate. We iterate based on feedback. And we feed back into that build process, and we build, and we test, and we iterate. And we make some changes, and then we build, and we test, and I think you know what comes next. We iterate. And eventually, we reach a point where we can say, actually, this will probably work. We have a solution. We've solved our problem. We have an answer to our initial

question. It might be a research question. It might be a question about individuals, or people, or a dynamic, or a set of working practices. We might have delivered on some kind of product, or some kind of service. Maybe we're delivering a service, a continual delivery of a service that can also work in this approach. But essentially, we've solved the thing that we intended to solve. We set out with an idea or a concept at the beginning of the first diamond.
Play video starting at :5:58 and follow transcript5:58
And in this second diamond, we work through this process of taking that synthesis of research and making meaning of it. Contextualizing it in a meaningful way, and building something that people are actually going to be able to use, that will add value to people's lives, some utility, it'll be usable, accessible. So what we're really thinking about here is delivering on concepts. And this is what interaction design is all about. But to get here, to get to this end point, we've got to think carefully about what we're doing. And we've got to follow this path of divergence and convergence. Open up to all the possibilities and permutations.
Play video starting at :6:31 and follow transcript6:31
Give yourself enough scope to understand the broad dynamic of the area, and then settle in on some core conceptual ideas. Open up to the idea that there might be more than one way to build this thing. There might be more than one technology solution. It might be about hardware. It might be about software. It might be about neither. It might be about improving working processes, but be willing to open up in this second diamond. And then again, follow that process of convergence to ensure that you're building and testing and iterating, and that something good is coming out from this process. And this is really what interaction design is all about.

We're now going to start talking about some research methods and techniques that allow us to capture bits of data that are interesting to us. We want to find out about what we're doing. We want to find out about people's experiences. We want to design positive experiences for human beings. We're going to start by looking at our users. Now, the first question we have to ask when looking at our users is what information do we need to design for these particular users or this particular use case? Our job is essentially about thinking about the nature and manifestation of activities that allows this person to achieve some given sets of goals or some task or some scenario.
Play video starting at ::45 and follow transcript0:45
Generally speaking, most people use a system with an intent to do something to achieve some set outcome. Our job as interaction designers is to think about this in lots of different ways and trying to understand the complexity of this use case. What information do we need to design for these particular users? Well, you'll be surprised how much data we can capture to help us to understand how we can impart certain experiences and ensure that people have positive experiences through design, through research, and through understanding of the users, their needs, their wants, and their desires. We might start out by looking at the user's motivations and saying well, the user has some things that they want to do. Now, we can generally speaking think of motivations as a need or desire supported by an action to achieve a goal. An example of this might be, I have a need to eat food or to drink or to do exercise.
Play video starting at :1:37 and follow transcript1:37

All those things are manifested through a set of activities. When we think about the activity of being able to eat and being able to survive, we have to purchase food products, we have to prepare those food products and cook them in a particular way and then we have to eat those products. We have to think about that set of actions, that cycle that allows us to substantiate that logical of being satiated, of being fed, of being nourished in a particular way. Also thinking about making sure that we can achieve these goals in a way in which we can achieve our own expectations too. It's not just enough to eat any type of food. I've preferences around things that I like to eat. I have a certain dietary restrictions maybe.
Play video starting at :2:14 and follow transcript2:14
I've got other manifestations of activities, so I might not have access to fresh produce all the time. You've got to think about all these different things that might be happening during the course of somebody trying to achieve a goal. Users motivation can vary from person to person. We have to think about these things in understanding and appreciating that complexity that's embedded in those experiences. We can move on now. We could think about more complex examples. We can think about what our user might need from our system.
Play video starting at :2:44 and follow transcript2:44
We're thinking about this one-to-one interaction, usually in this space. We're thinking about how user might come to our system, what they might do with our system to allow them to achieve certain things. If we're designing a system for a specific purpose, that makes it easy to understand what the user's actions and intents are. But it's not always the case that systems are designed with a specific purpose in mind. All systems can be misappropriated and misused to achieve other things, so we have to embed more complex experiences in our interaction design processes. What we really want to be thinking about, the question that should be in our minds when we're thinking about this, is how might the interactive possibilities manifest? What actions, what routes can people take for our system?
Play video starting at :3:25 and follow transcript3:25
How does the system facilitate certain types of actions and how can we ensure that those actions are meaningful and robust and allow the user to achieve the things that they want to achieve? Now he talks about this a lot during the modeling processes, looking at things that mental models and mappings and how we think about systems and how we interact. Things like the Gulf of Execution and the Gulf of Evaluation in relevant here. But we also want to ask interesting questions if we want to find out interesting bits of information about the manifestation of activities that allow users to achieve things. We want to look at what they're already doing, we want to look at manifestations of other systems, and then we want to ask specific questions around the design of our own system to ensure that we're building a system is relevant, it's contemporary, it meets our user's needs and wants and desires. In terms of our users it's not just about some intent and then selections. It's not just about clicking something or pressing a button or typing on a keyboard.
Play video starting at :4:21 and follow transcript4:21
Most interactions are more complex than that. We have to embed this level of interactivity in our systems and help to understand and appreciate how this complexity can manifest across the series of actions and the series of intents to achieve something that's much more complex than just trying to click on a button. It's not just about the motivation being achieved through a set of actions. In this case, we think about we have some goal and then we want to achieve that. We

have some motivations to do something. I'm motivated to click on a button, I'm motivated to interact with the system and therefore I can achieve my goal. There's actually some steps in between that we need to think about.
Play video starting at :4:56 and follow transcript4:56
We need to think about well, what are those steps and how complex are they? How difficult is it for users to engage with this? Based on things like the way in which they interact with things there. Preconceived understanding of the system, their learning, their memory, their capability to interact at that given moment. There's a lot of complexity that we might have to embed in our systems that takes these things into account. What we're essentially getting at here is that interaction is complex and nuanced. We'll look at some examples of how we can better understand our users and our use cases through understanding and exploring different methods for research, different methods for design, and different methods for evaluation.

In our last example, we looked at users and we thought about broadly what users are going to be doing with our system? Are we tried to understand those implicit set of interactions. We try to explicate that process by understanding what a user might want from a system and how we can capture that bit of information to allow us to understand how to build interest in user experiences. Now that's one way of looking at systems, but we also want to look at other things. One of the other things that we can look at is the task, the manifestation of activities that allows a user to come about with some end goal. They want to maybe read a book or they want to do a degree, for example, that's a really good example of a task and it's relevant. In this context we think about this manifestation of actions in a very, very specific way.
Play video starting at ::54 and follow transcript0:54
We can model these things and we've talked about modeling, but we also want to capture lots of rich, interesting bits of data around how people interact with things so that we can understand how, why people interact with systems, what they're trying to achieve, and what the manifestation of those activities might look like if you were to design or improve upon an existing design of a system. We can think about a task as a way in which we can achieve something. We want to achieve some intention, we want to achieve some outcome, and to do this, we have to complete a set of tasks, don't we? If you want to do a degree, we can't just do that as one set of steps or processes. We have to follow specific routes through the degree. We have to do examinations, coursework, lots of reading, watching videos, lots of discussions with our peers, talking to our peers and learning from each other. It's a really great way to learn and it's an experience that's really important to us in our learning journey.
Play video starting at :1:45 and follow transcript1:45
If we don't take part in those activities, we're actually selling ourselves short a little bit, aren't we? We're not really achieving what we potentially could. When we think about task as a manifestation of activities, possibly lots of activities, when we look at degree program for instance, we're really thinking about what are the actions, what are the steps, what are the stages involved? What path could I take through the system to allow me to achieve the thing that I want to achieve? Let's think about our tasks. Our task would give us some certain possibilities. We have some given objects that we can work with.
Play video starting at :2:18 and follow transcript2:18

If you're doing a degree, your objects probably books, reading material, most of this is digital, videos, etc. We've got lots of different objects in our system. We've got lots of different ways we can interact with our system. There are also interaction possibilities, we could choose to watch a video. We could choose to do some reading, we can do it online, offline. We can work in lots of different ways to substantiate those goals. It's not necessary that we have to follow a specific route to learn something, we can learn in lots of different ways.
Play video starting at :2:46 and follow transcript2:46
Again, there's experiences that are really dynamic and that's one of the challenges with working with data in HCI in understanding those dynamics sets of human processes and, of course, we have some views of the system as well, we have some views of how we perceive things, our perspectives, our conscious objects within the system. Thinking about those relationships between objects in the system is also relevant. Thinking about how can I supplement my learning from some reading, by watching some videos? How can I supplement my learning from watching videos, by doing discussions and so on and so forth. When we think about tasks, one of the easiest ways to conceptualize this is to actually break a task down into a series of formative steps. We can say, well, if I'm going to make a cup of tea, I'm going to follow a specific procedure all the time. My boiler kettle, get your cup out, put your teabag in, pour in the hot water, pour in the milk and so on and so forth.
Play video starting at :3:38 and follow transcript3:38
There's steps and stages and if we don't do this in the right order, that can create problems for us. It can create situations where we're not achieving the thing that we want to achieve. If I don't study enough, I'm not going to get a good degree classification. I need to make sure that I'm studying the right way, that I'm understanding the material, I'm absorbing the material and I'm engaging in the right way. When we think about task-based perspectives, we're often thinking about this very, very granular way of people interacting with systems. We often don't think about this at high level, we don't think about what we're doing. When we're driving, we don't think on changing gear, I'm speeding up, I'm slowing down, I'm maneuvering, I'm signaling, I'm doing all these other activities quite a lot of the time it's some of those things are just on autopilot.
Play video starting at :4:21 and follow transcript4:21
We don't really think about them. But when we are designing systems for users and we are designing systems to complete a set of tasks, we have to be very, very careful in thinking about all these steps and stages involved in the process. Some general considerations around cost-based perspectives. We cannot just assume that a task is a generic set of procedures. Tasks can be achieved in all manner of ways. When we work on tasks, when we work on achieving something, the user can take lots of different routes to a system. It's really ultimately about them getting to their goal.
Play video starting at :4:54 and follow transcript4:54
It's about them achieving the thing that they want to achieve. The process by which they do that can be quite dynamic and it can shift from person to person and even from interaction to interaction. I might be interacting with a keyboard today, tomorrow I might be using mostly my mouse, the day after I might be using touch-based interfaces. The manifestation of interactions can change and I can achieve the same things using lots of different types of tools. Again, this is nuance. Some tasks are really complex and some tasks are also dependent on other things. We can think about building up tasks through a set of procedures or a set of processes and that can

get really complex when you think about things like flying aircrafts or firing of missiles and things like that, they are very, very complex processes that involve lots of mathematics, lots of understanding of physics, of engineering possibilities.
Play video starting at :5:44 and follow transcript5:44
Designing systems isn't as simple as it seems.

So now that we've had some time to think about our users and think about the tasks that our users want to achieve, let's think about some broader considerations. Let's take this wider and let's scope out further. So let's think about the environment. Now, you might think that your environment is not really a necessary consideration, there is something that we can't control, something that we can't necessarily design for, and that's not necessarily true. In fact, a lot of my research is about building systems that are designed to work in different types of environments and embed that complexity of the environment in the design of the research and the design of the tools and technologies to support different modalities of working. So generally speaking, when we think about environmental considerations, we can start by thinking that most of us don't work in isolation. We have colleagues, we have friends, we interact with systems in different ways, we interact with systems often to talk to one another, to interact with one another.
Play video starting at :1: and follow transcript1:00
And so we can impart our understanding of the system as a set of dynamic social processes. We can think about computer systems as being augmented by social types of interactions. They're not limited by just some set of functional possibilities, are they? We can think about playing games with our friends, we can think about interacting using social media and chat based applications. So there are lots of possibilities and there are lots of permutations in both the design of our system to facilitate these actions and in thinking about the possibilities that the environment might impose upon us. Some constraints and restrictions, some challenging environments that might be difficult to work in. So context will generally impact how we interact with things.
Play video starting at :1:41 and follow transcript1:41
It will generally dictate what we're doing, if it's raining or if it's cold we might put a jacket on. These are simple examples of how context and contextual implications can affect us, but there are more broader implications in terms of how we interact with systems. So on a sunny day sometimes, it can be very difficult to interact with technology, you get glare on your screen, it might be really warm and so that affects your laptop, it can't cool down quickly enough. So we have to think about all of these things when we're designing and building systems. So when we think about the environment, we can start with the physical environment and we can think about the physical environment impacting us and changing the way that we do things. But we can also think about designing for physical environments. We can think about where is this system going to live?
Play video starting at :2:25 and follow transcript2:25
Who's going to use it? What sort of interaction possibilities are taking place? And how is the environment going to impact people when they're using systems? Now, a classic example is, have you ever tried using a touch screen device in the rain? It's a pretty challenging experience for most people. It's one of those problems where quite a lot of the time, people would just avoid

using the system in that case. But we could design for this, we could design experiences that would support this modality of work.
Play video starting at :2:52 and follow transcript2:52
So it's not to say that that particular constrain, that particular nuance or that particular subtlety around how we interact has to restrict the way that people do things. It's just a case of the system has been designed a particular way and in certain environmental conditions it causes problems. So this idea of all actions being one-to-one is quite dismissive of reality. We have these implicit set of interactions that we do with systems every day that we don't really think about, so typing, looking at the screen is something that we expect to do. But actually the complexity of systems means that we have to design for a whole manner of possibilities around interaction and possibilities around restrictions about how we can interact with system. So if I lose cell service, I can't make phone call, that's a generally appreciated understanding of the system. But, of course, nowadays we have Wi-Fi calling so that we can pass that data over a different type of network.
Play video starting at :3:44 and follow transcript3:44
So we have manifestations of systems that can support different types of interaction and can allow us to overcome problems and challenges. And this is what we should be aiming for as interaction designers. Not creating problems for our users, but ensuring that we can build systems that allow users to bypass problems and ensure that they can achieve the things that they want to achieve with our system, even in problematic scenarios. So, some classic examples of questions that you might want to ask yourself, do you cover your PIN when you use an ATM? So if you want to get money out using an ATM, you put your card in, you generally put your PIN in and then you decide how much money you want to get out and you get the money out and take your card back. Now, this type of interaction is something that most of us are familiar with, but most of us will also cover our PIN because there's a degree of privacy, we want security around our data and so that's a consideration of the environment. That's a consideration, that we're actually taking into account all the different things that are going on around us and then changing the way that we interact with the system based on a set of processes, based on our general understanding about the environment and the way in which these things are working.
Play video starting at :4:45 and follow transcript4:45
So you may also want to obscure the view to your phone when you're having a personal conversation and this is another example of adapting to the environment. Now, we can do this as individuals and our systems can also adapt to the environment. So many modern mobile phones now have adaptive backlighting controls where they can detect that there's quite a bit of sunshine and change the backlighting to allow you to still be able to read your emails and your text messages and be able to interact with your device. So this is a two way relationship, this is a two way thing. We can design systems that support different modalities of engagement and the systems can be designed in a particular way to support us and we can think about how systems can change and how our interactions with systems can change based on our understanding of the environment and the things that are going on around us. So when we think about the wider environment, there are also considerations to take into account. We have to conform to certain standards.
Play video starting at :5:38 and follow transcript5:38

Now, the obvious standard is the legal standard, but what about other context of use? Of course, we have to follow the law, but there are a lot of contextual implications around how we interact with systems. It might be considered rude to get my phone out and start texting people while I'm having dinner with some friends or some colleagues. So, again, we have to think about the context of use in this. And we have to think about these environmental considerations in terms of, what are the people in the room? Who are the people in the room? What does that dynamic of interaction look like?
Play video starting at :6:7 and follow transcript6:07
Am I going to interact with my phone the same way that I do with my smart TV or my laptop or my desktop PC or my touch-based interfaces or my games consoles. There's a whole set of considerations that take place and where we interact with systems and how we interact with systems and the things that we want to achieve all relevant in this consideration. It's not just enough to think about a human being and behaviors. It's not just enough to think about these as task based type interactions and it's not just enough to consider the environmental constraints. We have to consider all three of these things, and all the complex interactions that take place within these environments and how these things can have knock on effects on one another. This is a real challenge with interaction designing, thinking about all these different parts and how they can work together.

In this lecture we're going to talk about data. We're going to talk about how data can help us to make useful, valuable decisions. When we think about data, we often think about it as being representations of lots of numbers, and digits, and numerical representations, in orders of things. Processes that are quite dynamic and shifting all the time. Now, we want to contextualize data in our rich contexts, we want to understand what data actually means. Based on a given set of steps or scenario, and so we can do that. We can do that in HCI. We can understand, how data can be used to inform design decisions.
Play video starting at ::44 and follow transcript0:44
Now this can happen at both ends of the spectrum. We can think about this as a set of early design processes, to encourage us to build things like, really nice early prototypes, that meet our users needs. But we could also carry on using data, once our system's been published and billed. We can think about using things like, analytics to inform design decisions, and change the way that our system works. Based on some sets of feedback, based on how people interact with our system. Data can be used across the entire trajectory of actions that we perform, of course, this course. We can think about data as being useful at every step, and every stage of this process.
Play video starting at :1:19 and follow transcript1:19
We need data to inform. We need data to help us to reflect, and we also need data to help us to understand things, or understand things about things sometimes. It's not just about understanding x impacts y, is about understanding x impacts y, z, a, b, c, and so on and so forth. When we change something, when we look at data, when we reflect on activities, we're often thinking in this holistic wave. As well as thinking about granular level data. Now, the obvious question to ask is what sorts of data is going to enable us to make the right decisions? This is a tricky question that I've been trying to answer for a number of years now, and the research that I've done and certainly my colleagues in the same area, have the same sorts of problems.

Play video starting at :2:1 and follow transcript2:01
We have access to lots of rich types of data about our users, about interaction possibilities, about modalities of interaction. But how can we use this to make meaningful decisions? This is the really interesting question to ask. In fact, data can be used in lots of different ways. I'm going to describe different types of data to you. Then I'm going to let you think about how these data might manifest across a set of activities, and how we might use them to think it's a high-level, about understanding how systems used, and also thinking about those low-level interaction possibilities too. All data tends to boil down into one of two things.
Play video starting at :2:36 and follow transcript2:36
We either have quantitative data, which from the quantity you can probably guess what that's all about. It's about numbers, and figures, and orders and representations. We also have qualitative data, which is much more descriptive rich, contextually driven data about things like experiences, and scenarios. When we think about qualitative data, it tends to be much harder to quantify. Quantitative data, we often see things like spreadsheets or in databases. Qualitative data might have representations, as interviews, as logs of interviews, or transcripts, we could also think about it as on video footage. I guess this lectures is an example of a type of qualitative data, in the sense that we're getting video.
Play video starting at :3:18 and follow transcript3:18
You're getting some audio of me talking, you're getting some transcripts. All that data is being used to manifest across, a wider set of interaction possibilities essentially. Let's talk about quantitative data first. Now, quantitative data, often tells us about numerical representations of things. We can think about things in terms of numbers. Now examples of these might include things like, completion rates. How many people have been able to complete a task?
Play video starting at :3:42 and follow transcript3:42
For example, WE might be looking at more task time analysis. How long does it take somebody to do something, or we might be thinking about the order and structure of things. You might be thinking about, well, which steps did a person take? In the process. Maybe we think about a set of steps is one to five, and that's the way we'd imagine somebody goes to a system, that maybe they'd slightly two and then do three, and then four, and then they do five, and they'd finish with one. This could be another example of quantitative data. Anything that we can quantify, anything that boils down into numerical representations, or things like order, or any structure that we can impose on things.
Play video starting at :4:18 and follow transcript4:18
It's generally speaking what we think of as quantitative data. Now of course, there are benefits, and there are disadvantages of using any type of data to try to inform us about things that we want to do, and things that we want to achieve. The benefits of quantitative data, generally speaking, rely in the opportunities for standardization. We can normalize data. We have a starting point. If we think about metrics around quantitative data, we might think about height and weight as being specific metrics. We can measure somebody's height.
Play video starting at :4:49 and follow transcript4:49
We can measure that in a quantitative way, so we can compare people commonly, we can say this person is taller than somebody else. We can say this person is shorter than somebody else, and so on and so forth. We can make these interesting comparisons and we can do this across lots of people. We could have hundreds of thousands of people, to be measure and then we can

figure out average heights, and means, and medians, and modes, and all these other types of calculations. We can even calculate more advanced things. We can look at percentages of populations, and look at maybe the 99th or the 100th percentile for extreme values. We can look at the split of data.

Play video starting at :5:23 and follow transcript5:23

We can look at specific demographics. We can add framing to this and start to understand how people with different characteristics, might conform to certain rules or standards. The advantage of quantitative data really is, that we can compare it easily. That we can capture lots and lots of quantitative data. We can use it to inform us in a standardized way, through sets of metrics, usually. Now, generally speaking, you'll see quantitative data used to inform things based on statistics. Influential and descriptive statistics are really useful to help us make sense of data and help us to understand data.

Play video starting at :5:57 and follow transcript5:57

We can build models, we can build mathematical models that help us to understand what the data means and whether we've come about the data by chance, or whether there seems to be likely that there's some pattern or trend or some emergence from the data that suggests something interesting to you. Now, the obvious advantages that with quantitative data we have more generalizable characteristics. We can take that data and we can generalize across a population. We can capture lots of data, lots of people and we can generalize. We could calculate the average height of the population if we captured enough data, we could say, I'm quite happy to make that generalization now. The data informs us in that way, but it might not be particularly descriptive. It doesn't tell us the whole story, does it?

Play video starting at :6:39 and follow transcript6:39

This is the challenge with quantitative data. It doesn't tell us why people from a certain place in the world might be so tall or it doesn't tell us why certain characteristics and properties have come about. We don't get that understanding, that rich detailed understanding about what's underpinning that. This is where qualitative data comes in. Now with qualitative data, we have the opposite end of the spectrum. We're looking at contextually rich data about specific things. This is usually focused around human behavior.

Play video starting at :7:9 and follow transcript7:09

We've focused on what humans are doing, why they're doing things, how they're doing things, and it's generally speaking, a more dynamic sets of data. When we think about quantifiable data, it might be the size of clothes, we've got numeric representations or large and extra large and so on and so forth. With qualitative data, we might not have those codifications opportunities. We might not be able to say that this particular characteristic fits in this framing. We might have to spend quite a bit of time understanding what these things mean in a broader context. In this sense, the data's not as generalizable. It's more specific to individuals and interaction possibilities for that specific individual or that specific group of people, rather than those generalizable properties that we've talked about.

Play video starting at :7:54 and follow transcript7:54

Now, when we talk about the limitations in terms of generalizability, we also have to think about how difficult it can be to compare qualitative data. If I give you a rich account of how HCI works and the background of HCI, and modeling techniques, and all the other things that we're learning on this course. You're getting quite a lot of rich data, aren't you? But you probably

struggled to compare that to another course and say, "Well, what have I learned based on this course? If we were to compare it to another course?" If you were to do a math courses or a psychology course, it'll be very difficult for you to compare the videos and say, "That imparts more learning or that imparts less learning." Because they are dynamic.
Play video starting at :8:32 and follow transcript8:32
Because there's a lot of objects involved in these systems. There's a lot of different sets of processes. When we think about our rich account or something, when somebody's talking and describing things, there's lots of different ways that we can describe things. There's lots of different types of language that we can use. We can talk about things using different types of syntax and different types of semantics and that creates this embedded complexity. When people use certain words, they might not mean the same thing. Well, they might use different language to describe the same things.
Play video starting at :8:59 and follow transcript8:59
Again, quantitative data can be hard to compare and contrast, and we have to do a lot of work to untangle that complexity. Now, which method is better did we prefer quantitative or qualitative? You probably have your own views about this stuff, but it depends on who you ask. Of course it depends on the task, and it depends on the research agenda. The answer is that neither qualitative nor quantitative is better in general terms. But we can talk about them spurt for specific types of things that we want to measure, for specific types of things that we want to research and then we can understand their value, in those contexts. We can say that quantitative data is really good at representing characteristics around how people use the system, for example.
Play video starting at :9:42 and follow transcript9:42
We might be able to calculate things like the average time it takes to click on a button, which we've looked at before. That's an obvious case for quantitative data we want to compare and contrast. But we might also want to rich account of why a person uses one system over another. Now, we wouldn't be able to do that in a quantifiable way. We won't be able to quantify that as an experience. I guess we could look at statistics and data about people using systems, but that wouldn't tell us why. It wouldn't tell us how they engage with their systems and why they made a choice of using one system over another.
Play video starting at :10:14 and follow transcript10:14
Why that we use Windows over Linux. Why don't we use Linux over a Mac-based operating system, or Android or some other operating system that's out there. We all make these choices. We decide on the systems that we want to use. These choices are specific to individuals. There's a lot of contextual richness that's embedded in that. We're looking at these two things that we're thinking, Well, they have competing objectives.
Play video starting at :10:36 and follow transcript10:36
But that's not the case. Generally speaking, it's good practice to mix and match the approaches. Take a bit of quantitative stuff where it fits. Take some qualitative stuff, we're fits, and try to put them together. Try to use them in different ways to validate your assumptions. If you've done some task time analysis, you might want to interview your users afterwards and ask them about their experience with the system. That way you get some understanding about what they've done in terms of quantifiable metrics that we can measure and compare.
Play video starting at :11:5 and follow transcript11:05

But we also understand why people have interacted to certain way. Why they've clicked on a certain button, why they have some misunderstanding about how the system works. These two bits of data can be put together to allow us to draw more interesting conclusions. In this case, both quantitative and qualitative data can be rigorous and valuable insightful approaches. We can take both of these approaches, and it's not to say that one's more scientific than another, or that one's a more rigorous or one's more valuable than another. It really does depend on the contextual implications of what you're trying to find out.

I'd like to talk to you very briefly about something that is really quite funny, and something that we tend to see in natural environments quite a lot. I want to talk to you about the idea of a desire path. Now, when we think of a path, we can think of a route through a system or route through some set of systems, and this could be something like a walk home. We can think about the route that we take from work to home and then from home back to work again. Now most of these routes we spend a lot of time on the road, we're following certain conventions were following road patterns or roots for buses or trains. But in some cases, desire paths are all about subverting these routes. Let's imagine that we are walking along and we come to a point where there's a grassy verge or there's some area of grass.
Play video starting at ::53 and follow transcript0:53
Let's draw some grass out here. Let's imagine we've got a nice grassy here we have some trees. Now, imagine a scenario where our path is here, and we want to get to this point, we want to get to this area. We have two choices. We could follow a path that's been designed for us, and we can take a route, well, we could do what most people do, and we can actually cross this unintended path. This is what we mean by desire pots. When we talk about desire paths, we are talking about the most optimum route for individuals.
Play video starting at :1:25 and follow transcript1:25
Now this isn't just about geographic locations and this isn't just about physical parts. We can think about this in terms of systems design. We can think about the route through a system. Now I want you to imagine a ticket machine that you've used before, and we've probably overused ticket machines. Now these are quite complex constructs and there's a lot of things that we have to do. Let's think about designing a ticket machine to begin with. In a ticket machine, we have to have a method of payment, don't we?
Play video starting at :1:52 and follow transcript1:52
We'll just model this as Pound because that's a reasonable model of assumption. We could think about money payment systems of this and we could think about connotations of this. We could have payment through a card, so we could use card, we could use a debit card or credit card. We've got offshoots of this potentially, of things that we can do. How else could we pay? Well, we could pay with cash. If we really want it to be contemporary there are other options, aren't they?
Play video starting at :2:19 and follow transcript2:19
I've never seen a systems that could pay with a check that wouldn't work, I wouldn't imagine, but there's no reason that we couldn't pay wave cryptocurrency. That gives us another option. What's the challenge with designing a system like this, that has all these features and functionalities in? People are going to struggle, aren't they? Is going to create problems. The more inputs we have on a system the more options people have to pay, the more challenges it's

going to create about finding the right route through the system. Now we talked about this in a desire path of crossing a grassy verge to get to somewhere, and avoiding the path, taking the most optimum, the most efficient route, which from a graph theory perspective makes perfect sense.
Play video starting at :2:59 and follow transcript2:59
From a human perspective about saving energy, it makes perfect sense. But somebody's gone along and design that path for a reason, and that's been designed wrong in essence. We have designed that in a way in which we expect people to follow a certain behavior, and that patterns change based on a person's needs and wants and desires. It's a classic example of poor design. May the payment systems that you see will have these similar features. We can think about this desire path. We can think about a situation where people want to pay for something and they want to do it quickly.
Play video starting at :3:29 and follow transcript3:29
If you're paying for some parking, for instance, if you've popped in shopping center car park, and you want to pay for your parking, generally speaking, you want get that done as quickly as possible and get on with your day. Now, a classic case of understanding this comes in many forms and you can see lots of buttons and obscuring things and covering things over and crossing things out on the machines try to make it clearer. But one of the classic things I always say in these types of machines is the most used button tends to be the Help button. Now this suggests that we've got the design monk. If the Help button is a mangled mess on a ticket machine, you know that people are getting stuck. What we have to think about is simplifying that experience and making sure that that experience maps to the mental model that people have of the system, and making sure that that experience is a logical one. It's one that people can follow.
Play video starting at :4:19 and follow transcript4:19
Maybe we only have two payment systems. Maybe we have two ways of paying, and we label them cash, and card, and we settle on that as a construct. That's a perfectly valid design. Just be aware of this when you are designing systems that desire paths are a very real thing, and we can actually use these as a way to design new systems. We can think about what people are doing in terms of these types of interactions, we can look at existing systems, and we can take feedback from these existing systems to design the next system, the future of systems. This is a type of data that can be really useful. Don't be frightened of using this in your studies.
Play video starting at :4:56 and follow transcript4:56
Be aware of the fact that there might be problems with this, but it's a good starting point and it's a point that we can draw from, it's a point that we can expand upon and we can validate our assumptions down the line. But for now, it's a pretty reasonable starting point.

So we're going to finish up now we're talking about existing forms of data and how we can use this. Now this might seem a strange way to finish the topic we've talked about collecting original data for quite a lot of the topic. And now we're talking about using existing data, and you might wonder why I've chosen to cover this at the end of the topic instead of the start. But in fact this is perhaps the most useful type of information we have access to because if we're designing a new system, we're trying to learn lessons from our old systems. We're trying to understand how those systems could be designed in a better way, but what features of the systems already

work? Now in this lecture, I'm going to talk about two particular types of data, and I'm going to talk about why they're useful and why they can help to inform design decisions. So the first thing I'm going to talk about is log data, so typically these features as analytics.
Play video starting at ::57 and follow transcript0:57
And you might see them defined as other things, but essentially we're looking at log data as a means to give us some information about some things. So we could look at the website and we could look at the interactions in our website, and we could say, well generally speaking people would take a path through the website. So they'll land somewhere typically on the landing page, which is the first page here, it's our index dot HTML page usually, and then they'll take apart through the system. And then at some point in their journey they will leave the website, and they'll go into another website, and they're going to do other things, or they'll turn the computer off. Now we can use our analytics to understand what people are doing, so if they spend 30 seconds on the homepage they're probably spending a lot of that time looking at images or reading or engaging with the content. If they then spend 20 seconds on the next page, might be a short page with some basic content on, maybe they're just logging in or registering or doing something like that. If we then find that they spend 500 seconds on the next page, that's telling us something interesting, it's telling us that there's something important on that page, that they're spending a lot of time on now.
Play video starting at :1:59 and follow transcript1:59
Now we don't know if that's a good thing or a bad thing it could be the case that they're having a really positive experience. It could be the case that this is really good and these 500 seconds are rich engagement. They're engaging with something on the page, they're really getting into the reading, or watching videos, or they're engaging with the content in a really rich and informative way. It could be a bad thing, so that 500 seconds might be 499 seconds of them getting lost and then a second of them clicking the link to the next page. So again, we have to use these in a sensible way, and we have to understand what people are doing, and we have to break these things down into meaningful structures. Now we can do this in terms of the design of the website, can't we? We can say, well maybe I want to break some of these pages down if people are spending lots of time on a page, I might want to break that page into two pages.
Play video starting at :2:45 and follow transcript2:45
To try to break down that analysis and make sense of what's going on in that page and try to see where those 500 seconds are being spent. If I break that down, I then get a sense of where the positive and negative experiences are in the page and those analytics become more rich. Those logs that we've got become more rich in helping us to understand what people are doing and why they're doing things. The other thing that you might find this useful from log data is looking at exit points. So we talked about entry points, we've talked about our lending page, but if a lot of people going on to your site and then leaving from a particular page that might be telling you something useful. That might be telling you again, one of two things, it could be telling us that there's something positive or that there's something negative. The positive might be that they found the content that they want, and they're quite happy they're going to move on with their exercise or with their day now that they found the thing that they need.
Play video starting at :3:34 and follow transcript3:34
Or it could be they're leaving because they're having such a negative experience that in fact they're going to go to another website or access that content somewhere else. So we don't

really know and that's the challenge with this sort of data we can't encapsulate more rich data than we already have. But we might have is thousands or tens of thousands or hundreds of thousands of cases where people are doing these things. So we can build up generalizable properties around the population of people using these tools, and we can use that data in a meaningful way, but it doesn't give us that rich context that we want. So what else can we do? How else can we understand the existing context of use, and how can we use that information to give us a really interesting insights into what we could possibly do? Well, we could look at code reviews, we could look at examples of similar products, and we could gain some insights into what matters to users.

Play video starting at :4:22 and follow transcript4:22

So we could have a look, and we could make sense of things like reviews. So we could have a look at some reviews of some similar software let's say we take three software applications and doesn't have to be software, could be hardware or anything else. And we can look at the reviews, and we can see what's positive and negative and we can extract those features out, can't we? We can say, well actually there's two positives and one negative from their system or there's two negatives and one positive from this system or these systems are positive, so we want to take all the features. And what we can do is we can amalgamate these features into a set of constructs in our own system. So we can say, we're going to take the good parts of this system, the good part of this system and the good parts from this system, and we're going to impart all of them in our system. We're going to drag all these properties over, and we're going to use these to design an even better system.

Play video starting at :5:9 and follow transcript5:09

So we can take all this rich data, and we can create hopefully an even more positive experience for the users where they have the best possible experience that we can provide them. So we can use this interesting data that already exists, and it doesn't have to be reviews, it can be looking at code, it can be looking at properties of particular software attributes, it can be looking at analysis that other people have done. It could be looking at literature, it could be looking at market research, it could be looking at things like financial data, but there's lots of different sources of information out there that we can use to help us to understand. Help us to verify that we're doing things the right way and to validate our assumptions and say actually we had some assumptions that this feature was good, but they've told us in this system that it doesn't work. And so we've got a real problem there we have to think about how to carefully design this feature to make sure we encapsulate all the good parts, and we remove the negative parts. So don't be frightened to look at existing data, don't be frightened to look at how other people do things and try to learn from other people's mistakes. It's one of the easiest ways to design systems, don't make this mistake yourself, learn from other people's mistakes.

How are we going to collect this data from our users? In this topic, we are going to look at a number of different methods that you can use, and right now what I want to talk about is interviews. Interviews are incredibly powerful tool for us as interaction designers when we want to collect rich and interesting data from our potential users. Now, it's likely that your experience of interviews might be a scary one in that you may have only experienced an interview, in setting where you're trying to apply for a job. In which case, it might be a very high pressure

situation, a very formal situation, so you might think of interviews as fairly intimidating. However, interviews actually don't need to be like that. As interaction designers, the interviews that we are going to be running will be a little more informal and a little more conversational and that is the benefit of the interview.

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The fact is that an interview is basically just a conversation between two people, and we as interaction designers can benefit from that because our users are already trained in how to be interviewed, because our users have conversations with their friends and their family every day. This is really useful for us because it means that our participants are relaxed and are ready to think clearly and express themselves freely. Now, the types of conversations that I love are conversations where I ask someone questions and I get to learn a lot about them, about their thoughts and their opinions on things, and then they ask me a lot of questions and then I get to tell them exactly what I think on those matters too, those are my favorite conversations. The conversations I hate are the ones where I'm asking all of the questions and the person I'm talking to doesn't ask me any questions back. I find that really boring because I like sharing my opinions. Now this actually is a really bad trait for a interaction interviewer. Because as an interviewer, when we're having a conversation with someone, we actually need to remember that we only need to ask questions and then we need to stay quiet.

Play video starting at :2:17 and follow transcript2:17

It's really important as an interviewer that we don't share our own opinions or we don't try and talk for too long. That can be very frustrating if you're someone who likes a good conversation where you get to share your own opinions too. But given that I've just told you that interviews are so frustrating to do if you like talking, why am I trying to sell them to you? Why am I trying to convince you that as an interaction designer, there are beneficial method for collecting data. Well, it's largely due to the fact that interviews allow us to collect incredibly rich data from our participants. This is for a number of reasons. The first is that we are having a conversation with our participants verbally often.

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This means that there is no extra processing time needed for our participants to tell us exactly what they think. They don't need to have a thought and then think to themselves, how am I going to express this in a very formal way if I type it out or how can I make this shorter because I can't be bothered to write for very long. If you are asking people to write or type their responses, they can often find it difficult to do or they can often shorten their responses. If we are talking to someone, this means that the moment that our participant has a thought or a feeling on a matter, they can express it to us immediately and they don't need to worry about how they're presenting that to us. This means the data we get is nice and free and truly a bit more representative of what they think. The other reason that interviews are really beneficial is that they can be quite flexible. We can adapt what we're asking our users and we can listen to what our users are telling us, and then adapt our questions further.

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If we ask users just to simply fill out boxes or fill in, write in text areas, then that means that our users are very constrained to give us information that we have already asked for in the form of the questions that we've decided to use. In an interview however, if a user decides to go off on a tangent and discuss something ever so slightly different, then we are able to adapt as the

interviewer and we're able to follow that tangent if we think it sounds interesting. It may sound quite counter-intuitive that if you as an interviewer have laid out some nice questions, that it would be a good thing for your user to go off on a tangent and talk about things beyond what you've already decided you will talk about. But actually, this is one of the brilliant pieces of information we can get from an interview. From an interview we get the answers that our users will provide us to our questions, but we also get our users telling us what questions we should have asked. If a user appears to go off on a tangent after you've asked them a question, that's because the user is telling us, you should have asked me about this. That's the user telling us this thing is particularly important to me and that's a brilliant piece of information.
Play video starting at :5:15 and follow transcript5:15
As an interviewer, you need to make sure that you are truly listening to your participant when they tell you that. It's not enough just to say, "Okay, I've listened to their response to my question, onto the next thing." You also need to be doing some processing as you hear those answers and you need to decide whether or not the user wants to tell you something else. Because it's on these tangents that we as interaction designers, learn about the questions that we should have asked. Let's have a look at the different types of interviews that we can use because there are a number of options open to us as interaction designers. Interviews can occur across a spectrum of structure. We can have very high structured interviews or very low structured interviews.
Play video starting at :5:56 and follow transcript5:56
Low structured interviews are often known as open-ended question interviews. Along this structure, there is also a middle ground for semi-structured interviews. Why do we have all of these options? They have different benefits and different downsides. Let's explore what those are and when you might want to use them. On the low structured side, this is a beautiful way to get some really rich and interesting and detailed data from your users. It's often quite good when you are in the very, very early stages of a project, and you don't really know what you should be asking about.
Play video starting at :6:30 and follow transcript6:30
The way a low structured interview works is that an interviewer will arrive at the interview and they won't have a list of questions they want to ask. They'll just have a general topic they want to talk about. This is likely to be the product that they're working on, or the area in which you are going to be doing your interaction design. In these situations, the interviewer asks prompt questions to the users, and then a real natural conversation occurs. As the interviewer, you get the chance to listen to what the user is saying. Whenever they say something that you are particularly intrigued by, you can follow that particular topic of thought and ask further probing questions of your interviewee. Then you end up creating some wonderful rich data set of information that your user has told you is important to them, and you have decided it is interesting.
Play video starting at :7:21 and follow transcript7:21
It's a great way to get this unstructured data when you are in a situation where you don't really know what you should be asking about. This is one of the beautiful, beautiful things about interviews. At the other end of the spectrum, we have high structured interviews. In a high structured interview situation, the interviewer will turn up with a list of questions that they want to ask and an order in which they're going to ask them. Then will proceed during the interview to

ask those questions. There's still some flexibility in that the interviewer can let the interviewee talk for any amount of time to answer those questions. But it means that if the interviewee goes off on a tangent or tries to talk about something else beyond this question sheet, then the interviewer has to bring them back and ensure they are still answering these set questions.
Play video starting at :8:10 and follow transcript8:10
That goes against what I've been saying about interviews. That goes against this idea of collecting this rich unpredictable data from our users. But sometimes for interviews, we really want some consistent data. Perhaps we want to be in a situation where we can compare the answers between all of our participants. In a high structured situation, you can do that. You can be certain that the response to Question 3, for every single participant is a response to the same question. This is really beneficial because it means you can compare those answers, and you can look at how many participants found this particular thing interesting.
Play video starting at :8:47 and follow transcript8:47
How many participants found this particular thing difficult, and so on and so forth. High structured interview, while it doesn't allow for that space to get that rich of pieced data from our interviewees, it does allow for a lot of comparison, which can be very useful if you are trying to establish some kind of commonality between your participants. You can see that both the high structure and the low structure have their upsides and their downsides. Low structure is good for the rich data, but very difficult to compare. High structure is good for comparisons and creating an understanding patterns in your users, but it's not great for collecting that rich of piece data that sometimes your users provide. In the middle luckily, we have something called a semi-structured interview. These are really beneficial for taking the good bits of both of these sides and spectrum.
Play video starting at :9:41 and follow transcript9:41
In a semi-structured interview, the interviewer will turn up with a list of questions they want to ask, and then order that they want to ask them in. Much like the high structure situation. However, if after Question 2 the interviewee says something particularly intriguing, then the interviewer has the chance to say, oh, can you tell me more about that, or why do you say that? and asks some generally more probing questions, thus allowing that level of serendipitous data collection. It means that there is a structure and you can be sure that every participant will have generally been asked the same questions, which allows for some comparison. But you have given the interviewer permission to explore the area a bit more and allow themselves to be open to discovery when the user talks about something that might be a bit beyond what the questions are that you have on that sheet. You can see that there are benefits and downsides to all of the interview styles on this spectrum.
Play video starting at :10:43 and follow transcript10:43
It's up to you to decide what it is that you want to get from your interview, what data you want to collect, and therefore, which structure would be best for you to use. Now before I finish this introduction to interviews, there's one trap that I want to warn you about. This is a trap that I have seen so many interviewers fall into when they've been interviewing me. It's something that can really spoil an interview and can really ruin your chance of collecting good data. That trap is something that is just really human instinct. Now imagine that you are an interaction designer, and you are sitting in front of a user, or a participant, who you are asking questions of, and maybe you're asking them questions about a piece of software that you have been involved in

the design of, or they're telling you things about a topic that you yourself have researched quite a lot, because that is the area in which you're conducting your interaction design. Now imagine you asking them questions, and they begin to give you answers that you know to be technically wrong.
Play video starting at :11:47 and follow transcript11:47
You ask them about a feature of a product, and they tell you that the product doesn't have that feature, or they say, "Yeah, I never use that feature. It never works properly" or "I can't ever get that to work. I really hate that." It's only human nature, if you know the answer to try and help them. If someone tells you that they can't get something to work and you know that you have the information to help them to get it to work, It's very natural to want to say to them. "Oh, no. Here's how you get this feature to work.
Play video starting at :12:17 and follow transcript12:17
It does actually exist on this software product, you just need to do this step then this step." You might feel like you're being incredibly helpful at that point, but actually you're spoiling the interview. You're spoiling it for two different reasons, one, if someone tells you that they can't get something to work, then you immediate correct them and tell them how they can get it to work, then it's likely that your user will start to feel a bit stupid. They've just shared with you that they find something difficult, and you've essentially told them that they are in the wrong and here's how they can improve themselves. Whilst that might feel like you're being helpful, you're actually maybe making them feel a bit stupid, or making them feel a bit silly about themselves. This is a really bad mental state for our interviewee to be in, because if they're feeling like that, they might be feeling shy or sheepish. That means that they might not be likely to answer all of our questions for the rest of the interview, because they're worried that you will judge them again.
Play video starting at :13:14 and follow transcript13:14
The second reason that this is a bad thing for you to do as an interviewer is because, if a user is sitting in front of us telling us that a feature doesn't work, then they are telling us that it has been designed badly. If they are telling us this, then the data they are giving us is the most rich, wonderful, useful data that we could possibly ever hope for. If a user sits in front of us and tells us that they can't understand some things so they never use it and you correct them, then we don't learn anything else from them. However, if they tell us that this is how they feel, then you as an interviewer ask further probing questions, then we start to gather a lot of data about exactly why this feature isn't working, or what has been designed incorrectly to ensure that the users don't really understand how to use it. If a user expresses this to you, then instead of telling them how to use the feature, you need to be ready to say, "That's interesting. Why do you think that feature doesn't work?" Or "How would you expect this feature to work?"
Play video starting at :14:15 and follow transcript14:15
Or "How would you want to design this? How would you expect it to function?" That, however, is an incredibly difficult thing to do, because it means you have to ignore your impulse, your urge to help the person in front of you, and to help them to understand how to use something. It's likely that, if the user in front of you is telling you this thing doesn't work, then there are many other users out there who share the same feeling. You need to listen to this user, because they speak for many. As I said, this is such a common trap, and I have seen so many interviewers correct me as I tell them something that they perceive to be wrong. That means that I've

stopped sharing my thoughts, and they've missed out on some of the information from me, because I feel a bit silly then, and it's really hard for you to avoid falling into that trap.
Play video starting at :15:5 and follow transcript15:05
I want to give you a little task that you can try to help you to avoid that trap, or at least train you to be brave and sturdy enough to not correct your users if they tell you something that you perceive to be wrong. This is the exercise; I want you to imagine that you have been given a brief by a company to help them design a website. This website is on topic X. The company I've said to you as the interaction designer, we have lined up some potential users for you, and we'd like you to interview them on topic X. Topic X here has to be something that is incredibly important to you, maybe your favorite book or film, your favorite hobby or sport, and you are going to interview someone about that. I want you to find a friend, or a family member, who you can ask probing questions about that particular topic. It's extra hard if you could find someone who particularly doesn't like that thing that you do.
Play video starting at :16:5 and follow transcript16:05
I want you to ask them their opinions, their thoughts, ask them how they interpret it, and I want you to just listen to them tell you these things. It's very likely some of those thoughts are going to be things that you don't agree with, it's very likely that their interpretation will be wrong of certain things, but you need to be the interviewer, and that means that you need to be silent, and you need to just listen. It's incredibly difficult to hear someone that you love talk badly about something that you love, but you are going to need to just listen, and not share your own opinion. In that way, you will be beginning to train yourself as an interaction design interviewer. You'll be training yourself to listen to what your users are saying, without you needing to step in and correct them. It's going to be hard, but it's going to be useful. Good luck.

So now that I've hopefully convinced you, that interviews are a really useful tool as an interaction designer for learning more about our potential users. Let's look at what's actually involved in running an interview, because interviews really need to be carefully planned, in order to get the best data out of them. The first thing you need to think about is exactly, what type of interview you want to run. We spoke about the difference between high structure and low structure, and what the benefits of those were. It's important that when you're planning, you think about exactly what you're trying to do. If you're in an exploratory phase and you're very early on in that double diamond, you might want to think about using open ended interviews. That means that you will have a few questions that you want to explore, but you don't necessarily have an exact structure, you want to let the interviewee tell you what you need to ask.
Play video starting at ::58 and follow transcript0:58
If however, you're later on in the design process, and you have more of an idea about what you want to do, and you want to use the interview to confirm or explore some of your initial ideas. You might want to use a high structured interview situation, in which case you might need to have some of your questions planned out carefully. Now, when I'm talking about planning these questions, you need to ensure that the questions you're designing are not too complex, and that they don't involve a lot of jargon. It's very easy to think about something you want to explore, an idea you want to talk about with your interviewee, and it's very easy to accidentally join a few

different questions up in the same question. This can be really difficult for the interviewee to follow, and it's likely going to end up in the interviewee only answering one or two of those questions, which would ultimately be pretty useless for you. So it's important that if you read through your questions, and you realize that you've accidentally created one that has a lot of different sub components to it that you split it out, and that you ask those questions separately to your participant. That way you'll ensure that they have the proper amount of time, and memory to remember what question they have been asked, and therefore to give that useful information to you.

Play video starting at :2:14 and follow transcript2:14

Similarly, it's useful to remove jargon. You might have been working on a project with a lot of detail and a lot of time, and you might be really familiar with all of the terms that are involved in that topic, your interviewee might not have that level of vocabulary. And so, it's important that you look at the questions you're asking, and ensure that you are asking them in the plainest language you can to ensure that everyone understands what's being discussed. If you talk about a technical term and your interviewee doesn't understand that term, they may ask you to clarify which would be great, but they may not and they may make assumptions, meaning that you get some very strange answers from them. Now, I've just told you there that you need to make sure that you add questions, but I want to make make it clear that you don't need to just keep adding and adding questions. Interviews are ultimately a very expensive use of your time and your interviewees time and for that reason, you need to make sure that every question you are asking is there for a reason. It's very easy to make an assumption about the types of questions you think you should ask, maybe you've been asked them before in interviews, and you think that's the type of thing we should be talking to our interviewees about.

Play video starting at :3:30 and follow transcript3:30

But, I really encourage you to reflect on all of your questions, and think about exactly what it is you want to learn from this interview. It's quite likely that you put a few questions in there, that don't actually contribute to the ultimate goal of what you're trying to do. If you can remove those questions, then that will make you and your interviewee more comfortable. The less time you need to spend in this interview, the better and actually the more time you can contribute, the more time you can give to those other questions which might be more useful to you. As with every method that we are using, you need to make sure that you are reflecting on what you're doing. What I mean by that is that after every interview, I want you to stop and think about how well that interview went now. I'm not asking you to relive particularly embarrassing moments, or anything like that.

Play video starting at :4:22 and follow transcript4:22

What I'm asking you to do is to look at the questions you asked, look at the answers you were given, and reflect on whether or not they were all very useful. Reflect on whether or not some need to be added, some need to be removed. Whether or not you need to re-plan the interview entirely, because actually you didn't get towards the data you wanted to collect. What kind of ridiculous is, if you plan an interview and then you just go ahead and you talk to all of your participants, even if you realize that the interview isn't quite right. What is more sensible is to be iterative, is to reflect after every interview, and work out whether any changes need to be made. In that way, you will make sure that you are really collecting the data that you need. And then as

a final thought here, I just want you to reflect before you start doing your interviews, on whether or not interviews are exactly the right thing you need to do.
Play video starting at :5:13 and follow transcript5:13
As I said before, interviews are very expensive in terms of your time. They involve you and one other person collecting that data, and that's not a high yield of data considering the number of people involved. There are other methods that will get you more data more efficiently. Now, I'm not saying that you should never use interviews, but it's important that you can justify the use of an interview in this situation, because it's very easy to just jump to an interview and assume that you need to do it, without ensuring that that's definitely the right thing to do. Maybe, there are other options available to you. How are we going to go about finding participants for our interviews? What I'm not going to tell you about here, is exactly where you find pools of participants.
Play video starting at :5:58 and follow transcript5:58
It's likely that for the different projects that you'll be working on, you'll have different pools available to you. It might be that if you've been hired by a company, they will provide you with a series of users. It might be that you need to go out and find them. It might be that actually you have access to friends of friends, who would be appropriate. Now, let's discuss actually, what we mean by appropriate there. Who do we want in these interviews? How are we going to choose our participants?
Play video starting at :6:24 and follow transcript6:24
Well, the participants will likely need to have some sort of experience, help us represent the users and the stakeholders, who we are ultimately designing for. So we need to think about whether or not the participants need to have experience with a certain technology, whether they need to be representative of a certain sample of the population, whether they just need to have a simple interest in a particular topic. You need to ensure that you are choosing participants, who will ultimately give you data that will help you design for your end users, and your end stakeholders. Another thing that you need to consider when you're finding your participants, is exactly how you're going to reimburse them. It might be that informal settings, you can reimburse a participant in an interview by saying, that you will take part in one of their interviews. This is likely the case in your studies. That's a perfectly acceptable way to do things.
Play video starting at :7:17 and follow transcript7:17
However, you do need to ensure that you are definitely representative of the right sample of population, and that your friend who's doing your interview similarly, is actually a representative user. In more formal settings, reimbursement will likely look like money or vouchers. This is very important, because an interview is a very time consuming thing for someone to take part in. And by offering someone reimbursement, you are showing that you respect the fact that they have offered to help you out, and that you respect the fact that it is going to take up a lot of their time. So you need to ensure you understand, and have planned for how you will reimburse your participants. So now let's talk about the actual running of the interview. What things do you need to consider, and what's important to running a successful interview.
Play video starting at :8:4 and follow transcript8:04
The first thing you need to do, is ensure that both you and your participant are fully comfortable. It's very likely your participant will give you quite quick and sharp answers, if they need to use the toilet halfway through the interview. It's likely that you might miss something that you're

participant has said, if you are sitting on a rickety stool and you're a bit worried that it might collapse underneath you. It sounds obvious, but please make sure that everyone involved in the interview is completely and utterly comfortable. Comfort doesn't just extend to physical comfort, I'm also talking about comfort in terms of the interviewee, understanding what's involved. When you start your interview, you need to make it clear to your interviewee that you will be audio recording them. Now, I think this is a brilliant way to make sure that you get all the data you can possibly get from your interview, is to record the audio of what you and your interviewer you're discussing.

Play video starting at :8:56 and follow transcript8:56

However, if you do want to do this, you need to get consent from your interviewee, and you need to do this up front. You can't just do it at the end, you have to tell them straight away, I would like to audio record is that okay? You'll note that this is a question not an instruction, because it's okay for the participant to tell you actually, no, I'm not happy with the audio recording, but we can still continue with the interview. And that's fine, it just means that you'll have to rely on your notes a bit more. It's not going to be ideal, but it certainly doesn't mean you have to scrap the entire interview. Now, I think you still need to have notes even if you have that audio recording. And that's because this audio recording, could be a possibly very unwieldy thing.

Play video starting at :9:37 and follow transcript9:37

Whilst it will have really useful information, about the tone of voice that your participant had when they answered a question, and it will have the exact wording that the participant used, rather than what you managed to scribble down at the same time. Whilst it has those really beneficial things, it will also be a massive, massive file for you to have to navigate through. If your interview has gone on for an hour or two, then you are not going to want to listen to the entire thing back, it's going to be very tedious and very boring. And that's where these contemporaneous notes come into action. Because if you have notes, you can make a note every time someone says something of particular interest that you want to go back to the audio recording and listen to. Maybe at three minutes and 40, the participant said something that was a brilliant quote that you want to take back to the team. You can quickly scribble that down, and then you don't need to worry about memorizing three minutes and 40, three minutes and 40, three minutes and 40.

Play video starting at :10:33 and follow transcript10:33

If you're having to worry about remembering every time the participant said something particularly brilliant, then you are going to stop being able to ask the questions and listen to those answers, whilst the interview is happening. The notes mean, you can completely let that go from your mind. What's even better, is if you can get someone else to be there to take those notes. If someone else is responsible for making those notes that will help you navigate the audio recordings, you as the interviewer can just do one job and that job is to talk and ask questions and to listen. And you're more likely to notice when the interviewee says something that you maybe want to explore more, or more likely just to understand and respond to what the interview is saying. So that is the ideal setup, have an audio recording, someone taking notes, and so that you can fully commit and be an active listener in the interview session. Another part of the planning that you need to do, is you need to ensure you know exactly what questions are

definitely necessary, and which questions can be ignored if that needs to happen due to for example, time constraints.
Play video starting at :11:39 and follow transcript11:39
Now, you might be running a semi structured interview, and that means that you feel like you're quite free to choose which questions go into the interview and which ones don't. However, if you're working with a team, they might have a different idea about which questions are important. So you need to know that the team needs questions two and four to be answered, and everything else can be adaptable as you need to. You might be thinking to yourself well, I'm just not going to adapt. I'll just ask all of the questions until they're all done, and then that means I don't need to decide which ones are necessary, and which ones are less so. The reason you can't do that is because, that would make the interview an absolutely massive task, a massive undertaking with a very ambiguous and uncertain time frame, and that is another way to very much disrespect your interviewee. When you start your interview even before that, when you book it with your interviewee, you need to be able to tell them the exact amount of time the interview will take, and the exact amount of time that they will need to be there with you.
Play video starting at :12:40 and follow transcript12:40
If that means that you don't get to ask all of your questions, then that's on you to decide how you just navigate that. Maybe you ask your questions quicker, and you cut your interviewee off. Maybe, you have to decide which questions get lost but it's up to you to decide that, you cannot decide for your interviewee we'll just go over by 15 minutes that will be fine, that is not in your control. One final thing you need to convey to your participant, is that they are allowed to end the interview at any time. It might be that whilst you've designed a very thoughtful and well designed interview plan, that some of the questions you're asking are distressing or traumatic in some way to your interviewee, it's not up to you to decide whether or not that's the case, it's up to them. And if they decide they want to finish the interview, they need to feel confident they can do that without you continuing to interview them. If your interviewee tells you I need to finish this now, you are not allowed to challenge them and say why, but why?
Play video starting at :13:41 and follow transcript13:41
What's the wrong, what's happening? You need to respect that they need to finish, and you need to let them go. And you do need to make that clear to your interview at the start, so that they understand that they're allowed to do that. That is likely to put them at ease, and make them feel that they have some sort of control in the interview. Which is a really important thing for them to have. Finally, let's talk about some of the tools that you can use whilst you're in an interview. Now, whilst an interview is quite a verbal thing, that doesn't mean it has to exclusively be a conversation, you can use physical props to help with the interview.
Play video starting at :14:15 and follow transcript14:15
One thing I would encourage, is thinking about whether or not it would be useful for you as the interviewer, or for your interviewee to bring in artifact to the interview. Now by artifacts, I just mean a physical object, or a photo of something. I don't mean it has to be a historically important artifact. I'm not encouraging you to raid any local history museums. It just means, that you will have a prop in the interview that might help spark conversation and questions from you, or ideas and thoughts from the interviewee. Now, this could be something that either of you bring in. It might be that say, you're trying to design a system that will help people store their

memories in a way that is easily accessible through a computer, that you might ask your interviewee to bring in something that they would like to be able to store digitally.
Play video starting at :15:3 and follow transcript15:03
They might end up bringing a child's toy, or a photo or something like that. This type of information is really rich useful data for you. It tells you what's important to the user. And also, it means the participant has something exactly personal to them that they can talk about in the interview, and it might spark some thoughts from them, it might remind them of some memories. It's a useful way to generate more conversation. If you as an interviewer want to bring something in, you can as well. But just remember that artifacts are a useful way to embellish, and extend an interview.
Play video starting at :15:37 and follow transcript15:37
Another useful tool that you can bring is sketch paper. Whilst I've already told you that verbal communication is some of the quickest communication people can do, which is why it's useful to run interviews. Sometimes, there are things we cannot express just by talking about them. Sometimes we need to rely on our gestures, or we need to just express something visually by drawing it. If you have sketch paper and some pens available, then that means that the interview can continue, and you don't need to break flow while you scramble to find some notepaper, if the interviewee says they need to sketch something quickly. You'll notice I've said the interviewee needs to sketch something there, and that's another thing that you as the interviewer need to resist. If you have paper in front of you and a pen, it's important that you don't just start doing some designs in front of the interviewee, because then you're starting to commit to designs, and we are currently at the early stage of our discovery phase.
Play video starting at :16:31 and follow transcript16:31
We are at the early stage of that double diamond, and we shouldn't be committing to any designs. So if there is paper there, it's for the interviewee to use, not for you. One final tool that I want to encourage you to use in an interview, is the idea of an icebreaker. Now, an icebreaker is often something that we use in a very kind of embarrassing way at work, where we have to kind of go around the circle and say one interesting fact about ourselves, or something like that. I'm not encouraging you to do that, I'm just saying it's good at the start of an interview. If you start with a few kind of small talk questions that aren't related to the interview at all, that can put the interviewee at ease, it can make you familiar with one another and it can just help kind of ease into this interview process, which as we've discussed before, some people can see is quite intimidating. If you can set the tone with a few small talk questions, then that can really help put people at ease.
Play video starting at :17:29 and follow transcript17:29
Similarly, at the end of an interview, it's perfectly acceptable to start by asking some questions, or just having some chat at the end of the interview once you formally finished. So you can tell the interviewee, thanks for that the interview is now over, how are you feeling, how are you doing? And that might be really nice for the interviewee to kind of let go because again, they may have been on edge. Another thing that's beneficial here is that sometimes, interviewees can give you a bit more information at the end of an interview. Sometimes they can say, yeah, that was fine, I was really not looking forward to talking about so and so because, I've really struggled to use that feature in the past. That gives you a little bit more insight into the users

thought process, with regards to what you were talking about. And sometimes, you can get some really useful data at the end of an interview, when the interviewee is kind of relaxed.
Play video starting at :18:22 and follow transcript18:22
It's important though that you make it clear to the interviewee that you are still audio recording, and that if they would like to you can delete remove all of that data that you just collected at the end of the interview. But, hopefully, you have a rapport with them and hopefully they feel safe and secure to let you have that data, it's kind of like a secret extra bonus level of open ended questions at the end of the interview. So, when you're conducting your interview, remember that whilst it is primarily a verbal exchange, you can use some tools to enhance your interview, and give you that chance to get some extra data.

Now, interviews are not the only way that we can get in front of our users to ask them some questions. There are other options available to us. One of those options is to run a focus group. Now, a focus group is basically just like any other interview in that there is an interviewer and they are asking questions to someone. However, the distinction here is that the interviewer is not asking questions to a single interviewee, they're asking questions to an entire group of people. Now, focus groups are something that are commonly used in comedies, that's something that I've noticed anyway. And I think the reason for that is is that a focus group is a great environment for having lots of personalities working together or sparking off one another.
Play video starting at ::53 and follow transcript0:53
And the thing that makes that funny in a comedy is the same thing that makes that useful to us as interaction designers. So let's talk a bit about what a focus group even is before we talk about why we would use them. A focus group like I said, it's just like any other interview except the interviewer is asking multiple people the same questions and they're asking them at the same time in the same room. A focus group can range anywhere between 3-10 people. One of the benefits of a focus group is that you get an understanding about what ideas are commonly shared amongst the users. What do I mean by that? Well, in an interview situation, if we're asking an interviewee Eastern questions, we can't really be sure if their responses are representative of just them or if their responses are representative of a wider user group.
Play video starting at :1:44 and follow transcript1:44
In a focus group, we can get a bit more of an idea, a bit more of a sense about whether the issues being discussed are shared amongst many people or whether they are just niche problems. A focus group generally sits in the middle of the structure spectrum that is a focus group will usually involve the interviewer having a semi structured plan about what they plan to ask during the focus group. But and you might imagine this would be the case. It can be very difficult to go into a focus group with a very strict set of questions and expect everyone in the room to remain on target. So for that reason it is best to be prepared to be able to be a bit flexible with which questions you have to ask and which questions you don't have to ask. So why would we use a focus group? What are the benefits of using one as an interaction designer?
Play video starting at :2:36 and follow transcript2:36
Well, one of the main benefits is that by getting a group of people together, you allow them to spark ideas off of one another and remind each other of things that they wish they had said this is a really beneficial thing. Imagine all of the times that you have walked away from a

conversation and then a day later I thought, I should have said that. In a focus group, you're more likely to be reminded of those types of things that you wish you have mentioned. And so a focus group can be very beneficial for eliciting as much data and information as possible from our users. Another reason that a focus group is useful is that they can actually be a little less intimidating for users than an interview situation. Remember we spoke about how interviews can be quite a formal and perhaps intense scenario for people to be. In a focus group, however, the interviewee is not the focus of attention, they are just someone in a room full of people and for that reason they can be a little less intimidating.

Play video starting at :3:33 and follow transcript3:33

Another reason that focus groups are beneficial is that they can be a bit more efficient. Remember how we talked about in interviews being quite expensive because they involve one interviewer getting data from just one interviewee. In a focus group setting, however you get one interviewer and you can collect data from multiple different interviewees. And generally focus groups are just great at generating ideas because when people work together, when people collaborate then they can be more creative and they can maybe come up with more interesting innovative solutions to the problems that you are setting them. So I think I've convinced you hear that focus groups are more efficient than interviews and they can generate more ideas than interviews. So they seem kind of perfect. Why would we ever not use a focus group?

Play video starting at :4:22 and follow transcript4:22

Well, like everything, there are some downsides to a focus group. One of the major downsides of the focus group is the idea of groupthink group think is what happens when a group of people come together and their ideas and their opinions gradually converge into the same space. Now it might not be that there are ideas and opinions were the same before they started meeting. However, by being around the same people and by discussing ideas, it generally is the case that your ideas can converge to a sort of middle ground and that's really problematic. Remember what we're trying to do here, we are trying to collect data and information about our potential users and if we are collecting data from a lot of people we expect them to kind of represent a range of people. If everyone is slowly coming together on the same idea, then we may as well have just interviewed one person because we lose that sense of distinction and we lose that range of responses from people. So that's one real thing you have to be aware of when you are running a focus group, you need to ensure that people really can stay distinct and have their own opinions.

Play video starting at :5:34 and follow transcript5:34

Now this can be harder than it seems, partially because of social etiquette. Sometimes in a focus group, something you'll find is that one person will be much much louder than the rest. This happens in social situations, I'm sure you found that. But in a focus group it can be really problematic. If you're an interviewer and you are asking a group a lot of questions and one person thinks they know the right answer to those questions, then they might be louder with their answers and they might be more confident with their answers. In a focus group setting, you are very unlikely to be asking questions that have a right or a wrong answer, you're just interested in what people's opinions are. However, this single member of the focus group, if they are presenting their ideas as though they are the correct response, it can make other people in the group who have different ideas or different opinions feel a bit less confident about what they want to say and therefore they might end up not speaking up.

Play video starting at :6:30 and follow transcript6:30
So again, despite having a room full of people in front of you with potentially a room full of different opinions. If one person is being a lot louder than everyone else then you end up missing all of the opinions from the quieter, less confident people. It's very difficult as an interviewer to manage that sort of situation. You need to collect the data from this person, their opinions are still interesting and useful to you but you need to ensure that they are being presented to the group in such a way that doesn't intimidate the rest of the people because you still want to hear all of the other opinions from the group. So now we've discussed the potential benefits and the potential downsides of focus groups. Hopefully you're in a position to decide whether or not a focus group is the right thing for you and for your interaction design project if it is now, we need to talk about how you actually go about running one. There are a lot of parallels between the focus groups and the interviews that we discussed in the previous session.
Play video starting at :7:28 and follow transcript7:28
So a lot of the things we've learned there will be applicable here. One of the first things to note about a focus group is that you are trying to organize far more people than you would be trying to organize in an interview situation. So it's even more important to understand the logistics and the arrangements of the focus group and to communicate that very, very clearly to everyone involved. If someone doesn't get the message, then they might be delayed and they will hold up an entire room full of people. So it's important that you can make everything that you want your participants to abide by the timing's the information they need as clear as possible so that every participant has the best opportunity of showing up on time and so that you have the best opportunity of running a successful focus group. As with interviews, you need to tell people when your audio recording. And I think it's even more important in a focus group that you have, that audio recording, it's going to be incredibly difficult to take scribbled and written notes from a multitude of people in front of you.
Play video starting at :8:27 and follow transcript8:27
It's hard enough from one, let alone an entire room. As with semi structured interviews, it's important that you have a list of questions you want to ask, I mentioned before how it's likely that a focus group with so many people may slightly go off on a tangent and that's okay. But just as with the interviews, you need to understand which of the questions are necessary and which of the questions are nice to have. In that way you'll be able to plan and decide how your time is used within the focus group. As before, you might need to hurry people along, you might need to cut some questions and you'll need to do this all on the fly whilst the focus group is happening. Speaking of hurrying people along as well, this is something that is different in focus groups compared to interviews as I said in the previous slide, one of the issues with focus groups is that sometimes people really like to take up space and time when they are answering questions and that could mean that other people don't have a chance to talk. It's your job as the interviewer in the focus group to slowly hurry people up to manage people to ask people to wait their turn and do a whole host of things that might feel quite awkward to you from a social point of view.
Play video starting at :9:43 and follow transcript9:43
But they're necessary things for you to do because otherwise you'll be in a situation where one person talks and answers only one question for the entire session. You will need to ask people very politely to stop their answers now and let's see if we can hear from someone else in the

group very politely is the key thing here. It's important that you don't just jump in and interrupt, but you need to choose your time and choose your timing carefully to help people understand when they need to stop answering a question and give the floor to other people. This can be incredibly difficult to do, it's hard to do in a social setting and it can be even harder to do in a formal work setting. So one thing you might want to think about, if you are going to run a focus group is that you might want to hire a facilitator. A facilitator will have the skills and experience to know how to manage a focus group and how to do those difficult interactions when you need to ask someone to essentially stop talking. And for that reason you might want to think about getting a facilitator in because you yourself might not be trained or might not be confident in how to manage that situation.

Play video starting at :10:53 and follow transcript10:53

Now I'd like to just tell you a bit about a focus group that I was involved in and how I overcame some of those issues that I spoke about before, the problem of group think. The problem of one person's voice dominating the space and the problem of how to manage the group, how to ensure that everyone had a chance to think what I did in this situation was create a artifact, create a prop that would allow me to manage the focus group. So let me set the scene, I was asked to research a immersive theater production. This meant that after the participants, the theater goers had gone through this production, they were to come to me and we would have a discussion about what they had just experienced. Now this experience was very weird and wild. It was completely in the dark. It required participants to touch and feel things and to be guided by this piece of navigational equipment in their hand.

Play video starting at :11:49 and follow transcript11:49

It was quite a surreal experience. And so when participants came out of the theater experience, they were buzzing and ready to talk. I was aware though, that if I was going to collect any data or feedback from these participants. I would need to manage this group because it was very possible that one person would have a very strong opinion to this artistic experience that was distinct and different to another person in the group. So I created the zippy tablecloth. Now, one thing I should mention before I go on with this is that a lot of the participants I would be working with would have visual impairments. And so it wasn't going to be appropriate for me to be handing out flat cold touch screens to people, because that wasn't accessible to all of the participants that I was working with.

Play video starting at :12:31 and follow transcript12:31

Instead, I created a tangible interface. This interface was a tablecloth that sat over the table that I would sit at with my focus group. In front of each of the participants would be a zip that I had sewn into the tablecloth. This was no ordinary zip. I had also sewn in some conductive thread. This meant that I was able to, using a microcontroller determine exactly what position the zip was at, depending on how the participant has moved it. This meant I could record data about how people were responding to my questions electronically and I could also see how people were responding to the questions, which meant that I, as a facilitator could manage the group better.

Play video starting at :13:13 and follow transcript13:13

Let me tell you a bit about how it worked. I would ask a question to the group and the question would have a like at scale answer which meant that people could answer either, strongly agree, strongly disagree. Very likely, very unlikely you'll have seen these scale answers before. This is

something we'll talk a bit more about in the next session. Now, when people answered that question, they would move the zip to the specific position and then I would press a button to record that position. I could then specifically see exactly where everyone had moved their zip too, and I could understand exactly where the differences were in the group in front of me. Another element that I added was that I added a tone to that note that zip, this meant that everyone around the table could hear where everyone has positioned their zip regardless of whether or not they were able to see the position of the zip.
Play video starting at :14:5 and follow transcript14:05
This meant when it came to having that lovely fun focus group discussion about the piece, I was able to say to the person sitting, position three, I see that you really enjoyed that element, can you tell me about it? And whilst at the same time I was able to talk to the person at position one who ultimately probably would have been a bit quiet because they didn't enjoy it. But I could see that they hadn't and I had collected their opinion before, they had had a chance to either slip into groupthink and before they had had a chance to be intimidated by potentially the loud person sitting at table at that seat three. In this way I was able to use the focus group to the best of its abilities. I could facilitate a conversation between these four people who had just gone through a wild and surreal experience, but I was still able to keep things in control and I was still able to record everyone's response before they had had a chance to slip into group. Think so if you're ever going to be running a focus group, think about if there are any props or any tools that you can use to ensure that you avoid some of those potential downfalls. Because ultimately focus groups can be a really useful way of collecting that really interesting, innovative and creative data from your users.

Let's talk about questionnaires. Now, questionnaires are an interesting thing. They always create interesting scenarios where people react in one of two ways. They always want to talk about the statistical design and designing questionnaires are good, or they look at questionnaires and they say, "Actually, I hate questionnaires, I hate filling them in". I think we've all had the experience where somebody's come up to us in the street and said, "Could you fit in this questionnaire?" Or we've received questionnaires from unwanted emails and things, and we've said actually, that's really boring and I don't want to engage in that process. But actually they are a useful way of collecting data.
Play video starting at ::43 and follow transcript0:43
Now, there's different types of data we can collect with questionnaires and we'll have different things that we want to do. There's different types of analysis. We can generate different types of understanding about our users based on questionnaires and about their interactions and their habits, their patterns. We can ask all manner of things really with questionnaires. We have quite a bit of flexibility. When we talked about quantitative data before, you often see questionnaires that were able to standardize things and talk about quantitative data and you might see them in this category that's on the left here. Down this left side here, questions that looked like this.
Play video starting at :1:14 and follow transcript1:14
Now I know there's no questions there, but questions where you'd expect a generic response, where people tick a box or select a number. I might go tick, tick, or maybe a cross possibly. Then they might have numbers in fields as well. They decide based on some certain preferences or values, what they want to do. Now, this first type of questionnaire is what we call

a closed type of question. Closed questions are where we refine things around, specific things. We might ask you, "Do you like going to the shop, yes or no?"
Play video starting at :1:44 and follow transcript1:44
That's a closed question. There's only two possible responses. This example here on the bottom is three possible responses. We could say, one can be yes, one can be no, and one can be maybe. It gives people the opportunity to give a closed response. You pick from a number of choices. Now you often, sometimes with questionnaires, we'll say scales as well.
Play video starting at :2:5 and follow transcript2:05
You might see scales with closed questionnaires where you have what's called a Likert scale. Now a Likert scale is an interesting one and it's an interesting design construct. We'll talk about this a bit later. But Likert scales will give you options, so you might have an option of 1, 2, 3 like this, where you have to select one of the predefined options. You might say, I'm going to choose option 2 or I'm going to choose option 3. There's lots of different ways that we can deliver closed questionnaires but the idea behind these sorts of questions is essentially that we want to capture some data about a thing. We've got some preconceived notion about something, we've got some categories, or we can put things in order, or we can add some structure to stuff and then ask questions related to that.
Play video starting at :2:44 and follow transcript2:44
Now do you think that this is true or false? Do you think that these things are fake or real? Do you believe in these things? How much do you enjoy using these things? We can ask all manner of questions in this way, in this closed question type way. Now we can also do other things with questionnaire so we can ask for more open-ended questions. We don't have to focus on responses that are quantitative in nature.
Play video starting at :3:6 and follow transcript3:06
We can take qualitative responses from questionnaires. We could ask an open question. We could ask, for instance, "What is your favorite module in this course?" That might be a closed type questionnaire, but then we can follow it up so we can have another one on this side that said, "Tell me why that's your favorite module? Why is Sean's module your favorite module?" You could say because Sean's an amazing lecturer and he's funny and witty and I enjoy watching him. He's brilliant, he's amazing and all those other things that are clearly very true.
Play video starting at :3:35 and follow transcript3:35
With open-ended questions, what we're really getting at is that rich data, we want that contextually rich data, we want qualitative data that tells us a story about something, that tells us the why and the nature around those nuances that we're really interested in encapsulating. Now, there might be a good case for designing one or other, there might be a good case for designing both. But essentially questionnaire design is about making these choices, it's about deciding which of these two choices are you going to pick. Which is relevant for the type of question you're asking? As I talked about with the first type, with the one that's highlighted, with the closed questions in this example, what we're really getting at it's trying to refine our ideas. We've got some pre-conceived notion, we've got some idea about categories, about how things might rate, that's the intelligent questions we could ask. The other type of questionnaire, the open type of questionnaire, which is this one on the right here.
Play video starting at :4:26 and follow transcript4:26

These types of questions where you just expect a plain text response. These are much more focused on getting that rich data and trying to understand, how and why, getting a sense of what's going on in people's minds, what their own experiences are. You can get really deep level of understanding about things with these types of questionnaires. They're both valid and you often see questionnaires that utilize both types of techniques. You might be asking closed type of questions when you want to know a bit more about something, when you've already done some research and you said, "Well, actually I want to refine this further. I want a deeper understanding of these concepts". You can then jump over to the open style questionnaires and say, "Well, there's this really interesting thing, X, that we were talking about in here, and I want to know more about X, so tell me about X.
Play video starting at :5:7 and follow transcript5:07
Tell me about Y. Tell me about Z". What is it that is interesting to you? What is it that's valuable about these experiences? We can actually use both techniques to get at that rich and interesting data. Now in the next lecture, we will talk a little bit about some of the caveats around questionnaires and why they can be problematic. But as a basic techniques to capture data, questionnaires are pretty good.
Play video starting at :5:28 and follow transcript5:28
They're pretty easy to design, they're pretty easy to build into a set of strategies. You can go and collect lots and lots of data quite quickly, you can disseminate these to lots of people. They don't have to be on physically, so you don't necessarily have to hand somebody a questionnaire. You can distribute them via different platforms and you can go and collect lots and lots of data very quickly and easily. Now there are some problems with that and we'll talk about that in the next lecture. For now, I'd like you to think about questionnaire design and I'd like you to think about these two different techniques and try to appreciate that there is some complexity in both methods, and to understand that that complexity is embedded in the types of questions that you design. In the next lecture, we'll talk a little bit about why that's interesting and relevant.

In the last lecture, we talks about the basic structure of questionnaires and how we can ask questions. We looked at open-ended versus closed ended questions. We had closed-ended questions where we had these choices, and open-ended questions where we just expect some text-based responses. Now, they're both valid ways to capture data, but there are caveats around questionnaires that make them quite challenging to design for insertion scenarios. The first thing I'm going to talk about is the actual element of questionnaire design and asking the right questions. How do we know the right questions to ask? How do we know that we're asking the right questions?
Play video starting at ::43 and follow transcript0:43
Well, we can get lots of bits of information that feed into this. Maybe we've done some interviews before. Maybe we've got some understanding about our users from previous systems. Maybe we've encapsulated and elicited requirements from users from different places. Maybe we've already lost the questionnaire number, we're finding our understanding. We can have lots of data that feeds into questionnaires. But where does that go and how do we refine that?
Play video starting at :1:5 and follow transcript1:05

How can questionnaires give us the opportunity to refine things in an interesting and meaningful way? Well, actually there's a lot of subtleties around this and it's quite a complicated thing to design a good questionnaire that people will follow through. There are a few general rules. There are a few things to think about when you're designing questionnaires to get a sense of how to ensure you get the most number of people complete it, how you get valid data, and some of the things that you need to think about when you're designing these things as well, where there isn't an optimum solution or there isn't necessarily an easy pre-made solutions to these problems. In the previous example, I showed you some instances where we had some closed questions and we had some boxes and we had some options. You might see these as, I don't know, maybe elements could be disagree, agree, or a neutral response. We could have three responses, and that's DNA.

Play video starting at :1:54 and follow transcript1:54

We could have people who disagree, who don't like the system and this is a probably a fairly negative experience. Or we could have people agree who are having a positive experience. Or we could have people in the middle. Now, when we design these sorts of questions, how do we know how many responses to have? How do we know how many options to give people? This is particularly challenging. When you see Likert style questionnaires, you often see a number of responses.

Play video starting at :2:18 and follow transcript2:18

You might see these with three or with five, or maybe you've seen seven or 10. Now, generally speaking, these questionnaires tend to gravitate towards odd numbers. Why do you think that is? Why do you think that questionnaires have an odd number of responses? Why do you think three is a good number of responses for this type of exercise? Well, essentially when we look at this, we can see why calmly, because we give people the option to opt out. Essentially we can say, well, in this case, actually I don't have a preference one way or another.

Play video starting at :2:47 and follow transcript2:47

I'm a neutral respondent. I don't have a particularly strong feeling either way. Now, this can be valuable, but it can also cause problems. In some situations, you might find that if people aren't paying attention to your questionnaires or they're not particularly interested or involved, or maybe recruited the wrong type of participants so your sampling technique is wrong. You might find that they carry on going into the neutral areas they fit in the questionnaire. Now that's a problem because you're getting effectively invalid responses. You're asking people questions where the responses aren't valid.

Play video starting at :3:16 and follow transcript3:16

We're not actually measuring the thing that we're supposed to be measuring are we? If people aren't paying attention to the questionnaire and just taking some boxes, that's not really getting the desired result of interesting, rich, useful data. This is a problem we have to design for and you have to decide in your questionnaires whether or not you're going to give people the option. The other option is, you could give them two choices and you could also have an N/A response. You can have this on the side perhaps, and give them the N/A response and say, well, actually, it's not applicable to me. It's not that I don't have a response, but I just want to say, it's not applicable to me if you ask me a question that's not relevant to me or my needs or wants or desires, then I can just opt out of following that question and you can ignore the data for that participant. Now, that can also create problems.

Play video starting at :3:59 and follow transcript3:59
If you have lots of rich data about people and their opting in certain categories, like so. Then you get a situation where somebody opts out, that might obscure the data or it might heavily weigh the data in favor of particular types of responses where they have responded. Or it might actually change the number of responses you get. If you want to encapsulate a wide audience and you want to get lots of data about different types of people, it might be problematic to do if you give people the option to opt out and just say, I'm not going to respond to that question. Now, that's not the easiest thing in the world to design for, but we have another problem with these types of questionnaires that's even more problematic and even more difficult to design for. Now, the Likert style questionnaires are interesting because we can choose a number of responses. As I've said, we can have 3, 5, 7, 9.
Play video starting at :4:42 and follow transcript4:42
I'm going to stick with this odd number because I think it's a good way to go. I don't like even numbers because if you have even numbers, you give people the opportunity to have feedings that aren't particularly strong in either direction. You could have disagree, and you could have agree, or this could be strongly agree for instance, it could be an extreme on this boundary perimeter or strongly disagree. Then you could have weighting. You could actually decide to have weightings where we don't have a middle option, where we don't give people the option and they have to select one way or another which choice they're making. In this case, this could be a valid questionnaire design. For instance, if you're finding that lots of people are ticking in the middle or they're not showing a preference and you want to get a preference.
Play video starting at :5:20 and follow transcript5:20
But you can also be directing people. You can be saying, well, people who don't have strong feelings now will have to choose a positive or a negative response. They'll have to choose, disagree or agree, or strongly disagree or strongly agree. These extreme values become even more important then in this case, and it becomes more difficult to statistically analyze our data. What do we do? How do we solve this problem? Well, there's a general rule of thumb with Likert style questions that we can think about how involved a person is with a system.
Play video starting at :5:48 and follow transcript5:48
How much experience they have? How much knowledge they have? Do they really have strong feelings for these things? Is this a system that they're using a lot? If it is, then we might want to extend our Likert scale and we might want to use a larger scale to have a broader range of representation, a broader range of options. We could have options 1-10, for example, and all the options in between will give people a diverse range of opportunities to respond to things. We could have quite rich data responses in here that will give us even stronger values.
Play video starting at :6:17 and follow transcript6:17
You can probably already see what the problem with a large Likert scale is, with having lots of optional responses that data might be more difficult to analyze. It's also difficult to quantify what this means in real terms. If I put a negative or a positive response that's perhaps a seven or a three, or five, or nine. How do we quantify these things? Do these things cancel each other out? Is a four and a six, effectively a cancellation where we can say, well, they're both a middle value. Both are middle value, they don't care, or they're not interested or it's not applicable.
Play video starting at :6:50 and follow transcript6:50

We've got to ask all these difficult questions about our data and try to make sense of this and say, well, what does it actually mean? Questionnaire design is actually more complicated than it looks. It's one of these things where we have to really, really carefully design towards our audience. Some general rules, keep your questions relatively short. Don't give people 100 answers, or don't give people 100 options or 100 questions even. Try to think about minimizing this process. Keep the number of questions down and only ask things that are relevant and useful to the thing that you're trying to find out.
Play video starting at :7:22 and follow transcript7:22
Choose an appropriate Likert scale and make sure that it's relevant. If we choose 3, 5 or even 100, make sure that we've got a justifiable case for doing that. For choosing 100, it might be a system that people are using every single day of their lives for eight or 10 or 12 hours a day and so we've got really, really strong impactful experiences that we can encapsulate in this model. If this is the first time you've used a system a Likert scale of three or five might be more appropriate. When we're designing questionnaires, we also have to think about the data that we're going to get back and how we're going to analyze that. It might be a case of choosing not to have boundary values or not having middle values, forcing people to go one way or another depending on their own preference, or just making some general assertions about things. Saying, well, in fact, I'm only going to give people two options.
Play video starting at :8:10 and follow transcript8:10
They either like the system or they don't, and this might also be a valid case where we might decide that actually a questionnaire design doesn't have to ask really, really complicated things. We can do this through things like multivariate analysis, analyzing specific features of a system. If we're asking people about a preference around a button or a particular typeface, you might want to give them just two options and ask them to opt in to a choice as well as opt out. There's lots of different things we can do with questionnaires. Have a little think about this and we're going to go into some activities where we're going to try to design some questionnaires. We're going to do this a bit differently than I normally do it. We're going to try to design some really bad questionnaires. This should be fun.

I'd like you to imagine a scenario now. There is an office, and we have two people working in this office. Now, we want to observe them, so we want to get a sense of what they're doing, we want to know what their work looks like. We want to analyze this. We want to do some research around working environments in people who work in specific types of environment. This could be any office, they could be doing any type of job. But let's assume that they're interacting with another and doing a job of work that requires certain processes and procedures.
Play video starting at ::38 and follow transcript0:38
We want to make sense of what this is. I turn up in the office with my clipboard and I watch these people for a length of time, and I get a sense of what they're doing. I write about the things that they're doing. I might take some photos, I might capture some video evidence. I want some media just so that I can analyze some things later. Now this seems like a sensible thing to do, doesn't it? I get to see these people in their real world environment.
Play video starting at :1:1 and follow transcript1:01
I get to see how they're interacting, I get to see the day-to-day complexities, the context of things. I'm understanding not only what are they doing, but why they are doing things. I get to

see a general sense of things and people reacting to one another. This is a great way to collect data and this is a technique that I use a lot in my own research, it's quite useful, it's quite valuable. That rich data can be used to create really useful insights about how to build systems for people. But there's a classic problem with this. There's a real big issue that we have to contend with when we're doing observations of people.

Play video starting at :1:33 and follow transcript1:33

You might have already guessed what it is. What's the challenge when you observe people? Well, are you observing their real behavior or are you observing a simulation? Now in some cases, people might not be particularly interested that you're there with your clipboard watching them, but in most cases, I think this would worry people. In most cases people would see somebody observing them and they might change their behavior based on the observation. The patterns that they engage in the conversations that they have. Things that they do day-to-day may adapt based on the people around them.

Play video starting at :2:4 and follow transcript2:04

If somebody new comes into the office with a lab coat on and a clipboard, and they write down everything that they're doing all the time, that could make people uncomfortable and it could change their behavior. Now this can invalidate our data and it can be a real problem. It's something that we want to be aware of. This is actually called the Hawthorne effect. This has been observed before in science. This isn't a new phenomenon. This is something that most of us in ethnographic research are very well aware of.

Play video starting at :2:33 and follow transcript2:33

The Hawthorne effect is essentially the fact that people will change their behaviors based on the fact that they're being observed. This is a problem from a scientific perspective. This is a problem from a research perspective because we want to analyze people. We want to use these rich data capture methodologies. But we can't use them without having some impact on the way in which data is being formed is changed and adapted according to the fact that we are there. Just by being there we're changing the things that people do, we're changing their habits and their behaviors. If I was to turn up into this office environment with a clipboard, taking notes, these people might improve in productivity.

Play video starting at :3:11 and follow transcript3:11

They might suddenly become much more productive. They might work harder, they might take shorter breaks, they might change their behavior because they're frightened. They're frightened and they're not aware of why I'm there. They see me as a threat. They see me as somebody who might be reporting on them or somebody who might be to take that job. There could be a number of psychological effects that I'm having on these people. It's very important that we are understanding and appreciative of these things.

Play video starting at :3:34 and follow transcript3:34

If I was to send somebody else in, in six months time, their productivity might go up again. Now, it's very easy to see other things causing these types of activities as well. We could say that we are imparting some effects on people. Maybe I'm running an experiment. I've changed something in the office. Maybe I've changed the height of the tables or we brought everybody some nice new equipment to work with, and we want to see if that boosts productivity. Now that seems like a reasonable research question, doesn't it?

Play video starting at :4:3 and follow transcript4:03

We're going to change something in their environment and then we're going to see how it affects them. But actually what we're doing here is if I'm going in and observing this behavior, I'm also having an impact on what they're doing. If there productivity increases, that might be because of the change that we've made in the environment. It might be because of those nice standing desks that we bought, or there's really nice new MacBooks that they love using. Or it could just be because I'm there. Be aware of this. This is one of the caveats in this method.
Play video starting at :4:31 and follow transcript4:31
I'm not saying that this is a bad method to use. I use it in my research all the time. Ethnographic studies are really useful for getting rich contextual data. But just be aware of the fact that you can obscure your data in unintended ways and have unintended consequences on people's habits and behaviors. This separation of elements is something that's quite tricky to do. We'll talk a bit more about ethnographic methods in the next lecture.

So in the last lecture I framed ethnography as a set of constructs around observing other people, but it doesn't always work that way. In fact, there's an extension, there's an additional method that we can use and we can use this to be introspective. So it might not be the case that I'm walking into an environment, and I'm watching somebody else, and I'm observing what they're doing. Now this might be a traditional way to think of ethnography, and it's a method that's from the social sciences. It's a technique that's been used for a number of years to try to understand things like other cultures, or other places, places of work, it's been used to understand social conventions and to test these things. But actually it doesn't always have to happen in this way, and in fact we can think about autoethnography, we can think about a special type of ethnography. And when we say auto you've probably figured out what this is all about, but autoethnography is the study of one's self.
Play video starting at :1:2 and follow transcript1:02
So you can observe your own patterns in your own behaviors and you can be introspective, you can understand what you're doing from both your own perspective and from a scientifically valid perspective as well. There's lots of studies been done on autoethnography, about how we do things as people. And this idea of looking introspectively is really useful, it's a really nice way to do research, to think about well how do I write code? How do I learn about coding? How do I complete activities? How do I engage with things in the world? You can do these types of studies on yourself and on the communities and groups that you're in.
Play video starting at :1:37 and follow transcript1:37
So this doesn't have to be a process that you're observing other people, you could be observing yourself. Now in terms of ethnography we've talked about some of the challenges here, but what are the advantages of studying others or studying ourselves? Well, there are some clear advantages, aren't there? First of all, we want to get a sense of context, we want to understand what people are doing and why they're doing it, and we want to understand the day to day habits of people. Now this is perhaps the richest data collection method that we can use, we can really understand a lot from just watching people, from just spending some time with people, and understanding what communities of people do and how they react to one another based on different types of feedback, things like conversations, interactions on a day to day basis. Do people hold the door for one another? What is the culture of a place like?
Play video starting at :2:24 and follow transcript2:24

We can answer all these rich, interesting questions using ethnographic methods. Now this is a field study based method, so we tend to do this in the field. We can't really do an ethnographic study in our labs because people don't live in labs, people might work in labs, I guess you could do an ethnographic study of people working in a lab. But again, it's about that process of introspection, it's about looking inside the community of people, trying to understand things like social conventions. Conversations, what sort of information can they impart, what can we draw from interesting dialogues and interesting interactions that people have with other systems and with one another? And through these rich contextual understandings of what people are doing we can design really, really good systems that promote all the right properties, that understand the subtle connotations of what people are doing, and then we can embed them in our designs and our builds to ensure that we have higher quality software.

We're going to go back to a process that we covered earlier once. We're going to go back to user-centered design and we're going to incorporate some techniques. Do you think about generating and validating requirements? Now let's remember our user, let's never forget about our users. It's always a good thing to remember our users in the work that we do. Now, there are a few things you have to be aware of when you're thinking about designing systems and they're general mistakes that people tend to make. We often overgeneralize and think of broad definitions of users.
Play video starting at ::38 and follow transcript0:38
We tend to think of users as being a large set of people. We think about those large characteristics of groups of people. That can be problematic. It can make it really difficult to design for individual use cases where we're thinking about this overgeneralization property, thinking about large groups of people. It's important to be respectful of individual differences and understand that those individual differences will also generalize up. We can design from the top down and think about large groups of users. But we also, in some cases want to decide from the bottom up and think about these isolated scenarios and how these situations might generalize up to other groups of people.
Play video starting at :1:13 and follow transcript1:13
How that can help and support people who might have issues around things like accessibility or usability, or just in terms of improving the interaction design process. As an example, our users could be experts or novices or somewhere in between, as most users tend to be. Users could also be indifferent state. They could be engaged in different things. They could have different levels of cognitive load. That can be in social environments in isolated scenarios or they could be in complex contexts, whether using phones to manage a working sets of processes. When they're doing something physical alongside engaging attack.
Play video starting at :1:51 and follow transcript1:51
There's lots of complexity when we think about interaction design, there's often lots of things going on. It's very rare these days that people actually interact with systems in that very isolated linear way when they go and click a button. That's the only interaction they have over a system, or whether it sat in front of a computer screen for a long period of time, without other types of distractions or interactions going on. I have a good example of this. I'm often responding to emails and I'm on teams and having conversations and watching exams and coursework, all of

the same time. These processes tend to blend. My focus tends to switch between those activities.
Play video starting at :2:23 and follow transcript2:23
When you think about designing for somebody like me, you consider those [inaudible] and context too, and you consider me as being an individual with very specific needs. But my needs would also generalize up to people like other academics working in the digital domain. Other academics working in other fields, for instance, like psychology, may have similar constructs around how they work. It's very important to encapsulate these features in terms of how he designs systems. Of course, human beings have different expectations of reality. We will have different perceptions of the world, and that perception might change how we see things, it might change how we expect things to work. We've got to consider all these different factors.
Play video starting at :3: and follow transcript3:00
Now, a lot of people would say, well, I cannot design for everyone, is impossible to design for all types of people. In fact, there are universal design properties that are built around these ideas of providing access for everybody, for ensuring that the largest group of people have access to things. That they can use things in a meaningful way. In fact, there were lots of techniques that we can use to help us to design for more people. To build up from a small group of people to larger groups of people, we've got lots of techniques that we can borrow and use. Now the universal design practices are really interesting and I think that really valuable in thinking about ways in which we can facilitate actions and activities for large groups of people and also smaller groups of people who might have specific needs. By designing for small groups of people, we can actually extrapolate up and say, well, these groups of people have these specific needs, but that also might generalize to another group of people, or a larger group of people or, a subset of those people.
Play video starting at :3:53 and follow transcript3:53
These needs and wants and desires from our system can actually generalize across groups of people. We can find unexpected things in our system. We can design things into our system that we might not have initially thought about that by considering these complex use cases, we can actually build up a better system, a higher quality system. Now, there's always an argument about breadth versus depth with these types of things, and there was actually a good case of both. We can say that looking at those wide groups of users is important, but we also want to look at specific cases. There's also the middle ground, we can group users up into categories. We talked about this when we covered information, but we can do the same thing with understanding our users.
Play video starting at :4:31 and follow transcript4:31
We can think about categories of users in cases of use or use cases as we know them today. Then we can think about those properties and those characteristics. There's different ways to conceptualize these different types of activities. Now when we think about goals, we can think about a complete system that encapsulates the needs, wants, and desires of most of our users. Now, I would agree with him in saying that it's impossible to design for every possible type of user. But we want to reach a point of saturation where we've designed for the most possible used cases so we can. We've got the most usable system, that's a reasonable goal to have.
Play video starting at :5:4 and follow transcript5:04

We can say we've done all that we can in relation to these factors, in relation to this goal. This is the ultimate goal, and to do this, we have to keep our users in mind at that board level, but also at that specific level.

I'm going to talk to you about a technique I find really useful. We're going to talk about personas. Now, personas are a great way to think about users and in fact, they are not users. This is not what we're considering, we're actually thinking about fake users. We're creating people who don't really exist, in order to substantiate an understanding about contexts of use and that rich context, and how it can be useful to help us to design for both individual people and groups of people. Now, what we do with persona is we characterize individual properties of people, and we take wants and needs and desires, and we characterize them through a set of individuals. We create an individual who has a fake age, fake name, and lots of other fake characteristics that might model a type of user that would use our system.
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Now, this sounds a bit silly, but actually, it's a really useful technique to allow us to understand how that rich context can form better design decisions. What we ultimately want to do is design better systems, and by understanding our users through sets of properties and characteristics, and features, we can then think about our requirements and the needs and wants of uses in very specific ways. Now, when we think about properties, they are usually drawn from data or analytics, so you'll find that personas tend to come from large sets of data that you have about usage and use cases. This might be about large basic research that had been done in terms of few things like lost your questionnaires in surveys. Or it might be about the analytics of use cases, so it might be looking at how people currently use existing systems. But generally speaking, you can capture lots and lots of rich data from this, you can look at the types of people who walk in and out of the store, and what they buy, what their buying habits look like having navigated the shop. There's a lot of different features that you can take from, there is large data sets and try to understand how and why certain things are happening.
Play video starting at :1:55 and follow transcript1:55
Now, there are a few ways that you can generate personas and this is just one example of how we might do this, and capture data from lots of different places as we've talked about before, and this data can be used in different ways to generate certain types of personas, and I'll talk about a few examples. So episode one, we have clearly identified a group of people who do not enjoy using our system, and this is a classic case for a persona. We're thinking about people who have had problems with our system in the past, or maybe even people who don't use our system. Now, there's probably a very good reason for that. It's probably a very good reason that these people aren't using our system, or aren't having a good time with our system, or really struggling with certain aspects of our system. What we want to do is, we want to create a persona around this type of person. We want one individual that characterizes all these features of the individuals who are struggling with this system.
Play video starting at :2:42 and follow transcript2:42
We can take that, we can impart that knowledge that we found that people between 30 and 35 who are male, who are from this specific region who are using this specific type of system are having these specific types of problems. Now, this seems like a very specific way of looking at

individuals and saying, "Well, actually, that's not going to generalize up." It doesn't really make sense from our kind of data perspective to say, "Well, we've got all this rich data about all these different use cases and now, we're squashing that down into one individual." But actually those individual properties and characteristics, that context is what we're trying to get at. We're taking that data and we're trying to add some context to it, we're looking at what features these individuals share, and why they're having so many problems with our system. This essentially helps us to think about communities of people that might be isolated, or put off by our system. We're thinking about fixing things for these users and it benefits us to do this.
Play video starting at :3:34 and follow transcript3:34
We want a system that is usable by all, and if there's a certain group of people who are struggling with it, by encapsulating their properties in persona and thinking about the ways in which these people might use the system, we can start to understand the problems and challenges that they may be facing. Let's have a look at persona 2, and this is a slightly different example. We want to ensure we're designing the right type of system for a new emerging market, so maybe this is a new system that's never been built before. Or maybe we're setting a new product or service, or maybe we're selling somewhere that we haven't sold before. Or we're building some system for a group of people that doesn't currently exist. Or for a group of people who've never used these types of systems before, and they're a lot of different cases where this might come about. Again, we want to characterize features from our chosen demographics, so we're saying, "Well, what are the wants and needs of these individuals?
Play video starting at :4:20 and follow transcript4:20
What do these individuals look like? How can we encapsulate the rich features of these people who might want to use our system?" All sorts of things that they want to do with our system. "What are they currently doing with the existing systems?" We can take this data, we can take this contextual meaning and understanding about how they engage with things, what their needs and wants, and thoughts and desires are all about, and then try to encapsulate rich design contexts in our system, and think about, "What is the route through the system? How might they interact with the system? How might they experience certain types of problems and challenges with our system?"
Play video starting at :4:51 and follow transcript4:51
Again, that allows us to facilitate better designs and better development strategies. Now, let's look at persona 3. Now, this type of person really enjoys using our system. This is probably the type of case that we won't look at so much because we've already got a good use case for these people, so we understand what works for them, and they're quite happy that they're working well with the system. They're getting what they want out of it, so we don't necessarily have to worry too much about these individuals. But we probably do want to understand what's going on, why they're having such a good time with the system. We might want to use personas, in this case, to encapsulate characteristics, and figure out why these groups of people are having such a good time with the system, what's working really well for them.
Play video starting at :5:30 and follow transcript5:30
How can we take those attributes and actually facilitate that across a wider group of people, and ensure that everybody's having a good time, not just these isolated individuals who are having some kind of positive experience based on things like their previous position, or what they're trying to achieve from the system, interaction modalities? There could be a whole host of

reasons why a group of people are either having problems with the system or are having a really positive experience. What we want to do is encapsulate the understanding of context. "Why are these individuals having such a good time? Why are these individuals struggling? How can we better understand this as a user journey?" We spent a lot of time looking at systems and trying to understand how to build and develop systems, but we don't spend as much time as we should looking at users, and the user is what we're really interested in.
Play video starting at :6:17 and follow transcript6:17
Personas bring them to the front of our process, they say, "Well, this is our user and this is what we should be focusing on. We've got a good idea about a typical user in a specific context, and a specific set of scenarios." We want to bring them to the front of the process and say, "These are the people who are using our system in this particular way."

We've talked a lot about the personas, but let us now consider the scenarios by which we're going to judge these personas, and we're going to understand how these personas fit in a wide contexts of use. Now, we can get the most out of a persona by considering scenario based, contextual implications. We're not just using this in isolation thinking about a general user. We're thinking about a user in relation to several activities, in relation to a set of things that they want to achieve with our system. We can think about the steps and processes, and we can imagine this person using our system at every step and every state, and thinking about the problems, the challenges, positives and negatives of our system in relation to that specific user and their specific wants, needs, and desires. I'll start out by how not to do this. We can say Sean is tall, I'm evidently tall.
Play video starting at ::58 and follow transcript0:58
I hit my head on door frames a lot. That's something that happens because I'm tall. I don't always look out, I'm not always aware what's going on around me, and so sometimes I hit my head on the door frame and I have a sore head. Now, the solution to this might seem obvious. We will just remove all doors, but that's a terrible way to solve a dynamic problem. We need to think about problems in relation to how can we encapsulate those rich set of features that those people have and make sure that with facilitating the right types of activities and the right types of actions to ensure that these things don't happen. You can probably think of a whole manner of things that you could use to solve this as a problem.
Play video starting at :1:32 and follow transcript1:32
It can be warning signs, it could be changing the shape of doors, it could be having steps up and down through doorways. There's a whole host of ways that you can solve this as a problem, but removing the doors is not the solution because that creates lots of other problems for other types of users. When you're designing personas and thinking about scenarios, don't just consider in isolation. If you're going to make a change to your system, that's going to affect lots of people. Again, use this as an opportunity to be reflective. Think about this process of verification and validation. Is this going to affect other people?
Play video starting at :2:3 and follow transcript2:03
Is this meeting our needs, our wants, and desires as lots of the people and as individual groups of people? Let's look at a more concrete example. We've got our example here. Sally is 38 years old and has two small children. Sally wants to navigate the aisles of the store with a

double buggy and she cannot. This is a problem. We understand that her store is not wide enough, or that the products that we have are spaced in a way in which it's very difficult for somebody to navigate this shop.
Play video starting at :2:29 and follow transcript2:29
We can think about this in terms of what actions could be performed, and how might this change the way in which people interact. We can have wider arrows and that will help a lot of people. That will help people who are walking double file, it will help people who have certain things in their arms, or maybe they're struggling to navigate around corners and gaps. When we design a system that facilitates some actions for one group of people, we're actually thinking about the impact that it has on much larger group of people. But we're doing that on an individual level, and we're thinking about those individual people with those individual personas and those individual scenarios. How is this going to impact each one of these people, and what's the impact likely to be? If you're increasing the width of the arrows, we can actually facilitate lots of positive interactions.
Play video starting at :3:12 and follow transcript3:12
We can create a situation where our store is more accessible by people, are more likely to spend some time here, and are more likely to buy products from us. That's a good thing as a positive impact to this. Now, conceptually, the more we know about our users, the easier it is to facilitate intelligent, rational design decisions. If we have access to more data and we can think about these personas and scenarios in really detailed ways, we can actually do a lot with them. We can make really interesting decisions based on all these different types of feedback. Now, of course, there are mixed considerations with this. Some of our users might be in a situation where they have needs that compete with other types of users.
Play video starting at :3:50 and follow transcript3:50
They may have a situation where some of our users do not like strobe lighting, and some enjoy it. What do we do? We're making a film or we're creating some interactive graphics. We've got some people who would like this ideas, some people who don't. We've got to make some considerations and we've got to decide what the best way of designing this system is, so you facilitate all those different people and those competing objectives. Well, ultimately, we have to look at this as a cost/benefit analysis, and we have to say, well, ultimately, we have to compromise, we have to find alternatives in some cases. There are two real situations that we can take here.
Play video starting at :4:21 and follow transcript4:21
We can compromise and we can go down the middle, or we can think about alternatives where we can facilitate both types of users in different types of ways. Now, of course, there may be extreme cases here. If somebody has photosensitive epilepsy and we have shed lighting in whatever we're producing, that can create real problems that could cause somebody type of seizure. It can be really damaging to that group of individuals. In that case, we might actually want to remove this features entirely so as we don't create harm in our system. But these considerations are complex and there isn't an easy and simple solution. We have to really think about these things in a robust and meaningful way.
Play video starting at :4:55 and follow transcript4:55
Again, going back to what I said earlier, the more information, the more data, the more resources we have access to, the more personas we generate, the more scenarios we consider,

the better our decision-making process can be. Why do we do this? What is the point to do this? Well, it's all about the user story. It's about thinking about a set of interactions. It' s more than just a click button, I click a link, I scroll up and down, I navigate around a page and I turn computer off. Interaction design is obviously more complex than that.
Play video starting at :5:25 and follow transcript5:25
Of course, so are human beings. People are complex creatures, and complexity is embedded in our lives. We cannot ignore this, instead, let's embrace it and let's use this as an opportunity to create interesting design challenges for us that we can solve to create more robust, meaningful experiences for human beings, because ultimately, we want to give everybody the best possible experience. We can do that by understanding our users and embedding that complexity in our setup design processes.

Last time around we looked at the double diamond process. We explored some constructs around interaction design of the basic components that actually make up interaction design. In this topic, we are going to spend quite a bit of time talking about modeling and why modeling is useful and how it helps us to underpin the work that we do. Helps us to understand both systems in human beings in relation to the things that they're trying to achieve. You can think of this as a goal-oriented sum activities. But we can also look at this in terms of things like system architectures and understand how to build an effective system based on a number of constructs. Let's go back and think about what we did last time.
Play video starting at ::47 and follow transcript0:47
Last time we looked at the double diamond approach, which looks something like this. [inaudible] diamonds. But essentially the principle behind this was we were thinking about modeling a design process. We we're thinking about moving through a set of constructs and allowing us to facilitate a set of interactions that allowed us to move across this axis and think about as a project's moving in half linear way, what's happening, how is our understanding changing? But also thinking about moving in other directions as well. Thinking about that process of convergence and divergence of opening up, closing down, narrowing scope, opening up scope again and never rain back down and thinking about that refinement of ideas and concepts to what should hopefully be a set of good ideas and good concepts. The idea behind this is essentially that we are converging. I'll use that word again.
Play video starting at :1:36 and follow transcript1:36
We are converging on a self-designed constructs that works for us, that works for our circumstances, and allows us to be successful in designing systems. We don't want to think of things after the event and think, I should have done it this way or I should have changed the way that I worked based on these sets of features or parameters. What we really want to be doing is making sure that we're doing things right the first time and that we were not causing ourselves problems down the line. If we start to design wrong, if we start to design with the wrong ideas or concepts in mind. If we design without thinking about what the user is capable of achieving, or thinking about the user's expectations or that reality that we create problems for ourselves. How can models help us? Well, we've already seen an example of a model that might work.
Play video starting at :2:14 and follow transcript2:14

Now, let's think of some other ways that we can model things. How do we model human beings? What is a model for human being? Well, we could start very simply. We can say, I'm going to draw a stick man. My awful drawing skills are apparent once again, turn on the toll that tries to make sense of what I'm doing even though my drawing is so bad that [inaudible] can't even understand what I'm doing in all of the time. But this is a model of a human being.
Play video starting at :2:37 and follow transcript2:37
This is a model that we could use to think about things like interactions with the real world. What does this person do? What happens to this person based on events? We can think about things like functional modeling. So we can have a function that's performed on or performed by our individual. We can think of this as a set of modeling structures to allow us to embed an understanding of how this person interacts with the world. Now, there's a stylistic version of this that probably most of us have seen before.
Play video starting at :3:7 and follow transcript3:07
When we model user interactions and when we think about the user as a set of component parts and things that they do and things that they can achieve, then we might be familiar with seeing a structure that looks something like this. We have a series of boxes. Then we have some text underneath that describes the events that are happening in those boxes. This is a technique that's quite popular in HCI around storyboarding and it's persona driven. It's driven around the idea of having an individual or a series of individuals interacting in different ways in each step and each stage. They might be driving a car. Apologies again for my awful drawing skills.
Play video starting at :3:49 and follow transcript3:49
They might be writing a letter, for instance, so you might have a letter and a pen, and they might be scribbling away at some letter. We can see there that there's a set of interactions. We can see that modeling process and our assumption is that it's linear, so it's happening left to right, possibly even top to bottom, so we might have another set of frames down here that covers the next spectrum of activities. But the idea behind this is essentially thinking about the ways in which people can do things and thinking about the ways in which systems can allow people to do things. This human centric way of working is really important to us. It's really important that we understand how human beings work, how systems work, and how they can work together. Now, modeling is the strategy behind thinking about these things and we've looked at one model of human beings.
Play video starting at :4:34 and follow transcript4:34
We looked at another extension of that model and looking at how human beings might exist in these frames, these keyframes, these key events that happen within a set of circumstances. But we could also model a human being under another set of interactions or another set of components. We can think about component parts, we can think about an arm, individual legs, we can think about a head, and the body. Those are individual component parts and we can model a system based upon that. We can say, well, an arm is a part of the system, so just draw a square around the arm and a leg is another part of the system. We can model each one of these things. We can model them in different ways and we could describe the different types of functionality, so an arm, I could wave, for example, that's what an arm can do.
Play video starting at :5:20 and follow transcript5:20

That's a functionality. We can think about an example of a modeling system that we've probably already seen that allows us to model this, that allows us to describe these sets of processes, and it actually has two component parts to it. It's a part of UML. If you've done Agile software projects with me, you might have seen this type of UML diagram before and you probably already know what I'm talking about, so now I'll split it into two, and I'm talking about actions. I'm talking about this idea of attributes and operations in a class diagram. Thinking about an arm, we can think about what an arm can do, and that's a good example of one type of representation of modeling of a system, of a functional aspect of the system. We're thinking about what the functional components of the system or the technical components of the system?

Play video starting at :6:1 and follow transcript6:01

We can model all of these things too. It doesn't have to be about an implicit set of interactions. It doesn't have to be about things that mental models and cognitive models. We can use modeling in lots of different ways. That's what we're going to cover through the rest of this topic. We'll look at some examples. We'll have a go at trying to model some things in our own lives.

Play video starting at :6:18 and follow transcript6:18

Then we'll think more broadly about activities and environments that interact with one another over a given set of circumstances and periods. Thinking about not just this idea of one-to-one interactions, this person does this thing. I wave with my hand, it's thinking about what does that mean in the broad spectrum of activities. We'll look at some more concrete examples of that and how modeling can help us and how we can design, and how we can model systems and human beings in a really effective way.

Now that we've had a chance to look at modeling in our own lives, let's think about some more examples of things that we could look at. These might have been examples we've already covered and discussed, but let's think about how modeling can be used for other types of things. Let's use a traditional structure, one that we're most familiar with, one that most of us either live in or frequent to a certain degree. Let's have a look around how this can be modeled. Let's think about the idea of a house where people live. House would possibly be modeled in this way. We could say that's a house, a very primitive house or being.

Play video starting at ::42 and follow transcript0:42

In fact, I'll use my tool to make it look a little bit better. That's my house. I'm going to have two upstairs windows. I'm going to have a door. It's very big door, isn't it? Most houses don't have doors that big. But I might have a tree in my garden, who knows, and I've got lots of other things there.

Play video starting at :1:2 and follow transcript1:02

Maybe I've got a path or a driveway depending on where we live, maybe I've got a picket fence, outlets, let's draw some perimeter around this. That's one way to model a house. It's very simple model of a house. Probably, not all that effective radius describing what the house is capable of doing, but it gives us a broad understanding about what a house might look like. We've got a sense now of where this sits in the periphery of a system. We can model that, and we can model that alongside other objects. We can model alongside a human being, and we could look at how human being might interact with that.

Play video starting at :1:31 and follow transcript1:31
We're talking about maybe somebody's journey to and from work for instance, that might be something that we might want to model. There are other examples of things that we could model with that primitive view of how a system looks. Another view of a house that we might all be familiar with if we bought homes, or if we've tried to make extensions to a home, so changed our homes in some ways, is a floor plan. You may have a room by room view of what house looks like. Describing each individual room in terms of size, in terms of objects and proximity to one another, we can have a series of views here, and let's say my house has three rooms, very limited house will be in that room circular. Well, it could be, you never know, stranger things have happened. Some people like to design houses like that.
Play video starting at :2:12 and follow transcript2:12
Maybe that's my swimming pool in my garden. Let's go with it. Let's work with the constraints of the system we have in place. That's my swimming pool in my garden, and I have some features around it. Let's imagine some landscaping going on. Maybe I've got some trees in there. Some of the bits and pieces.
Play video starting at :2:30 and follow transcript2:30
This is my house, it has two rooms. I have a swimming pool in a garden. We can model this and we could describe features of the house. We could say this area is, I don't know, four meters maybe. Then we could describe this area is three meters. It's very small house actually a small room, but it gives us a sense of the size of objects in relation to one another. I can imagine things in that place.
Play video starting at :2:50 and follow transcript2:50
Now, we can imagine where the sofa would go, and maybe you have corner sofa over there. Maybe I'll have a television, maybe I've already put in here. Could be much huge television that's actually twice or three times the size of my sofa. Very big television, looks like it's about three meters big, that would be great for watching football on. But in the constraints of the system, we can model different types of things. We can think about, well, how do these rooms fit together? What is the interactivity going to look like?
Play video starting at :3:17 and follow transcript3:17
What's it going to look like if I put the door here, how is that going to affect things? I'll just highlight where I'm at. The moment is not clear. Let's imagine I've gotten a door here, and that door opens up. I've got a nice wide door, I come to my television now there. I understand from this modeling process that there's certain restrictions here, the same constraints. It allows me to better understand my system, and a set of interactions, I can think about how I might live in this environment.
Play video starting at :3:42 and follow transcript3:42
Where would my object go? Where would all the things that I own go? Where are my possessions? Does it fit my needs and wants? I could also model this in a function of why. I can have a list of things down the side telling me things that are good and bad, for instance. If I'm looking for a new home, I might say things that positive and things that are negative.
Play video starting at :3:59 and follow transcript3:59
The main features of the house, it might be limited in space. It might be most spacious depending on the proximity of the rooms and things like that. Again, it's about thinking about modeling this in relation to user-oriented goals. If I want to swim every morning, having a

swimming pool, of course, makes a lot of sense. If I never swim and I pick a house that has a swimming pool in that, that's not really quite right for me. It doesn't really make sense because I've still got the maintenance and the upkeep of that isn't adding utility. As you can see, modeling gives us some value in helping us to describe the systems and the human beings that work together.
Play video starting at :4:34 and follow transcript4:34
We can think about those implicit set of interactions, and start to explicate some of those interesting things are happening and try to make sense of how is this going to work in a situation? How can we improve on things? How can we model in the right way and ensure that we have a good understanding, a good functional understanding, a good technical understanding, a good user-oriented or user-centered understanding of interactions and interaction possibilities? There's a whole host of ways that we could use modeling to benefit others, to help us to understand, to design, to evaluate, and to conceptualize all these things in a meaningful way. Again, modeling is something that's really useful, and we'll have a look at some more techniques that we can use as we go through. But I'd like you to keep this in mind and think about ways in which you can model things. Think about home. Maybe you could come up with some more examples of how you might model a home.
Play video starting at :5:19 and follow transcript5:19
Maybe you could look at some functional examples. Maybe you could think about it in relation to all the other things that are happening around you. You can model it as part of the electric grid, for instance. Come up with some more examples, have a chat, join us in discussion forums. Let's just talk about this idea of modeling, and how it can add value and utility to what we're already trying to achieve.

Let's have a look at some more modeling techniques and let's move on to trying to model systems. We've looked at some examples of types of things you might want to model houses, we've looked at an example of a UML diagram which is all about understanding classes and attributes and operations. Now let's move on to thinking about the whole process of actually building a system and how we can model that. In fact, we've already seen some examples of this. You've been building systems as you've gone through the course. Of course, you've got some understanding about how to do these different things. You might have been using agile, where you might have been using one or the other software development methodologies.
Play video starting at ::42 and follow transcript0:42
There's quite a lot out there. Some more contemporary, some less contemporary but they all model the way in which we can work in order to substantiate some sets of achievements. Essentially to build something, we have to do some things, and so we have to follow some steps or processes. Let's have a look at what these processes might look like in relation to modeling a system or modeling a set of systems that allows us to achieve something. Let's have a look at a model that I quite like one of my favorites. In fact, and let's have a look at a couple of different models this time around. I think we can look at the differences between them, but we can also look at how their variation might not really matter in practical terms.
Play video starting at :1:19 and follow transcript1:19
Let's think about modeling a system around the Spiral model. This is an interesting one, and it's a different way of thinking about things than most people are aware of or familiar with. When we

think about things, we tend to think in a linear way. We think across an axis, so we go left to right, top to bottom. That makes sense for most people because that's the way that we read, that's the way that we navigate the web. Generally speaking, we do things that way and so it makes sense. Actually the spiral model embeds a different way of thinking.
Play video starting at :1:47 and follow transcript1:47
There's essentially four steps to the spiral model and you move between them in a linked way. We can think about four steps or stages. We'll break this down into four. During those steps we've got Steps 1, Step 2, Step 3, and Step 4. Now, what happens is you have this spiral and I tend to draw out from the middle but actually, you might not think about it this way. You might think about it as it's going in. This idea of a spiral is essentially thinking about this process of moving through these steps of iterations to moving between these steps sometimes.
Play video starting at :2:21 and follow transcript2:21
Iteration in-between the steps is important. It's not just enough to think about. We talked about design as an example of this, so thinking about design as an isolated set of constraints is not really the right way to think about systems because design should be feeding in from somewhere else. We should begin bits of data coming in, we should be looking at scenarios of use, trying to contextualize that and then using that to feed into new, better designs of systems, and so design is iterative, it's formative, it's not something that is linear in the sense that we do it and then we stop and we forget about it. We will do it in a linear way. We'll design, evaluate, test, iterate and this idea of iteration it and makes it non-linear with cyclic, we come back on ourselves. The spiral model is a good way to embed this way of thinking in our systems.
Play video starting at :3:6 and follow transcript3:06
We start out by identifying objectives of Step 1. This is where we think about, well, we've got a research question, and when we think back to our double diamond model, that was another way that we could think about things, our requirements, our understanding, our research was really about identifying objectives. Things that people want to deal with the system, how are they going to achieve those things? What does that look like in the manifestation? When we got to the second diamond in the double diamond design process, we're actually thinking about implementation. It's thinking about these objectives in implicit and explicit ways. Thinking about, well, what objectives are people have in a general scope?
Play video starting at :3:39 and follow transcript3:39
What things do they want to achieve with our system? Then what's the manifestation of activities around that, that allowed them to achieve the thing? We start out with identifying objectives in Panel 1, we then move over to Panel 2 and think about performing a risk analysis. Understanding what risks we have with our system and what could go wrong. Most of us should be building contingency into our system. I like to model this as an X, as a cross. You might want to do it as a red cross, if that's your way of thinking.
Play video starting at :4:7 and follow transcript4:07
Risks are essentially things that can go wrong, so I like to have this visually modeled as across. But performing risk analysis is something that's an ongoing process. As we get more information, we can think about risk as being this implicit thing that becomes more explicit as we get more information. The data allows us to understand actually, there might be a COVID pandemic coming up. With the infection rate and the rate increasing, we might be subject to these things and it might have never happened, or it might not have happened in a 100 years or

so, but again, we need to model this in our system and understand that actually this is becoming a more realistic risk than we thought it would be, and now we need to think about how we're going to act. Whether we're going to pertain resources in our systems around certain activities to allow us to overcome those risks. Our next step is down here in panel Number 3, which I've labeled.

Play video starting at :4:56 and follow transcript4:56

I'm just going to highlight where we are for you, just so you know. We are in panel Number 3. Essentially down here with thinking about development and testing. This is a modality we're all familiar with, all systems designers, developers. Thinking about building things is essentially fairly straightforward. I like to model this as a tick, and I've got red which is not right really because that's building a broken system and we don't want to do that. I'm going to model that as a green tick.

Play video starting at :5:20 and follow transcript5:20

Building something developing and testing, we want our test to pass generally speaking, so it'd be nice if we got lots of ticks in there and said, okay, we're finding that this is working, that is producing something meaningful and has value. Then finally thinking about review and evaluation. Again, this is a question mark, I think it's about, well, we're going back to the idea of objectives and thinking about whether or not we're meeting our objectives. Do our aims match what we say? Are we evaluating in a contextually driven way? Does it make sense the way that things are working? This is one way that we could model a system.

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We could think about modeling a system as a set of interaction activities, how we build things, how we test things out, we evaluate things. Now let's have a look at another way we can model this type of system. It doesn't have to be this way, we could model in other ways so we can model in other interesting and valuable way. We can think about other types of development strategies and it's not to say that any one of these is better than another, what we're doing here is just showing you the different ways of modeling. It's not a case of, I pick this methodology over that methodology, each project is different, it really depends on the resources and the people you have, what things you're trying to build, how much flexibility you might have in a contractual obligation, for instance. There's a lot of things that can change the way that you work. Let's have a look at one more way of modeling systems, design and development.

Play video starting at :6:39 and follow transcript6:39

In this case we'll use one, we've probably all seen before, one that's really popular. We're going to look at the waterfall method. We can think about modeling these steps in this more linear way. We can think about, well actually, this is a slightly more linear process, when we think about the waterfall methodology. We can think about moving down in terms of fidelity, in terms of increasing our functionality, and through each step, and each state, and each stage. What we're hopefully doing is converging again on our design goals, on the things that we want to achieve. Now this way of thinking is quite an old way of thinking, but for a long, long time it was used in software development and it worked really well and there's some projects where you have a really rigid focus, where this is actually a good way of designing and developing systems.

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It's not to say that this is fundamentally flawed, it's just to say that it has a certain application space and although lots of people have moved on to agile, that doesn't mean that the waterfall method or iterative variants thereof aren't useful in helping to think about how we build systems. We usually start out at the beginning with some requirements. Again, we'll model this with a question mark. What does a user want to do? What do they want to achieve? We then move on to the design. I like to think of design as being a set of constructs around, well, how are we actually going to prototype?

Play video starting at :7:52 and follow transcript7:52

How are we going to create variants of this system and smaller versions and then scale that up over time. What's typically happening is we're improving over time you might have a crossed there, we might have crossed there and finally we get the right design and it's a plus, it's a big tick. Then we think about implementation, which is much more about thinking about the functionality of the system. How am I going to build this thing? We might have some pseudo code or some actual code, we might compile some code and we might run some code, we might do some things, we might render some things, and that process is all about implementation so typically this looks like software. What comes out here is usually software. Whereas before we have an architecture, we have a structure, we have some component parts.

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Here we're thinking about software, we're thinking about delivering on some realistic things, some tangible elements. Then we might move down into verification, which is usually, again that tick box that, okay, are we meeting our requirements? Let's go back. Let's think about these things that we've thought about before. Validate them against some expectations. Are we actually doing the thing we said we'd be doing? If these are the requirements, does our system match each one of those requirements in a meaningful way?

Play video starting at :8:58 and follow transcript8:58

Then at some point in time we will go through this process where we wonder off and we go into this process of maintenance. This is supporting for bugs, making sure that new versions of the system are working with new operating systems, with new hardware, with new software, just ensuring that the system doesn't fall down and crash and explode and cause chaos. This is a fairly linear set of processes. We can think of it as top to bottom, and as we move down, our structure would look something like this, on our screen now we're now going through that. But what we'll also be thinking about is, our design solution is somewhere down the bottom here. This is where we've solved the problem here and over time, we're gradually developing our understanding. We're also at this point in the process thinking about each step.

Play video starting at :9:42 and follow transcript9:42

It's not just about understanding how we're going to build the system, but how are we going to understand our use case and our users. Each one of these stages, it's not just about delivering, it's not just about implementing the system, but also thinking about the design and development and validating as we go. Many models of this many extensions of the waterfall model nowadays look at ways in which we can actually feed back, and making sure that we can look back and reflect back on the things that we've done, and make sure that we're doing things in the right way. If we spent a long time on design, if we spent a long time on requirements, and then we get that wrong, there's very little opportunity to go back in what's the traditional model of this. More adaptive models, more reflective models might be more suitable if we're working in an area

where there's a degree of uncertainty. But again here modeling is one of those things that's quite complicated and we can model things in different ways. We can model humans in different ways, we can model systems in different ways and they all have intrinsic value for a certain set of processes.
Play video starting at :10:35 and follow transcript10:35
It's about understanding what these models work for, why they don't work and applying them in the right context. That's what we want you to understand by the end of this topic. That these models are all different, but they all add value in understanding, that they all have utility but in the same sense that we should use them in a way in which it's meaningful. We shouldn't just stick to a model and say that's what I'm going to do, we should be flexible, we should be adaptive, we want models that are flexible and adaptive too, so we have to work in that way as well. Hopefully that makes sense and we'll move on and we'll look at some more examples of different types of modeling we can do around the user-centered design aspects. We'll talk about how we can verify, whether we can validate things based on certain types of models and how they can be used in a much more iterative way than some of the examples that we've seen so far.

Models can be very useful if we're trying to model the behavior that we as interaction designers should take. What steps should we do and how will we get to a finished product. We can also model the user themselves. Now what we're about to go through here may seem slightly like we're reducing the user to aesthetic boxes and systems, and that can be quite counter intuitive to what we're trying to do as a interaction designers. As interaction designers were trying to understand that all humans are different, and that our users will have diverse sets of needs and wants from what we're making. However, if we start to model our uses, this can be a really useful first step to ensure that the designs that we are producing actually will work for the users in the first place. That any of the obvious flaws or problems we might already be able to catch if we put our designs through a model, if we assess what we are thinking, and what we are creating against these models.
Play video starting at :1:10 and follow transcript1:10
Let's have a look at what I mean by modeling a human. Before we start that, what I want to do is, I want to discuss how we might model a computer. What do I mean by model? Model is example of something. It is a reduced form of an object in the world. If we were to model a computer, for example, one way we could do this is by splitting a computer up into various parts that have different roles and responsibilities. Here I want to reduce the computer to a set of inputs, outputs, and processing power.
Play video starting at :1:48 and follow transcript1:48
What inputs might be model a computer to have? If we think about a standard laptop or a desktop, we might say that a computer to has a keyboard input or it has a mouse input, but that's not thinking about what computers means these days. These days we will have tiny computers in our pockets. What inputs does a smartphone have, for example? A smartphone has an input of a touchscreen that's quite similar to a keyboard. Let's think about other ways that we input data into smartphone. Another method is that we regularly talk into our smartphones, so that's a form of audio input for our smartphone.
Play video starting at :2:26 and follow transcript2:26

Sometimes if we're playing games, are smartphones might be able to track our gestures as well. Again, smartphones are a pretty boring form of computer these days. Computers are all around us and the types of inputs they take are very diverse. In VR, for example, our head movements are being tracked if we're wearing a form of head-tracking VR system. Even our washing machines are computers these days, and the inputs that they have are things such as buttons that we are pressing or even wake senses within the machine. Just because your first thoughts for input might be about buttons, and keyboards, and mouse, doesn't mean that that's what they always restricted to do. Computers these days can sense light levels, they can sense sound levels, some can even sense smell.

Play video starting at :3:10 and follow transcript3:10

Computers can be very diversely modeled in terms of what their inputs are. Now let's do the same for outputs. What outputs does a computer have? If we think about our standard one, a computer will have a screen that shows us information and it can often output audio. There's a quite basic. If we go back to those instances of other types of computer, for example, our phone emits haptic output. Haptic means buzzes, or shaking, or anything that we can sense with our hands, anything that we can sense through feel.

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If we think about that, washing machine example, one of the outputs of a washing machine is the correct amount of water and the correct amount of spinning of a motor. There are a number of different types of outputs that we can have as well. The final aspect we want to model here is the processor. This is an important aspect of a computer, exactly how quickly can I produce responses to some of the inputs that we are giving it? How much data can it store? For example. The ideas of its RAM, its CPU.

Play video starting at :4:16 and follow transcript4:16

Things relating to the processing and memory. These are also important parts of our model of a computer. Why have I modeled the computer in this way? What I want to do now, is I want us to consider the opposite. I want us to consider how a user, how human would be modeled using these same three categories. Let's firstly consider the inputs that a user has. A user doesn't come with a keyboard and a mouse.

Play video starting at :4:45 and follow transcript4:45

A user's inputs are slightly different to this. One way to think about user inputs is to think about the senses that the user might I have, for instance, our eyes are a form of input. This means that we can take in visual data. The types of inputs that you can give a user, showing a user information on a screen, for example, or LCD display. Another form of sensory input is listening. We are able to take it audio cues. This means that we can give it user input by giving them sound information.

Play video starting at :5:22 and follow transcript5:22

Just like I was talking about with regards to our mobile phones being able to take in sound information, we as humans can do the same thing. Another form of input that we can give the user is that haptic information. A user is able to not only feel that their hands, but feel all the way along their body, and this is something that we are seeing more regularly with regards to wearable devices. Wearable devices are those that we can wear passively on our person and don't necessarily need to hold or actively pay attention to. The key with these wearable devices that we don't pay attention to is that they need a way to grab our attention and give us

information. If when we were designing, for example, a smartwatch, we assumed that the user would always be able to take input via the visual channel. Then it's likely that the user would miss a lot of the information we were trying to tell them.
Play video starting at :6:13 and follow transcript6:13
However, if we think about the other forms of input that the user can take, we might realize that haptic is a better idea. Haptic input would mean that the watch would buzz to get the user's attention. This would mean that the user wouldn't need to be paying attention to the screen all of the time and we would be able to get their attention as and when we needed it. You can see in this way, the human actually has a lot of different types of inputs. Therefore, as interaction designers, we have a lot of different opportunities for designing things that can convey and give information to our uses. Now, let's talk about outputs. Interaction design is a two-way thing.
Play video starting at :6:51 and follow transcript6:51
We're not just talking about what technology can tell the user, we also need to think about what the user can tell technology. What type of outputs does our user have? Well, our user has motoric outputs, motor outputs, the ability to move. This means that our user can press buttons on a keyboard, or a mouse, or a washing machine. They can manipulate physical objects to give information to technology. Users also have a bit more passive information they can give us via technology. One of those things is movements.
Play video starting at :7:26 and follow transcript7:26
Now, by passive, this is information that the user doesn't actively think. I need to press this button, I need to move my finger over to press this button." It's something that the user does automatically. You'll notice as I'm talking to you now that I'm gesturing, and this is a very natural thing for me to do. It's a natural thing for a lot of humans to do. By using this movement data, we can create experiences and interactions that are a bit more naturalistic for humans. For example, if we think about VR again, we are often has hand-held units that you provide information for the game or the experience that you're interacting with currently.
Play video starting at :8:4 and follow transcript8:04
Rather than asking the user to sit in VR and move a joystick, which is a bit more artificial, we can ask the user to just do their natural movements. What would you ordinarily be pointing that? How do you ordinary go to pick something up? We can take this type of output data from the user and we can process it and use it in more naturalistic way if we'd asked our user to press buttons. One even more passive form of output that our use it and give us is physiological data. Physiological data is information about us that we are just producing without thinking at all. It's often reflex data, or just information that our body is producing as a natural secondary output for our normal everyday production.
Play video starting at :8:51 and follow transcript8:51
Why do I mean by that? Well, physiological data is information about our heart rate, for example. Perhaps the level of sweat on our skin or perhaps our pupil dilation. All of this information can be used by technology to adapt, and respond to the way that the user is feeling, for example. This is something we will talk about in a later topic. But important thing to note here is that when we think about what outputs the user happens, we don't limit it just to them actively trying to press a screen or press a button. Yes, that's one full of output.
Play video starting at :9:28 and follow transcript9:28

But there are multiple levels of this type of output that we can expect from a user. Maybe we can think a bit more creatively when we're thinking about what we can ask our user to do when they interact with our products. The final aspect of the human that we need to model and we need to consider is that processing power. Now, obviously as humans, we don't have central processing units or RAM, but we didn't have memory. This is something that is incredibly important in interaction design. It may seem like it isn't. When we're talking about interaction, we're often talking about this idea of input and output.

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But between the input and output, there is some processing that goes on within the user's head. We need to consider exactly what the user is capable of when we're designing these interactions. When we think about memory, we need to be aware that most users have a limited memory. That we can't expect them to immediately memorize everything at the click of a finger. We can't expect them to be able to recall things from very long ago. We need to know that humans have limited memory. Now, there are lots of different theories about exactly how much a human can memorize, and there are some that have been used well, and some that have been used incorrectly.

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You may know something about a human being able to store seven plus or minus two chunks in their memory. I encourage you to read about that if that's something that interests you. However, it's important to realize that this statistic is often very poorly used. The idea of a chunk is something that needs to be carefully thought through and it's very easy for someone who doesn't really understand the theory to just assume they know what a chunk is, and then assume they can give a user this amount of data to memorize. What's a bit more important and a bit more applicable is for you to think about realistically whether or not you're asking the user to use their memory. If you are, are you setting realistic goals? Some things to think about when you are assessing that for yourself are the different types of memory that a user has.

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Sometimes we might be asking our users to access their long-term memory. This means asking users to recall something that they have stored for a long time ago. A perfect example of this is a password. Every time you are asked to type a password, the user is being asked by the interaction designer to use their long-term memory. Think about whether or not it's reasonable every time you ask the user to do that. There's also the idea of short-term memory, or working memory. This is memory that we use as and when we are interacting with things.

Play video starting at :12:6 and follow transcript12:06

This might mean that we show the user some piece of information and we ask them in a later screen, for example, to just recall that information. Or we ask them to read a number that we've texted them on their phone and type it into their computer to prove that that phone and that account are owned by the same user. Whilst it might not seem like there's much memory involved in looking at something on your phone and then typing it into a computer. There is a halfway point between these two devices where the user has to store that number in their memory. Are you asking them to store something that is absurdly long and no user has a chance of remembering or have you designed it in such a way that will help the user to remember? We'll look at some case studies of this at a later date. Another thing to think about with memory is that there are different types of sensory memory.

Play video starting at :12:59 and follow transcript12:59
It's likely that if suddenly the video stops now, you would still be able to hear my voice. You would be able to recall the sound of it, and you might be able to hear if I started reading off numbers, for example, 18, 4, 87. You will be storing that in an auditory loop. It's also possible that if we stop the video now, you would have a visual memory of me. You might be able to see me on screen, even if the screen went black right now. This is not a guarantee, there are people that lack both types of these memories. But this is something that we need to think about when we are asking people to memorize things.
Play video starting at :13:36 and follow transcript13:36
Are we asking them to store it in the auditory memory or their visual memory? Again, there is a lot more reading to be done around this area if you are particularly interested in the cognitive processes. But what I want to do for you here, is just to highlight that there are these different opportunities for asking users to use memory in different ways. Now, I've just gone through a high level idea about how we Button model the user and I'm not the first to do this. In 1983, the authors Card, Moran and Newell created something called The Model Human Processor. Their attempt was to simulate a user model in the same way that you think about a computer model. You can see that they have split up the user into these stages that I was just talking about; the idea of inputs and outputs, and the possible various cognitive memory paths that this data can travel from between output and input or input and output.
Play video starting at :14:33 and follow transcript14:33
Now, this might look like, as I said at the beginning, quite a reductionist way to look at a user. But when we can start to think about how we can use this model in the early design phases, we start to realize that we can apply it in ways to help us as designers think and design more sensibly for the humans that will be at the other end. It's very easy during the design process to think about how things should look, or just the aesthetics of something, or to simply transplant an idea from one field into another without considering how it will actually be used by a user. Whilst I am going to encourage you to get involved with real life participants and users. Thinking about our humans as these models can help us avoid some early mistakes. Let's have a little bit of a look about how we can use these models. One thing that this can help us do is realize that, whilst all users are individual and every user will have completely different desires, wants, abilities, there are some things that are going to be generally common ground between our users, and if we want our products to be used by humans at the end of it, we can think about these models and realize what the limitations of this human user is before we start designing.
Play video starting at :15:50 and follow transcript15:50
For example, it allows us to think about the possibilities of that human's inputs and outputs. As I said before, if you were thinking, "Well, we'll just stick some buttons on it," or, "We will just ask a user to click on something." Yes, that's a fair interaction type, but it's important that you think back to this model and think about the variety of different input and output types that a user really has. Maybe it might help you to be a bit more innovative and explore new and interesting interaction opportunities once you realize that there are these variety of options. Another thing to realize is that this model will help us understand what the limitations of our users are. What is a realistic expectation for the number of inputs that we can also use it to make? For example, if you were designing a game and you wanted a user to use a game controller and press lots of different buttons at the same time to create a very impressive combo.

Play video starting at :16:47 and follow transcript16:47
Well, you need to be aware that the user has limited functionality of their hands, they won't be able to press every single button at the same time that there is a limit to the speed at which they can press these buttons for example. If you're aware of what the limitations of a general human user are, then you can start to design and ensure that whilst you might be trying to make things interesting or challenging, you can show that you're not making them impossible. Another important way to use this model is to realize that it does not represent every human. There are lots of humans out there who will have limitations on these various aspects of input, cognitive processing, and output. We'll go into this in a bit more detail when we discuss accessibility and inclusive design, but for now, it's important that you use the model and realize that whilst a lot of humans will have this base level input, output, and cognitive power, some users will vary from this quite distinctly, and so it's important that when you are modeling how use may interact with your product, that you think about what it would mean for a user to have a slightly different model to be missing this certain aspect or to have an extra ability in this certain area. A model can help us think about this before we even start talking to users. Although I would like to stress again, it's incredibly important that you don't just rely on models, talking to the end users themselves is always going to be the most important part of interaction design.
Play video starting at :18:15 and follow transcript18:15
Let's look at some examples of how we can use this model. As I said, it can stop us designing impossible interactions. This is something that I think affects me probably the most when I'm on websites, for example, and I am asked to enter a password or choose a password. Often the requirements for setting a password are absolutely assaulting. Some of them will require three numbers, no lowercase letters, only uppercase, and two types of punctuation. Whereas another one tells me that I have to be limited to 16 characters only, and I'm not allowed to include any English words in this, and another one will have entirely different requirements again. Can you see what the problem is here?
Play video starting at :18:55 and follow transcript18:55
If we look back at our model, we've already spoken about the fact that users have generally a limited capacity for memory, and that asking people to recall things is going to be quite a difficult challenge. When people design password requirements they have such astonishing distinguished features that they will need to meet, this is an example of pretty impossible design. If someone who was designing this stopped to consider the model of the user and stopped to consider exactly what strain they are putting on the memory of the user, they might avoid those design decisions, impossible design decisions that create experiences that users hate and create experiences that users have to work around, sometimes in secure ways by using password managers, and sometimes in insecure ways by simply using the same password over and over and over again. If the interaction designer had thought about this at the start, they might be able to avoid that level of insecurity in the system. Not only do models allow us to avoid impossible design decisions, they also allow us to make informed and sensible design decisions. As I alluded to them, asking a user to recall a piece of information is actually quite a difficult thing to do. Having to go back into your memory and pluck an idea out from the back of your mind is actually sometimes quite a difficult thing.
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Something that is easier than recalling it is recognizing information. Think about this, if you had ordered something from a site that you order from regularly and you need to go back and look at that order, you are often given the opportunity to recognize that order, that is, you'll go to that website and it will show you a list of your past orders. This is great, it means that you can scroll through the list and find that past order and then discover what that product name is because your friend is desperately asked you way you got this product from and what it was called so they could find it. If, on the other hand, this website required you to use recall, the way it would work would be that it would show you a past order page and it would simply have a search box that said type in what order is that you're looking for, then you would have to rack your brains and try and to remember what the product was called, try and remember some key words about it, try and remember maybe the date that you ordered it on, and then you'd have to enter that information. That is a lot harder to do than simply giving the user the opportunity to recognize information. You'll realize there are a lot of opportunities for you as an interaction designer to choose the recognize path over the recall path. When you think about the model of the human and how hard it is to ask them to dredge information up from long-term memory, rather than recognizing information, you'll start to think about better designs and making things more usable and better interactions for users.
Play video starting at :21:49 and follow transcript21:49
Whilst it may seem like the model is asking us to reduce all humans down to the same set of basic inputs and outputs, actually, what it's doing is it's ensuring that we make some very early good decisions, that we avoid making impossible designs and that we consider that humans do have a bit more possibility than maybe we might initially realize, and it might give us a bit more of an impetus to start exploring different innovative interaction designs.

So in this final exercise, we're going to be looking at another way that we can model. So we've done quite a bit of modeling already, we're going to do a bit more here, and we're going to look at modeling something, it's a little bit different to what we've seen before. And then I'll ask you to have a read of a paper which covers some basic modeling concepts from quite a long time ago now, actually from the 80s, where we were looking at modeling the idea of typing, which is something that will be a much more familiar construct to most of you. But let's go outside our comfort zone a bit here and choose something that's a bit different just to try to muddle it and think about the ways in which we can model these things and how they behave, what the interactions look like in different perspectives we can take on modeling because this is what this is all about. It's about trying to get as much information out of this process as possible and make sense of the world around us to try to to some degree compartmentalize things and talk about them in terms of their component parts. But also thinking about structures and how things are placed together, how things are organized in this implicit and explicit structure. And can we explicate these things and start to understand what it actually means in the context of interaction design.
Play video starting at :1:11 and follow transcript1:11
So let's think about the system that's designed to be interactive with. Let's think about a system that's designed for input and output, and let's think about that system in relation to how it works and the permutations of possibilities for modeling in terms of thinking about this as a system. And also thinking about those component parts that we talked about. So let's have a look at an

example here, and I'm going to give you an example of something that you probably recognize as I start to sketch things out and I'll leave it with you to try and figure out what we're modeling here. There's some notes there and we've got some lines. So that might give you a clue about what we're interested in modeling here. Hopefully I've done that right.

Play video starting at :1:49 and follow transcript1:49

But the essence of this is thinking about how we can model something which is designed to be interacted with, you know, without interaction, this type of system just wouldn't work. It wouldn't be very effective. So for those of you who haven't figured it out, this is an instrument and we're talking about guitar here, so we're talking about a guitar in standard tuning. So this is a tuning that most of us will be familiar with if you've ever played a guitar before, that's not to say that it's the only configuration and in fact, other models might look at different tunings and if we're talking about different instruments, there might be tuned in different ways. So a base would typically only have four strings of a guitar would normally have six. So this is a mapping of a guitar essentially that we're looking at here, but not, or guitars and model this way. So again, we've got this complexity about, well, we can model things in these explicit ways, but even then, this configuration might only fit some of the types of guitars that we talk about.

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So, a bass guitar, for instance, probably wouldn't be configured in this way. You tend to have a bass guitars, 12 semi tones lower. It's the same tuning, although it's a different tuning really. It's four of the six strings are used in the same way and it's tuned in the same way usually. But again, you're only using small component parts of that, and they're different In terms of the they're actually 1270 tons of parts of an octave apart between the two instruments. So they both called guitars. We can't necessarily model these things in the same way, They are quite different.

Play video starting at :3:11 and follow transcript3:11

And so we've got these distinctions of an acoustic guitar or an electric guitar and then a bass guitar, which is typically thinking about a lower tuned instrument. So, we've got this configuration. Now, we've got the tuning which is really important to determine how it sounds. But we also have other things that we have to think about here. So we can model it according to the tuning, but we could also think about component parts we could think about. Well, okay, these are the possible notes that would be played if it was an open string. But what about if we press down on different frets?

Play video starting at :3:39 and follow transcript3:39

So we typically have a grid that looks something like this. Once again, Apologies for my terrible drawing skills, but each one of those strings would have a different configuration. So if you were to press down on the individual strings, you get a certain note. And as you move through, if you were to press down on the 12th fret, for instance, on the E string that will give you an E, but it's again one octave higher, so it's 12 semi tones higher. And that will give you the opportunity then to play the same note, but in a different configuration, in a different key or actually in the same key, but in a different semi tone or a different octave. So it's a much higher pitched noise, it's a much higher frequency noise. And so again, you know, there's different configurations here.

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There's different ways that you can play, you can play on the top string open or the bottom string open. So there's lots of different ways to achieve the same things. But those would be different

notes, even though they both eat, they sound different to the year because they're in a different frequency. And so we can think about an instrument like this has been an interesting type of configuration that we can model. Now we don't just model notes in this top system. Do if those of you have seen tabulated before will be familiar with seeing something that looks like this and it would have tabs. So you might play a string, something like that on the instrument.
Play video starting at :4:47 and follow transcript4:47
Or you may have something that looks like This. Open one, open one, something like that where you play the open string and then you'd move down to what will be the B. I presume based on this notation, play the first fret on the b string. So that's given a configuration is giving an instruction. But it's also a digital sound that you can model. You can model that digital sound in a particular way. You might want to model that as a analog to digital conversion.
Play video starting at :5:13 and follow transcript5:13
You can model it as Midi, you could look at the frequency of the noises and model those frequencies. You can look at things like tempo, so the beats per minute would be another way that you can model this type of system. So again, we've got lots of different configurations that we can work with and lots of different ways that we can actually model systems based on what it is we're trying to understand. So what do we really want to understand about music? Well, let's assume this is just the musical configuration and we're not too fast about digital interpretations or squashing signals down or compression or Iki or any of those complex things. We just want to be able to play a song. So can I play a chord from this configuration?
Play video starting at :5:47 and follow transcript5:47
But I probably could. So what configuration for a court we use, let's use open one. Open 23 So you can tell from the root notes what's being played. But in this case you can also tell that this is a particular type of cord and anybody who can guess which chord. This is well done. But this is another configuration. So this is playing the notes together and when the notes are played together, they have combinations which sound nice together.
Play video starting at :6:14 and follow transcript6:14
And so we can play chords like this. We can play a C chord, we could play an E chord an accord. Maybe we're playing a minor, we could play a G. And then A D. So we've got all kinds of configurations of things that we could play with this instrument. So in a sense, although it's limited in terms of, we might have 20 four frets usually on a guitar. And then we've got six different interpretations of default sounds for each string.
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We can think of them as a default sound because if you don't press anything and just pluck the string, It will still make a noise. So you've got 24 times six possibilities that we could play here. And so the instrument itself is constrained. And we can model each one of these sounds, we can model them on the fretboard like we've done here, we can model them as chord combinations, like we've done here. Or we can model them in other ways. We can model them as digital sounds, we can model them as frequencies as pitch, as tempo, as all these different features of musical tom bro. So what we're dealing with here is essentially a set of features that we try to model according to a set of constructs that are important to us.
Play video starting at :7:14 and follow transcript7:14
We want to make sense of the world around us and when we think about something like a guitar, we can think about what it sounds like that. That sound can be broken down into

component parts so we can think of them as individual notes that we play. We could think of them as combinations of notes that we play according to a set of configurations. So if we play a chord combination of C E N A for instance, that's made up of smaller component parts. So each one of those chord combinations has a structure. So you'd have different hand positions for those and then you might have differences in how you can actually play these things. So you can play a C chord in a number of different ways, you might use to play in one configuration or in another and they have slightly different sounds, although they're both B C chord combinations.

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And that creates some complexity in terms of interaction design and thinking about how do I move between these different configurations? How do I move between these component parts and how do I ensure that this movement is meaningful to me? But I've got some structure. So how do I go from a C to an E. Well that's interesting because that's a particular set of interactions that you can think about now. When we start to model these structures and we put these things together, we can think about things like the shape of my hands. So a C shape is typically like this and any shape is like that, so they're fairly similar.

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That's not going to be a difficult transition for me to do. So you'd expect a low rate of error with things like that you'd expect, it wouldn't be that difficult to move between those chord combinations. Now, if you think about A G and A D. For instance, those are much more difficult. So you think about a G pattern, my hands like that? I've got fingers all over the place, I've got fingers at the top and the bottom and something is in between as well. Now, if I'm moving from a G, let's say to A D.

Play video starting at :8:50 and follow transcript8:50

Where I'm at the top of the instrument and I'm moving from having my hands basically all over the instrument to the top. That's a much more involved movement pattern because I have to move all of my fingers to change that shape with my hand. So thinking about that as a set of interactions is much more complex. And so if I'm going, for instance, from a G to D. That becomes really complex. And so in that case I probably expect to make more errors. I'd probably expect that I might have issues with things that things might not go right according to my plan.

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Maybe I won't move as quickly so I won't be as efficient. So thinking about things like effectiveness, can I do it? Yes, in both of these cases and my efficient, I might not be that efficient if I'm not an expert. So if I've not played guitar for a number of years and I'm not a very good guitar player. Moving from a G. to A. D might be hard from it, depending on my level of expertise, whereas a C to any would probably be easy even for a novice. So again, we're looking at this in terms of modeling according to different types of things, we might want to model according to behaviors.

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So interactions, we've talked about, we might want to model according to a set of systems and how they interact with one another. But when we model the systems, we also want to think about their component parts. What is each system made up of and how is that actually going to change the way that this whole system is impacting something else that is doing. And so

modeling is really about understanding and appreciating the complexity of these overarching processes, but also breaking things down into smaller component parts. So we can make sense of them, we can make sense of how they work together and we can start to improve on our set of processes. This is essentially what we're getting at. We're thinking about how can we improve our processes?
Play video starting at :10:27 and follow transcript10:27
How can we ensure that we are looking in the right places for problems and how can we ensure that we're converging on interesting design solutions that solve those problems? Is this instrument designed in the right way to solve a problem? Well, given that millions of people worldwide play this instrument to some degree of success or maybe failure, depending on how good you are a plane, we can understand and appreciate that there are different connotations to this and it's successful in some cases for others, it's not, but generally speaking, it must work reasonably well if you've got so many people capable of playing this instrument. So we could say that that's a reasonably successful case. And when we model it, we can break that down into component parts and understand why they're successful. And we could even break it down into really, really small, granular level elements and make sense of that as a set of larger processes. So what does it mean when I play a note?
Play video starting at :11:14 and follow transcript11:14
What does it mean when I play a chord? What does it mean when I play a whole piece of music alongside other people, can I keep up with them? Kind of playing the same key? All of those questions are really what we're trying to answer. Makes sense of. So we're going to move on into the next exercise and we're going to read a really important paper. And this paper will give you a sense of how we can model typing and thinking about typing and all those subtle constructs around that from the perspective of modeling.
Play video starting at :11:39 and follow transcript11:39
Good luck

In this example, we're going to be thinking about something called Norman's Action Cycle. Now, Donald Norman is a pioneering individual in the HCI field. Somebody who has been heavily involved in the research around the psychology of design and the understanding of design concepts in relation to interaction and interaction with systems. Donald Norman really set the stage for a lot of the work that we do nowadays. He certainly a hero of mine and individual with a great deal of knowledge. Somebody who really enabled us to think about systems in a different way. Think about designing and developing systems for purpose where we were able to actually substantiate a larger body of knowledge and drawing lots of evidence from different fields to help us to better understand the user and the user's interactions and the nature of those interactions.
Play video starting at ::54 and follow transcript0:54
What it means to interact with system and how we can better think about this and conceptualized this as a set of processes. That's essentially what modeling is. We're thinking about it as processes, some things that we want to achieve based on some given criteria. Now, let's have a look at Norman's Action Cycle. Now, this is a very simple model, so we can think about two existing entities within our state. We can think about the user. That could be you or it could be some other individual who's using your system.

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Of course, when we think about interaction design, we also think about systems. We have user and a system and there's a set of interactions there, isn't it? We can think of this as a directed interaction, or we could think of it as two-way interaction. We can think of this in a very simple way as input and output. If I type some keys on a keyboard, the system and I'll put some data to me. I can see that there's been some keys typed on the screen. Something's changed, some states changed in my system, and so I'm able to recognize that an action is taken place.
Play video starting at :1:53 and follow transcript1:53
Now, this is one way to think about this, but actually Norman's Action Cycle takes a slightly different view of this. In this particular type of interaction cycle, we think about things as being cyclic. We think of a set of interactions as following a set of patterns. Now, the input output really is not cyclic, it is just one input, one output. It's not really a good way to think about interaction design because there's very few cases where we have one input on a system and that's it. We then walk away from that system. We very rarely achieve our goals that way.
Play video starting at :2:23 and follow transcript2:23
Let's think about how we could model this in another way. Norman came up with this concept around two things. The user to the system and the system to the user in the same way. But actually thinking about these as implicit steps. When the user is doing something with the system, there is a process of execution. They are executing some steps or stages. They are imparting some given transaction with the system.
Play video starting at :2:50 and follow transcript2:50
It might be a click or it might be pointing at something, might be changing something on the screen. It can be typing, swiping, all kinds of manner of interactions can be modeled in this way. But there's also another process [inaudible] that goes after you've done something. You don't just type some things in and then get up and walk off. You need a screen, you need other devices to see what's going on, to understand that something's actually changed or something's happened. That's what this model is all about. It's about thinking about that process of evaluation.
Play video starting at :3:18 and follow transcript3:18
We evaluate the state of the system based on our execution. We're essentially thinking about, well, we do something and then we wait and we see what happens. Now, in an ideal world, in a perfect scenario, that process of execution will feed back with evaluation, and then we'll move on to the next state of execution and evaluation. If we have to do this more than once, there's a problem. If I have to type a key and then evaluate and realize I've typed the wrong key. I have to then perform another execution. I have to press the Backspace or scroll to the key that I've pressed the wrong and press 'Delete', or select it with my mouse and delete it that way.
Play video starting at :3:55 and follow transcript3:55
Then I have to go through that clause again. I have to execute the same instruction and evaluate that until I get it right. You can see how systems design could impact these practices and how it might be problematic if you're getting lots of executions for the same goal, for the same sets of processes. The more that we have to do as a user to correct the system, the more problematic our scenarios will be. This is a nice way to model interaction design, and it's a very simple model. But we'll look at some examples of why it's useful and how it helps us to think about systems in more concrete terms in the next few examples.

So this idea behind gulfs is a really interesting one. And we think of kind of gulfs as being a gap, a gap between two things that are what we are thinking about in terms of our own perception and what the system is allowing us to do. So when we talk about this gulf of execution and gulf of evaluation, these are two things that are really, really important in this idea of modeling. So when we think about gulf of execution, there's a gap between what we want to do and what we can do. So we're thinking about the system in relation to the actions that correspond to intentions and intentions in turn, how they correspond to actions. So when I do something, I expect that that behavior is going to create some change of state in the system. So we can think about enabling the person to be able to do a given task based on some input.

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So my intention as a user is to achieve some things. And then I'll make some choices. I'll have some interaction possibilities, I'll interact with the system, and that will then facilitate something else happening. So there's usually a series of steps and stages involved in this. And this is what we mean by the gulf. So if you imagine we have some physical system, so this could be any manifestation of a type of system. And in fact, physical systems can be a whole host of things.

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So we're not going to describe what these are. We'll just use a nice generalization because there's a number of different cases we could use here. And then we imagine that there's a gap between that and something else. So what's the other thing that there's a gap between? We've got our physical systems. We know that we want to do a thing, and we've got this process by which we achieve that. So if we think about this as a process of execution, we want to execute, or execution, in terms of thinking about this as a gulf, then what are we going for?

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What are we aiming for here at the other side? What's the gap between these two things, essentially, is what I'm asking. And in this case, it's our goal. So we're thinking about achieving the thing that we want to achieve. I want to achieve something. I have this gulf of execution here. It's this gap.

Play video starting at :2:15 and follow transcript2:15

I like to think of it as a kind of water-based gulf. Imagine that this is two banks meeting together, and they're relatively close, but there's a gap, and you have to swim between them. So it's a nice way to think about it. There's a gap between the physical system, a manifestation of the human being, the activities, the interaction possibilities, and the goals in which we want to achieve something. We want to get some task done. Now, this is essentially what we mean by the gulf of execution. Now there's a second gulf in this model.

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There's a second gulf that we're interested in focusing on. And this is the gulf of evaluation. Now, it might seem strange to you that I've presented this second, because evaluation should happen first, you might think. And in fact, you'd be incorrect in thinking about evaluation happening first. So let's keep our physical system, and let's keep our goals. This is a good way to think about things. And let's imagine that there's still a gulf here.

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So we're going back the other way. If you remember the model that we had before, we're going backwards now. So we've crossed the gulf once, and we're crossing back across now. We're

trying to get back to the other side. So we've executed some actions. Now, what's happening from here is the data that we're thinking about is moving back in the other direction. So in terms of the gulf of evaluation, what kind of things might we consider here?
Play video starting at :3:31 and follow transcript3:31
What kind of things are interesting to us when we evaluate? How do we evaluate feedback from a system? Well, we're looking at things, aren't we? We're observing things. We're taking in as much data as we possibly can to make an educated decision about what we're going to do next. So we're thinking about, well, what's happened here? What's the state that the system's in?
Play video starting at :3:49 and follow transcript3:49
Is it in a viable state for me to proceed? Do I have to make changes? Do I have to make adaptations? And generally speaking, a good system, a proper system design that's valid, that's reliable, that's consistent, should be giving you reasonable feedback in a meaningful way, in a timely manner, in a way in which you can actually interpret that feedback and say, okay, I understand that. We've all seen examples of horrible error messages where you get error 5620B. That's not particularly meaningful, is it? And this is where we have problems with systems.
Play video starting at :4:19 and follow transcript4:19
This is where we're suddenly in this state where we're trying to figure out what's going on, and we have to perform more and more actions to get to that point of gulf of evaluation where, again, we're still not sure what's going on with the system, so we're performing more actions, and you get lost in the system through this set of processes. So what we really want to see is the system working in the way it's intended. But even if it doesn't work in the way it's intended, we can still get some useful feedback from the system. So when you're designing, think about this. Think about there is a gap here. There is a big gap, and it might be a chasm depending on how well you've designed your system or how difficult the task is. It's very important to think about these two gulfs, the gulf of evaluation and the gulf of execution, going backwards and forwards through these different states in order to substantiate some larger goal, to achieve something with our system.
Play video starting at :5:7 and follow transcript5:07
This is essentially what we're thinking about with this way of modeling.

So let's carry on thinking about this in context. Now let's contextualized this gulf of execution and gulf of evaluation. And we're going to use two examples for this. We're going to use two gain based examples, hopefully games that we're all familiar with but maybe if you're not, you want to go and have a look at the rules. So let's use the first example. Let's imagine I have a chess piece here and it could be any piece, it doesn't really matter peace. I just draw generic piece there because it could be a queen and king of porn for each state and chest.
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We think about there's a gulf of evaluation and execution things that we can do. We've got a limited number of moves from each piece with maybe the exception of the queen who can kind of move anywhere. We've got a number of steps and stages and at each stage we have to make a decision, we have to think about what we're going to do, don't we? It's no good to just sit there and hope that we can win a game of chess. We have to think very critically sometimes about the next move that we're going to make. So this is a good example of one of these cases.

This is a good example of where that thought process is mapping to two of these things in one way in which it's really complex and challenging and another way in which is really simple.
Play video starting at :1:15 and follow transcript1:15
So let's think about the action of moving the chess piece. We move the chest piece from one place on the board to another. So we go from there and we move to another place could be anywhere on the board. Doesn't really matter where we go, what we do, but that process is informed by a decision. So the actual execution of that task is not particularly difficult is it, to pick up a piece and to move it across the chessboard. It's not a difficult thing to do as a set of motor skills, it's not particularly challenging. But cognitively we have to assess the state of the board, we have to look at where all the pieces are.
Play video starting at :1:48 and follow transcript1:48
We have to look at our opponent's piece. We have to recognize where we are in the game, what sort of actions we might want to take whether or not we want to take a risk or whether we're going to play safe. There's all these considerations that we have to take into account when we're playing a game of chess. So this is an example whether the physical movement pattern is relatively simple. But thinking about that process of evaluation and understanding whether or not the manifestation of the movies a correct one. Am I going to lose my queen, is this going to cause me to lose the game? That's where the investment is in the user's part.
Play video starting at :2:20 and follow transcript2:20
That's where they're thinking about this process and thinking it's very important that as a designer you're conceptualizing and focusing on these elements. So you're not focusing on the manifestation of moving a piece across the board that's a simple thing to do. Everybody can achieve that. We're thinking about the relationship between that evaluation cause so if I stop and I pause and I think about a move to make that doesn't mean that I'm struggling to physically make that move. It means that I'm evaluating the situation. I'm trying to understand what are the subtle connotations of the move, what consequences are going to come from that? So I'm not confused by the system in this state.
Play video starting at :2:57 and follow transcript2:57
I just have a certain understanding about the elements in front of me. So let's look at another example. So let's imagine a kind of 2D Platformer. So you're playing a 2D platformer and you have a little stick character and he's running through a field and he has to jump over a thing at the right time. Otherwise the bad guy gets him. Let's imagine there's a bad guy up here, this is my bad guy and in this case there is a set of actions that are really important. So I have to do the right thing at the right time.
Play video starting at :3:25 and follow transcript3:25
I have to make sure that my character scrolls along and I run along and I jump at the right time just as the bad guys is away from me so I can jump over him will jump under him or about him on the head and jump over. I have to time that process right. So it's very important that the execution of actions in this state are thought about very carefully. Otherwise my character will end up falling off the world or or lose a life or some XP or some health or something. So this guy is a bad guy right here and I've got to be really careful. I've got to time this motion very carefully. Now I know what I'm going to do in my mind.
Play video starting at :4: and follow transcript4:00

Don't I know I'm jumping over the bad guy. If I've played this game before, I've got a general sense of I probably don't want to get hit by the bad guy. Maybe if it's the first time I'm playing, I might not realize that but if he's killed me 100 times already, I probably want to make sure I can jump over his head and move on to the next level. So again, this is thinking about two sets of process is the process of evaluation, which is very important in our game of chess. We have to make sure that we're doing things right and a process of execution, which is very important in R2D platformer where I have to make sure I avoid the bad guy. So in this case we're thinking about these things as being more or less important depending on the context. So it's very important that you design these systems to think about these implicit things within our system and make sure that your mapping things in the right way.
Play video starting at :4:46 and follow transcript4:46
And make sure that you are considering the right cases for context of use and that you understand that there are subtle connotations here that not every system is the same. And thinking about these two things in terms of the gulf in terms of the gap between the user's expectations and the reality is really important

Let's move on now and let's think about some other models that might exist in the world around us. We've talked about designing and modeling systems in particular ways and thinking about user interactions in some implicit and explicit ways. Now, let's think about something called the mental model. Now, we've got our user here. This is our user. This could be me. We could go back to our previous example of me being very cold and putting my jacket on.
Play video starting at ::32 and follow transcript0:32
Now, I'm still cold actually, so I'm going to put the heating on. I want to raise the temperature. I want to increase the temperature of the room so that I don't feel cold anymore. I want it to be a nice temperature. My mental model of this is that I need to be warmer. I might look at a thermostat and I might go up to the thermostat and I might say, with this thermostat, it's set at 12 degrees, let's imagine. Maybe 12 degrees is there, possibly not, I don't know.
Play video starting at ::58 and follow transcript0:58
It could be anywhere on the thermostat really. How do you design a thermostat? There's a good question. Now, I know that I want the temperature to be 20 degrees because I feel warm at 20 degrees. Let's assume I want to get to here. Now, the reasonable assumption will be that I turn the knob to 20, and then I go and sit down, and I'd wait for it to heat up. In fact, lots of people don't do this.
Play video starting at :1:20 and follow transcript1:20
It's an interesting convention in that people will often turn the thermostat more further round in order to substantiate an increase in the heat efficiency of the room, so it's heat up the room quicker effectively. Now, we all know from a physics perspective, this doesn't make much sense. But actually the user's mental model in this case probably does make sense to them. They're thinking about, well, I want to be warmer, 30 degrees is hotter than 20 degrees. If I turn it to 30, I'll probably get warmer faster. Now, it doesn't make sense when you understand how heat and things like that work. But from the perspective of the user, they've got an understanding at a very base level, and then mental model is telling them this is the right way to interact with the system.
Play video starting at :2:5 and follow transcript2:05

As another example, as a perhaps less silly example, if I was to have some text on the page and I was to write that text here. Let's just write some nonsense. It doesn't have to be real text. That text was underlined, so you don't know what that text says because that's not English or any language that I'm aware of. What do you think that does? What you think that the intention of that is? You're probably looking at that without any language or construct without any instructions, and you'd probably say that's a hyperlink, wouldn't you?
Play video starting at :2:36 and follow transcript2:36
You'd say, "I think I can probably kick on that. Even though I don't speak this weird language that you've just defined, I still understand it. I've still got some convention around." Well actually, that's probably a link, that's probably a hyperlink. It's underlined, its written in blue, and so I know that links are always underlined and always written in blue. Now, that's not to say that all things that exist on the web are links because of these two properties. But my mental model is telling me that because I've seen this before and I understand how this works, then that's a reasonable assumption for me to make, and I'm quite happy to make that assumption, and I'm quite happy to click on that thing.
Play video starting at :3:12 and follow transcript3:12
If it goes wrong, then it goes wrong, but my expectation is that. It might be around another type of perceptions. I might be looking in the top right corner for these things that I see all the time on my system. I might be looking up in the top right corner and I might not see anywhere. That's perhaps because they're over in the left side, because the minimize, the maximize, and the close button could be anywhere on the screen. But my mental model is telling me that I know where they exist. I've got an expectation based on my own experiences that in fact those things should be there.
Play video starting at :3:43 and follow transcript3:43
I'm looking up there and I'm not seeing them because they're not there. Again, you might see logos in the top left corner of the page. You might have some logo that says, "This is our business." Sometimes you see that, sometimes you don't. You might expect to see tabs along the top, and they could be along the side, so there may be tabs along the top or side. Maybe we're going down this right side, that would be quite unusual, would it? I don't think I've seen many sites that use that convention.
Play video starting at :4:8 and follow transcript4:08
But again, we're now changing the way that we think about things, and we're actually imposing on the user our own model for design. We're saying, "Well, I understand that you have a mental model and I understand that all the other systems out there are design this way, but I'm going to change my design." Now, this is a very difficult thing to do and you can become unstuck very quickly with this if you start to try to impose things on users and say, "Well, actually, my design model is better than your mental model." You have to find ways to give users clues if you're going to do this and you have to have a very good case for doing this. Be careful, and make sure that you understand people's mental models. This can happen through things like usability studies and heuristic evaluations, but it's important not to break convention unless you have a very good reason to do so.

Let's finish up by looking at some extensions of this model and try to understand how this model might fit a set of processes or stages. Let's start with a view of the world now. We all know what

the world is hopefully. I'll draw as a cloud and I know, generally speaking, cloud-based architectures of Internet-based technologies. But today we're going to say that the world lives in Cloud. Because this is fuzzy, it's unknown. The world is full of all of interesting things that are happening all the time.
Play video starting at ::37 and follow transcript0:37
We don't necessarily know what's happening at any given stage. All this complexity is coming from the world and we don't know what's going to happen next, as we've seen recently. Now, there are a number of steps and stages in this that we can think about in relation to our goal. Imagine our goal is here and we're trying to get up to the top goal. We're trying to achieve something. We want to tick a box and say yes, that's my goal, that's the thing I'd like to achieve. Now through this seven stage process that I'm going to talk about, one of them is ideas and concepts that again is a genius idea I think can really helps us to conceptualize systems, is thinking about there are actually seven steps that we could be out here.
Play video starting at :1:12 and follow transcript1:12
We've looked before at evaluation and execution. Essentially there we only really have two states, don't we? I suppose you could think of it as four states because there's those intermediary states between. But ultimately, this is a slightly more robust model and it's thinking about breaking that down into an even more granular level sets of interactions and possibilities. We'll start out with one, and we'll start with the first thing that we do. What's the first thing we do when we get data input from the world. What do we do?
Play video starting at :1:40 and follow transcript1:40
What things do we do when we see something, when we hear something, when our senses are triggered. What's the first thing that we do? Well, we have some degree of perceptions. We perceive what's going on around us. We take all that information in. We use it to make decisions, don't we? If I see something's happening, I react to that situation based on my own sensory input data.
Play video starting at :2: and follow transcript2:00
My perception of the world is really important and it's the first thing I do in any sets of interaction possibilities. Before I act, I have to actually acknowledge what's going on and make sense of in my own head. First of all, I perceive what's going on and just absorb it in a very blend way. What's the next step then? The next step is to think about what I'm going to do based off of this information. I'm thinking about step 2 as being something really important too. I'm thinking about this as my own interpretation of what's going on.
Play video starting at :2:29 and follow transcript2:29
We're all individuals, we all have individual differences, and this is important, this is a really important distinction to make. Not everybody reacts the same way to things. We could all have the same amount of input data. We could all be in the same room. Some of us might feel hot, some of us might feel cold. Our own perception of what's going on around is going to change based on that sensory data alone and that's not changing. It's just our interpretation of things is changing.
Play video starting at :2:53 and follow transcript2:53
We're changing the way that we interpret things based on our perception, based on our understanding of this sensory data. Then finally, we need to move on step 3 before we achieve our goal We've got one thing that we need to do before we get to our goal. What's that thing?

Well, it's probably fairly obvious by now because we've already talked about this quite a bit. That it's the evaluation of our interpretations. We can't get to our goal until we've actually evaluated our interpretations and say, well, what does this really mean? Should I react to this?
Play video starting at :3:21 and follow transcript3:21
If I'm cold, maybe I should put a jacket on. Maybe that's a good idea. We're evaluating our interpretations, were making decisions. We're thinking about, well, what's the next step? What's the next stage to change this? How do I know that I'm making the right choice? Well, I work through a process of evaluation data, and then I get to my goal hopefully.
Play video starting at :3:40 and follow transcript3:40
I move on from my goal and I think, well now I'm moving through this barrier, I'm moving back along this trajectory, and I'm moving back towards reacting to the world. I'm thinking about, the world is giving me some information. I've processed that information, I've absorbed that information and I'm reacting to that information. I've made a decision, I've made a choice at this point. Step number 4. This is a really important step, is our intention to act. Now I've made a decision.
Play video starting at :4:9 and follow transcript4:09
I need that jacket, I'm freezing, I'm cold. It's horrible in here. I know everybody else is feeling warm. The people won't let me put the heating on so I'm going to have to put a jacket on and I need an extra there. I've got an intention to act. I'm going to go and get my jacket. To do that, to warm myself up to work through that process, I have to impart some interesting things.
Play video starting at :4:31 and follow transcript4:31
Don't I have to do some stuff? I have to think about, well, what's the next step? What's the next stage? Number 5, what do I do at stage 5? Well, actually at stage 5, I perform a sequence of actions. That sequence of actions is conceptual first of all, isn't it? I know that I need to get up.
Play video starting at :4:47 and follow transcript4:47
I know that I need to go into the other room to get my jacket. I need to put my jacket on. I need to zip my jacket up. There's a number of steps and stages that will allow me to actually achieve that thing I want. I'm not instantly going to get warm as soon as I take the first step, as soon as I get up out of my chair. There's a sequence of actions involved at step number 5, which is very important. Then finally, we have to do one more thing and you probably know what this is.
Play video starting at :5:10 and follow transcript5:10
Given that evaluation is a last step before we got to our goal, this time around, we are actually looking at execution. We're thinking about step 6 as being our execution of our actions. Of course, that feeds back into the real world. We take that cycle and we go back into the real world. We start at one and we say, okay, I've got some data, I've got something to work with. I've got some perception of what's going on around me. We then think about interpreting that perception.
Play video starting at :5:38 and follow transcript5:38
I know how to react based on this. I know that this data is valid. I know that I am actually cold. I've spent some time to think about this. I'm freezing. I'm really [inaudible] I'm shaking. It's very bad.
Play video starting at :5:50 and follow transcript5:50
I know that I'm going to have to perform some action. Then I think about, well, what are my options? What my options here? Well, maybe I'll put the heating on mechanism choice, maybe

I'll put a jacket on. That seems like a reasonable choice. Then I've got an intention to act over [inaudible] I'm back over here, over on this other side now. I'm back on step 4, my intention to act I'm thinking about, well, what am I going to do now?
Play video starting at :6:11 and follow transcript6:11
How I'm going to act? I've got that evaluative data that I've been able to use to understand and appreciate that data that I'm getting from the world. Now I've got my intention to act. I've got my sequence of actions that I need to achieve the thing that I want to achieve and then I executed. I go, okay, step 6, we're finishing up. We're moving back into the world. Now, we're going to go back through that process again.
Play video starting at :6:33 and follow transcript6:33
I've put my jacket on, I've sat back down. I'll give it ten minutes. I'll perceive the data back to step 1. I'm perceiving again because I've had my jacket on for ten minutes. Now I should start warming up. I'm interpreting that perception. I'm saying, well, I'm not getting any warmer.
Play video starting at :6:47 and follow transcript6:47
Evaluation of interpretation comes at step three. Actually, I'm probably going to have to put the heating on here. I'm at that point now I think, yeah, if I'm going to achieve the thing I wanted to achieve and stop being cold, I have to put the heating on as well as have my jacket on and maybe also present at the jump room. My intention to act. I turn up the thermostat in this process, and I go back into the world again. Hopefully, I'm nice and warm and cozy. This is a seven stage Action Cycle.
Play video starting at :7:12 and follow transcript7:12
Hopefully this gives you a sense of why all this stuff's useful. How we can think about people's perceptions, how we can model people's perceptions, and why it's useful in thinking about interaction design possibilities with systems. How people evaluate things and how they execute their actions. It's actually more difficult than it sounds. Having a look at this model, we can actually see that complexity in play.

So let's keep going with the modeling process and let's carry on talking about ways in which we can model systems. Now I'm going to talk to you about a really, really important model in HCI, a model that is kind of the foundations for a lot of the work that we do. And it's really important that you understand that this model is one that's been established for a number of years and has really underpinned the work that's allowed us to achieve some really interesting predictive modeling in HCI. So a good number of years ago now, actually back in 1954, we all were introduced to the idea of something called Fitts' model. And it's actually Fitts' law. So this is a law, which means that it holds true in all cases. And it's a predictive model that's designed around thinking about the way in which we model human movement in relation to objects on the screen.
Play video starting at :1:4 and follow transcript1:04
So things like human-computer interaction, ergonomics, how we design devices, how we think about designing user interfaces, is all motivated by this idea of, well, actually there's a series of interactions that we have to do to allow us to achieve some things. So we're thinking about what do we need to do? We want to model the actions and activities that allow us to undertake a set of implicit processes to achieve some goal, to achieve something. The idea is that this could be very simple. Maybe we have a simple example of a button, and we want to model that button in

terms of a set of interactions. So let's have a look at that. Let's have a look at how that might work.
Play video starting at :1:39 and follow transcript1:39
So imagine I've got a button like this, it's really small, middle of the screen. That's going to be hard to get to, and it's going to be hard to click on. I may have a large button like this that's thin but tall, or I may have a wide button that looks something like that, that again is very wide but also not very tall. And we think about these things in terms of things like proximity, how far away my target is from the things, or how far away do I have to go to click on this. And we're also thinking about what trajectory we're taking. So are we going left to right, top to bottom? There's a lot of things to consider here, and that's not even considering the fact that we might be using different modalities of interaction.
Play video starting at :2:21 and follow transcript2:21
So this is not to say that I'm using a mouse or a keyboard or any other device. In fact, I could be using lots of different things, and what's so remarkable about this model that we're going to talk about is the fact that it works in almost all situations. So we can look at this in terms of different types of interaction. If you're on the bus, this should still work. If you're getting jostled about, if you're holding onto your phone with one hand, if you're on a laptop, a desktop, on an iPad, all these different devices, you should still get the same sort of interaction. And this even applies across different demographics of users. So this has been done in lots of cases, looking at using people's feet to interact with things, looking at children, looking at older generations of people interacting with systems.
Play video starting at :3:1 and follow transcript3:01
And this rule holds true in most cases. So we've got this really interesting situation where we've got this really robust model that's generated around lots of data and lots of understanding about how people interact with things. And so in terms of modeling, this is perhaps the gold standard. We're thinking about something that's really, really important and valuable and useful that allows us to achieve some really, really interesting analysis and predictions. We essentially want to be able to take this and to predict some interesting things based on some parameters. So what are our parameters? How do we define these things?
Play video starting at :3:33 and follow transcript3:33
How do we make these things work in terms of understanding this as a model? Well, in fact, it's really simple. Actually, the initial model that was proposed was a really, really simple model. And it looks something like this. So I'll introduce it to you. So let's say ID. Now, again, we're using acronyms.
Play video starting at :3:48 and follow transcript3:48
It's just something about computer science that's so strange. But this ID metric is essentially what we're measuring. So we've got a name for something. We've called it ID, and ID just stands for Index of Difficulty. So it's how hard it is to do something, essentially. Now we've got a standard value of log2. This doesn't change.
Play video starting at :4:6 and follow transcript4:06
And then we've got two parameters that we feed into this function. So we've got two things that we want to work with. Now, in terms of the parameters we've got, you can probably guess, based on the buttons example, what I'm going to put here. But you might not guess the metrics imposed on them. So we've got a factor on the top and a factor on the bottom. Now, the factor

on the top is essentially looking at something that we call 2D. Now, this isn't a 2D object or 2D as we've seen it before.
Play video starting at :4:33 and follow transcript4:33
2D is actually a measure of something. It's a measure of something fairly specific. So what are we thinking about? What is this 2D metric for? Two times something. So we're doing two times the distance to the center of a target. Now, what's important to recognize is that if we have a button like this on the page, we are navigating to the center of the target, not the edges.
Play video starting at :4:55 and follow transcript4:55
So this is a common misconception about Fitts's Law. And actually, you want to go here. You want to go in this middle bit. You want to go from wherever you are to that middle bit. So what we're calculating here is a standard metric. It's important to remember this. We are working to the center of the target.
Play video starting at :5:11 and follow transcript5:11
And the distance from the center of the target that we originate at. So if our origin is in the corner of the screen in the top right corner, we might be far away from that. And in fact, my button is almost central. So in fact, any of the corners would be about equidistant away. It's probably more positioned to the left than the right. But generally speaking, this is going to be quite difficult to click on from any of those extreme trajectories. Of course, if I'm here, where I've marked my line from, so if I was starting from here, that's not a particularly long distance, is it, to get into the middle.
Play video starting at :5:40 and follow transcript5:40
We don't have to go very far. And so what we do is we do two times this metric to determine how long it takes to get to the center of the object. Doesn't have to be a button. Could be anything. And then we divide it by something. So we divide it by this value under here. So if you imagine this is the width of the target from here to here.
Play video starting at :5:59 and follow transcript5:59
So the width is essentially this distance here going across the button from left to right. We take this width target and we put it into this function. And what this essentially gives us is the opportunity to then model this type of interaction. Wherever we start from, we can plug this into our model and it will give us a nice metric that we can use. We could be up here. We could be over there. It doesn't really matter.
Play video starting at :6:21 and follow transcript6:21
We can go from any of those targets to the center of a button. So this model is quite robust and it allows us to think about the types of interactions that people might have with systems and it allows us to model them. Now there is also an extension class of this, which I'm going to talk about briefly. I'm not going to get into the more complex models in this lecture, because I think you can read about these things and learn about them. And you can also read about some of the research criteria on extensions to this, looking at this in different types of environments, because it's quite interesting to apply this to different contexts. When we think about laws, they generally apply across a wide variety of activities. So it's nice to look at this in relation to other things and see, does it work or doesn't it work?
Play video starting at :6:59 and follow transcript6:59
And a lot of HCI research and literature focuses on this as a premise. So we're going to take our old value that we worked out and we're going to generate a new metric here. So we're going to

have our metric of IP. Now this is an internet protocol, it's not a networking course, we're actually measuring something different. We're measuring index of performance. So this is going to take our ID value. So our ID again is log2, and it's log2 of a function 2D, distance to the target over the width.
Play video starting at :7:32 and follow transcript7:32
So we're going to plug this into our new function that we've got, and we've got our new function which is index of performance in bits per second, of course, because we're computer scientists. How else would we work things out? And we're looking at this measure of performance as two metrics, our ID metric, which we've got up here, which is taking this, and we're looking at over something. And you can probably guess what this metric we're looking at over is. And it's MT. Can you guess from those letters what it might be? So we're actually looking at something called movement time, and this is actually a seconds value.
Play video starting at :8:5 and follow transcript8:05
So it's per second. So we're thinking about looking at movement time in selecting the target. So this is essentially, and Fitz talks about this quite a bit, he mentions it as the average rate of information generated by a series of movements is the average information per movement divided by the time per movement. So it's quite a more complex model, and you can really think of this as something very simple. It's throughput. When we think about throughput through a network, it makes sense to think about it in this way. And again, for this type of analogy, it's quite a nice way to think about systems.
Play video starting at :8:38 and follow transcript8:38
So it gives us that opportunity to think about things, but we can also model things in there around characteristics, so whether the model is accurate enough and if the calculation is quite right. And we can do more advanced things if you want to. We can create correlations and regression analysis and try to express these things under different parameters and different fields. And if you look at the extended version of this, you'll see that there are actually extensions to this that provide even more descriptive and even more parameterized characteristics that we can feed into this model and give us a really good description of how people work. But hopefully this was a good enough introduction to Fitz's law, and you've seen a couple of models, a couple of basic models around this. Now I'm going to ask you to look at a few more examples of things to try to formalize your understanding and to try to apply this to different contexts, which will be really interesting to see what the outcomes of these exercises are. So good luck, and I look forward to hearing about what you're doing in the next set of exercises.

Okay, so we've looked at some modeling techniques and we've looked at some examples of how we might model a system if we were developing our own model. Let's have a look at some pre-established models and try to understand how they work and why they're useful for us. So let's have a look at one of the oldest and most important models in HCI. Let's have a look at GOMS. Now GOMS are all about four key constructs in our system and we're thinking about modeling a system as a set of interactions. We're thinking about those interactive processes and building those up over time. So we start out formally with a goal. We have something that we want to achieve. Now this is a fairly simple example of something that we could model. We could model a goal and we have a checkbox there that says yes or no and we tick the box

based on whether or not we can achieve a thing. And this is a very simple way to model systems and lots of systems are modeled in this way. So we could say my goal is to ensure that the user is able to achieve something. So what we're really focused on with the goal is what the user intends to accomplish. There are a set of ideas in the user's mind about what they want to do and we're basically evaluating whether or not the person can achieve those things through this system. Now it gets a bit more complicated as we go through. So our G is a simple one. Our O is slightly more complicated than G and these are building on one another. So as we extend the G to the O we can see that there's some movement, there's some relationship between the two things.

Play video starting at :1:31 and follow transcript1:31

So the O stands for operators and the operators are actions that are performed to reach a goal. So when we think about the example of this idea of a checkbox and we tick it, in fact there is an action there isn't there. So most of us have used web browsers, who've done things with checkboxes before, we understand that if you tick the checkbox that changes its state. It changes the behaviour of the checkbox, the checkbox is now ticked based on a click. So what we're really thinking about with operators is actions that are performed to allow us to substantiate a goal. So we have a thing that we want to achieve and we have a step in that process that allows us to achieve that thing. Now this is fine for modelling simple activities but how many activities can you think of where there is just one thing that we want to do, where there's just one thing that we want to achieve and we can do that through one step? There probably aren't that many are there? So in fact we actually need to build up these operators, we need to build up these functions if you will, so we can think about it as a function, a simple function, and we need to build these up and think about how we can build up a set of functions to allow us to achieve a thing. So if I wanted to stand up that's a simple thing but if I wanted to walk that's a bit more complicated so standing up is one action, walking I have to put my left foot forward then my right foot forward then my left foot again, if I tried to go with my left foot twice that wouldn't work I'd fall over, a bit of a silly example but it's really thinking about building up knowledge over time. So when we look at our methods, when we look at our M in this we're thinking about the sequence of operators that accomplish a goal. So this might be lots of functions, it could be a function on X and that could be an input to a function on Y for example and then we could keep going this depending on how many steps and stages we need. So we could have lots of different things that we do, we could have lots of operators involved in our method, we could think of the method as being the class and operators being elements that belong to that class if that's a nice way of thinking for you, it's essentially thinking about there are steps and stages to allow me to achieve a thing.

Play video starting at :3:33 and follow transcript3:33

So what's this S for then? We've thought about goals, things that we want to achieve, tick boxes, we've thought about operators as actions that are performed to reach a goal, we've thought about methods as a sequence of operators to achieve a goal, so building up this idea that we're still trying to achieve the goal but now we're building on lots of operators in our methods. Well selections are just another extension of this so we're thinking about well actually there are lots of ways that we can achieve things and in fact when we achieve things we do that in different ways generally speaking so it's unlikely that the way that you do things is going to be the way that I do things. When we think about the example of walking, your gait, your stance,

your stride, all of those things are different, there's individual differences that we have to think about. So when we're thinking about this idea of selections what we're really thinking about is a set of processes. We want to think about the fact that there can be more than one method available to accomplish a goal. Now this is generally speaking used to think about the sequence of actions wherein a user will have a choice, so the user will have a choice of more than one way to achieve a goal and they might follow a particular path to allow them to do that. So if we start with our situation we have our selection so we'll put our selection in the middle here just because it's easier to see and through this selection we might have some choices.
Play video starting at :4:51 and follow transcript4:51
So I might have method one and we might have method two. So let's just say there are two ways of doing this thing, that there could be an infinite number, there could be hundreds, there could be ten, it doesn't really matter how many methods we have but there are choices to be made here, that's the important distinction, that's the important thing to recognise. And in method one and method two we have variations on how people do things and that's fine too, we already understand that there are a sequence of operators involved in these, so we could have operator one, two and three, so we have one, two and three. And we could have the same in our method A2 or our method one, we could have a sequence of operators in both. Now in both cases we might have a situation where they're able to achieve their goal, but there is a selection in place to decide which path they take to achieve that particular goal. So in either situation it comes out that they can do a thing, that's fine, we can tick it and say yes the goal's been achieved, but there is some individual differences about the sequence or the selection of actions in relation to choosing method one or choosing method two. Now what's really important to recognise here is that modelling systems is something that can be clear, but it can also be quite vague in terms of how we think about the ways in which somebody can achieve an activity. So when we think about a path through the system it might be path one, where we've got method two here to achieve the goal, or it might be the alternate path, and either are valid. But what we're really doing here is building up a knowledge repository aren't we? We are thinking about achieving this goal at the end, which is really important to us, we want to do something, we're thinking about a selection, a set of choices that we're going to make to allow us to achieve that goal, and then we've got different approaches or different methods which are essentially a larger version of our operators. So these are our operators here, operator one, these are our methods, which are essentially just a parent class of our operators, so we can think of them as containing lots of operators, you might think of it as a set maybe, might be a nice way to think about it, operator one, operator two, operator three, and so on and so forth. And then we can even think about our selection as being a set of options for our methodologies. So we could say our methods are, essentially we only have two method choices here, so we've got method one and method two, we might have more, but each one of these things builds up on the other, and eventually we can hopefully achieve our goal, which is really important. So hopefully that's a reasonable introduction to GOMS and you understand what we're talking about when we talk about goals, operators, methods, and selections. In the next activity we'll look at a more substantial example where this has been extended to a particular type of activity.

So with these activities we've kind of drove down to that really sort of granular level now and we're going deeper and deeper in terms of these examples. So in the last example we looked at

gums and we looked at a way to model systems based on four key properties. So we're looking for goals, things that people want to achieve. We're looking at operators, actions that are involved in achieving that thing. Then we're moving on to more specific examples that we're thinking about methods which are essentially a set of actions that we want to use to allow us to achieve our thing. And then finally, selections when we think about the different methods and choices. And then we choose a path through our system based on our own preference, our own understanding, our own way of working with the system.

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And so we build up these examples over time. We start with a goal which is something really specific. We think about those small sets of actions that allow us to achieve and think. And as we build those up over time we can start to understand the manifestation of the users activities and how that might affect the way that they use our system. Now in the gums example, we had GOM and S and we're going to look at a paper which covers another example or an extension to this, which looks at gums in relation to K L and M. Now K L and M stand for different things as well. So at this point in time we've got a quite substantial acronym to deal with.

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And this is the nature of computer science. We like acronyms. For some reason, we also like analogies, which again, a lot of these things are so you might find that there are acronyms and analogies in a lot of these courses, but it's just a good way to describe things. It's a nice easy way to remember things. So when we think about this, we think about what's called the keystroke level model. And this is about thinking about things at a very, very granular level. And it's something that we should already be quite familiar with as a construct.

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It's really about typing. So we're looking at operators. So we're looking at our hope in relation to this type of system. And we're saying, well the operators in this are the key stroke level actions. It's pressing a key or moving a mouse or or maybe it's scrolling through through the middle button on your mouse or the middle scroll wheel through your mouse or right clicking or doing all these other types of activities. We can do this at a very granular level. What this does is it allows us to understand, first of all those subtle interactions, those really small interaction that we do on a regular basis and then extrapolate up from that and think about models that would sustain those activities.

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So thinking about a set of those activities, Thinking about building upwards like we did in the previous example, can we actually understand how people use our system based on some expected level of input? So this is modeled through a system and you'll be reading the paper about this and understanding this in a bit more detail. But we'll introduce the paper now just so that you get a general understanding of what it's all about and why it's useful, why it's relevant. So this paper was really about understanding the keystroke level model and thinking about the standard set of operators for use in our guns model. So we're looking at things from a kind of execution time basis. And we're looking at lots of experimental data. So we went and grabbed lots of data, looking at how long it takes people to do things and then trying to model that in the system.

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So this is what we're doing in modeling all the time. We're looking we're trying to take that information, we're trying to make that information general Izabal, we're trying to take those properties, those characteristics and features of interaction. And we're trying to understand how they work in relation to a set of activities. So with this what we're really interested in is looking at this kind of keystroke level model. We're looking at a very very small set of processes and then building up over time and understanding how they can help us to do things. So we can think of a keystroke as being one of the things that we might model. So this could be very simply pressing a key on a keyboard.

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So you have a keyboard here and each one of these keys will take a certain amount of time to press. There might also be a certain rate of error with specific types of keys. Now we can characterize this according to a user set, we can say, well Germany speaking, a particular type of use will take a particular length of time to click on a button or to press something. Some corresponding key. This key could be anything but essentially we've got a set of interactions here where they're going to click on these buttons or or do something in relation to an action. So this is a key stroke. This is a keystroke we're thinking about this has an estimated time associated to it.

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And you can look in the paper about the specific times for these things. But what we're really interested in is modeling, we press a key. That's a fairly simple interaction to model, isn't it? It's not going to change drastically. I'm not going to press a key from up here and kind of her hands going to come down on the keyboard. My hands are over the keys and I'm pressing. So we've got this set of interactions where the only difference really is going to be how far away the keys from my finger.

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So there might be a slight decrease in how long it takes, a slight increase in how long it takes in relation to how far away the keys from my finger. But this is something that we can model with quite a degree of accuracy. The keys aren't moving about and they're not changing. So it's easy for us to model that we can say, well, all typists have to do the same thing. All typists generally speaking, type the same way, assuming they've had some basic training. So pressing the W key, for example is going to take about the same length of time and we can model this through data. Now we have other types of interactions that we model in this.

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So we have pointing, which again, most of us should be familiar with. Most of us have mice or some other device that we can point with. We point to things that we click on things. So automatically you can see how these things might start to build up a set of interactions. Point and click is a reasonable assumption for a set of interactions. And so we can think about pointing as an activity. And we can actually build these things up.

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So you could say pressing a key would require pointing first. Maybe maybe have to point at something and then type something into a text box so we can add these things up to get a set of interaction straightaway. We could say, well pointing and pressing a key. Those are two separate things you have to point and click on that thing before you can press the key. So we might have another step in there, we might have B which is pressing or releasing the mouse

button and we could add this on, we can add this onto our model. So we could actually this isn't quite right. We don't just press a button.
Play video starting at :6:3 and follow transcript6:03
It's not that simple unfortunately. So we've actually got three steps in this model that we want to achieve. So we have to first of all think about our B and B stands for in this model, press or release the mouse button. We have to think about that in relation to the other activities. So a keystroke and pointing. So we're going to point first aren't we? So B is not the right one.
Play video starting at :6:24 and follow transcript6:24
We actually want to go back and change that. As we model. This will start to understand a bit more about how this works but we want to do R B first which is appointing, we want to do R B. Which is our press already some amount of buttons. So we've pointed and clicked and now we want to type keystrokes and that might be a number of key strokes actually. So we might have, it's very rare that you type just one letter into a field isn't it? So we might have a number of things.
Play video starting at :6:47 and follow transcript6:47
So that might be plus key plus key plus key and so on and so forth. Might be pressing lots of keys. So now we've built up a model of how long we think it takes somebody to do something and that's what this is all about really. This this keystroke level model is thinking about things like keying, pointing, there's tasks around homing as well, which is homing the hands to the keyboard of the mouse. So this is another step, we actually have to step away from the computer to understand this. So we have to think about when we home, we're actually not just pressing a key, are we were moving might be moving from a mouse to a keyboard, we might be moving from a keyboard to a mouse. So again, we have to add this into our model and it's already quite complex enough on the screen, so I'm not going to keep adding these things on, but we get a sense of how things work.
Play video starting at :7:30 and follow transcript7:30
Now, there's another set of constructs that we want to think about in relation to this, isn't there? There's another set of steps and stages that are involved in this. We've not gone quite deep enough to actually model the way that people work, because there's something that we haven't thought about yet. So we have to mentally prepare and this is our M this is our M value, so this is described in the model as routine thinking or perception, so it actually takes us some time to react to something. So if you think about the example of driving, when we drive, we have a set of interactions, we might turn the wheel, we might change gears, we might break, we might stop, we might accelerate, but all of those actions are actually based on what's happening around us, so when we think about the activities that happening around us, we have a reaction time. So this is what we're really thinking about with this end value. We're thinking about mentally preparing to do a set of tasks or an individual task maybe.
Play video starting at :8:18 and follow transcript8:18
But this this mental load, it will depend on the nature of the task and the activity of the task. So things that are quite difficult and quite cognitively challenging. We might have a high value for this. We might take a while to think about something if we're reading something for instance and then copying it and typing it out on our keyboard, that might take us a few moments to actually read and process that information. So again, we have to build this into our model and of course there's responding as well which is an important construct. There's a lot in this model to go

through. There's actually a really, really granular level of construct and I'm not going to go through all of them in this lecture.
Play video starting at :8:52 and follow transcript8:52
It's good if you read the paper and have to think about this and then we'll move on to an activity where we actually try to model some systems. But what this really is essentially about is thinking about building up all these sequences of activities. So in the last lecture, I talked about sequence of functions. And again, that there's different wrappings, but there's different routes through the system too. So we could go through the system in different directions. We could take different paths, we could follow different models of interaction and we could take different methods, we can make different selections. So as we build these things up over our guns model.
Play video starting at :9:22 and follow transcript9:22
And now what we know about our keystroke level model, our KLM, we can think about gums, keystroke level modeling and we can start to think about things as interaction at a very, very granular level. Now this is beneficial in a number of ways. But one of the real benefits of this is thinking about modeling our users in relation to our system without actually having to test with our users. So we can take these models and we can extrapolate certain amounts of data from them. We can think about how long it takes a person to get to something. And then we can think about optimizing those sets of processes without even having our users in front of us. Hopefully you can see the advantage of this model.
Play video starting at :9:58 and follow transcript9:58
And as you work through this paper, you will get a better understanding about how to model things, how these models are useful and how we can use these models to actually model existing systems or potential systems that we want to design.

So in this lecture we're going to introduce the topic of design. Now, we're moving along in this double diamond process. And we're thinking about design as a set of things that we want to achieve. There's some things that we want to do. We need to think about how to build a system. And so the design constructs that are really, really important thing that we need to spend a bit of time one to figure out what the right way of doing things is. So where do we begin?
Play video starting at ::31 and follow transcript0:31
Where do we begin with design? Well, this is a tricky one, because when we think of design, we often think of this kind of open world where we can do lots of things and there are lots of possibilities. There are assuming the infinite number of possibilities. And so we have this real challenge in designing systems that can do all the things that we want them to do. But we have to have some constraints too. Do we need something to work with? And we do have some good basis for working with different types of design constructs in HCI?
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So we're going to talk about what these are, and we're going to talk about why they're useful. So as a general tip, there are lots of pieces of guidance out there. We can use to get us started and to ensure that we're able to design in a rigorous and systematic way and to converge on our design goals, because ultimately that's what we want to do. We want to design a system that's usable, we want to design a system that people are going to enjoy using. Now, we tend to refer to some of these guidelines that we've seen in terms of design as either rules or guidelines.

Now, these are sometimes used interchangeably in the literature, and you might find that they mean specific things to specific sub communities of people. But generally speaking, you'll see rules and guidelines has been the two things that we're going to see quite a lot of.
Play video starting at :1:37 and follow transcript1:37
You'll also see the words standards pop up as well, which is a much more formalized way of working. It's a much more formal set of structures. But these are essentially the three things that are going to be interesting to us through this topic. Now, we can also look back. We can look back at what we've done and we can understand a lot from the things that we've already covered in relation to how to design new types of systems. So we must also reflect on the lessons we've already learned. We've looked a lot of modeling, we've looked a lot encapsulating those rich bits of data in different ways.
Play video starting at :2:7 and follow transcript2:07
So there's lessons that we've already learned a really important to us. We've learned a lot about different types of data capture and how we can use data so we can't forget about that stuff, but we need to impart in new and interesting ways and rely on new and interesting frameworks to help us to move along in this way. And to use different sources of information to validate our understanding and our findings. So you may also see the term design patterns pop up. Now, this is an interesting one. This is a fairly recent term in terms of it being used to describe different ways of doing things different ways of doing design. And design patterns tend to focus at different levels of abstraction.
Play video starting at :2:43 and follow transcript2:43
So we tend to go up and down depending on the level of complexity of the systems we're working with. Design patterns can be for very small components of systems, but they can also be for larger set of constructs around how to design things like navigation. Now, in terms of design patterns, they can both be generalizable and individualistic, so they might point to specific things, or they might point to more general practices and general ways of doing things.

I want to talk to you very briefly about collating data. I want to talk about why this is useful, and actually what we really want is information. There's different levels to this, and we can think about it is essentially as three different types of representations of data. I'm going to draw a triangle here just to give you a sense of the three different levels. We start at the bottom with some data. This is rich, there's usually lots of this, and we throw a lot of this away. This is one of the reasons we start at the bottom.
Play video starting at ::35 and follow transcript0:35
The data is the thing that we're getting out. The data is the thing that we're trying to understand and add meaning to. We've got some data, we want to do something with it. What we essentially want to do is to convert it into information. We want to say, "Okay, how can this data be useful?" Eventually, at some point in time, we may even want to do other things with the date. We may want to convert it into things like Knowledge Management Systems or KMSs, which describe other ways to use data in more meaningful ways.
Play video starting at :1:1 and follow transcript1:01
But generally speaking, what we have right here is this information space. We're thinking about capturing our data, and taking it to a stage at which we've got some usable information. We've got some information about our user that we can actually use to generate other bits of

information. What we essentially want to do is we want to describe things like the user situation. We want to describe things like, how do we define functional elements of our system that are going to meet a user's requirements? That's a good question, and it's a question that should be informed with the information that we have available to us. If you've asked the right questions, if we've got hold of the right data, and we've categorized it into meaningful categories, into meaningful chunks and structures, we should be able to figure these things out.
Play video starting at :1:45 and follow transcript1:45
In the same sense, we might also want to look at technical specifications of systems to determine whether or not we are building the right system, whether the system meets the user's needs. This could be a number of things, it could be thinking about the speed of the processor, or it could be looking at the display devices, if there's enough information displayed on the screen. There's a whole host of technical requirements that might also come from these requirements elicitation exercises. But what we're essentially getting at here is trying to understand how this information can be used in a meaningful way to build systems that are useful and usable for human beings.

So let's start talking about requirements now and understanding our users requirements. So where do we begin, where do we start by understanding what are users wants and needs are? Well, there are lots of different places that we can identify this information. And we've actually already looked at quite a few techniques for capturing these bits of information, and we can do this in a lot of different ways. So first of all, we can think about our requirements as a set of needs, so I use this. Needs are important to us, we want to design a system that facilitates the actions that the user wants to achieve. So the user has some goals in mind, they have some things that they want to do with our system and fundamentally for a good quality system.
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We want to make sure that we can build a system that does all of the things that they want them to be able to do. Now when we facilitate this set of actions, there's a manifestation of a positive outcome if we do this properly. So if we build a really good system I use as well, enjoy using it and keep using it will continue to use it over time. And that benefits us if we're putting out either a product or service, there's a commercial benefit. But there's also some value in understanding that we've done the right things and that the user is getting some value out of that thing that we've built now. We also have to consider users once, so there are other things that sit outside of the periphery of specific requirements. That would also affect how a user engages with our system, so we have to consider our users perceptions and the world around us.
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And consider that each uses different and those individual differences are something that we want to tease out. When we understand our requirements, this is what we want to get out, we've got lots of different stakeholders with lots of different goals in mind. Things that they want to achieve with the system, but there are also other factors around that. So we looked at things like structuring information to make sure that people can get access to it. They can understand where the information stored and how they can retrieve those bits of information. But we also have to think about things like aesthetics and the general user experience, or what we might describe in some cases as the customer journey. There is a set of trajectories of actions through a system where we have to make sure that we're meeting the user's needs all of the time.

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Not just at the point of any kind of interaction, we want to make sure that over time, that the system is building up. And improving on the experience to a point where a user can achieve their goals or to get some intended outcome out of the system. So how can we do this, we can use many of the techniques we've already seen. So we could handle questionnaires and we can ask questions about our system. We could do observations and we can observe people using similar systems, or using systems that are like our systems, or competitive based systems. And we can also build prototypes and ask people to work through specific scenarios and do things like usability tests. To ensure that the system is meeting those needs, and that we're validating our requirements.
Play video starting at :2:43 and follow transcript2:43
So this is very much an ongoing process and it's something you have to think about over time. You don't just generate requirements and then forget about them, you want to make sure that you're validating these things as you go. And ensuring that your system is encapsulating all that rich complexity that's embedded within those requirements. And those systems that you want to design and build for fundamentally, we elicit required by asking, watching, exploring and a whole manner of other things. But we also need to consider a wide range of people, we need to consider different stakeholders and how they might interact with the system. So we want to ensure that people have different levels of access, now this might be looking at different types of users. So we could categorize them according to things like experience with the system.
Play video starting at :3:23 and follow transcript3:23
And so we could say, well we want novice users to have these features and we want expert users to have this other sets of features. But it also uses in the middle there, so we want to design for everybody and we want to make sure that our system encapsulates that rich complexity in that set of design contracts. General considerations are requirements, generally speaking, need to be quite robust, meaningful and agreed upon as a formalized set of standards. So it's not enough to just say I've got these high level considerations in terms of requirements. They have to be specific things that we can measure, that we can validate, that we can verify. So we want to make sure that we could go back, and consider how successful we've been in relation to each of the requirements that we've defined. And as we change these over time, we want to ensure that we're converging on our goals.

Let's consider how we might capture requirements and how we might embed them in a set of structures that we can do things like verification and validation according to some predefined criteria. This process of formalization could be done in a lot of different ways and you can use different strategies to formalize requirements. We've seen some examples of these. If you've done UML, if you put these use cases and cost diagrams. They are essentially formalized sets of requirements seeing insisted of modeling techniques. Again, we can model other types of characteristic properties and features of our systems in lots of different ways. There isn't just one more way to achieve these things.
Play video starting at ::46 and follow transcript0:46
Now, the traditional way to do this is to think of things as checklist. To just say, well, we can check these things off as we go, we can verify and validate based on a tick or across, we can say this has been done, well, this hasn't been achieved based on some feedback. Check list is a

good way to stop thinking about formal requirements and understanding. Well, we have a series of things that we want to achieve and then we can tick them off or cross them depending on how successful we've been. We might even want to put a number next to those things and say, well, there's a percentage value or there's some other value that gives us an indication as to how successful we think in relation to these individual requirements, these things that we characterize at a granular level. Generally speaking, you'll see requirements defined in two different ways and I'll talk about these now. The first thing that you see, requirements defined as functional requirements.

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We can think of functions as things we can do, things that we can achieve. Function might be to move my arm or my leg, or to pick up a cup, or to move across the room. Those are all different examples of functions. You can think of function requirements in a general way and say function requirements are cases where I want to do a thing. Now, generally speaking, functional requirements tend to focus on inputs and outputs. We do think we get something back from the system. It's a very linear, systematic way of thinking about requirements.

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You may also consider this in terms of questions as what can the system do? What can I achieve with the system? If I come to the system and I have instances of inputs, what outputs can I expect to get back? These are valid in the context of what I'm trying to achieve. These functional requirements are very much driven around things that you want to do and focus on this element of input and outputs, a very traditional way to think of computer systems. We also have to think about non-functional requirements. These typically specify the underlying architecture, the environment of the system, and they tend to focus on operation of a system or quality.

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We're often thinking about things that on tangible is easily, all the things that we can touch upon and say, this is a specific functionality, this is a specific feature. Examples of this might be things like usability or reliability. How would you measure the reliability of the system that we're using right now for example. You probably implies quite a few sets of metrics on that would need because you'd be thinking about being able to use it in different scenarios, the access to the equipment, the security of the equipment, and things like usability as well. Thinking about, well, it's usable? It depends on the context of the person. As we know, usability is a number of features, a number of things that we want to measure.

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No functional requirements tend to be much high level things, they're much more broad in terms of the scope of things that they consider. There is also important to consider these requirements elicitation and consider things like quality metrics, reliability, usability, and these examples as important in embedding requirements of our system and understanding how to validate our system against uses expectations. It's not just enough to produce something that's functionally relevant. We also have to consider these other factors that relate to how human beings engage with computer systems. How do we capture requirements? Now, this is an interesting question. Well, it depends.

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We can capture requirements in lots of different ways and we've already seen examples of things that they said, used cases, interviews, we can do observations, we can take notes on what people are doing, we can capture video footage. We could capture an audio footage, create transcripts of things. There are also different models for generating requirements-based media about understanding our users wants and needs. When you think about our users wants and needs, they quite complex, but we can get to them in lots of different ways. Any and all of these techniques are valid, but it depends on the context, it depends on the activity, and it depends on the thing you're trying to get at. The base of information that you wouldn't know about, is the rich contexts attached to that, and you have to go into that and decide which of these techniques is fit for purpose. How do we specify requirements?
Play video starting at :4:38 and follow transcript4:38
I've already talked about the Intel providing checklist of things that you might want to achieve very early level. This is probably a valid way to validate requirements. As we go, our validation process gets a bit more complex. You want to verify and we want to validate based on a number of different factors, based on a number of human-centered design approaches. What you really want to measure is critical success criteria against some things. When you specify requirements, you need to be clear exactly why it is that you're specifying? What exactly is this need?
Play video starting at :5:7 and follow transcript5:07
What exactly is this one? Why you see manifestation of activities that would allow this user to come about with an end goal. You've got to think about this in a really linear granular way to ensure that you can go back and check on these things. If you have vague requirements, if you have things that aren't easily defined or aren't easily captured and encapsulated within the techniques that you're using. It's going to be very difficult to say that the system has been successful or is systems failed, and you find yourself creating problems. Make sure that when you specify this is formalized, this is structured, and these things that you can measure, you can go back and look at this simple enough that they're broken down into specific components of things that you can look at. The tick-box example is one example of that, whether things are very specific, you can say yes or no, binary decisions.
Play video starting at :5:51 and follow transcript5:51
But you also want to think about what can measure these things? All these things in relation to other things. Have we talked about them in relation to the things that we want to achieve based on some other factors? Maybe this is a time-based representation of things. Well, maybe we're thinking about functionality as a set of core constructs of our system and then breaking it down into really granular level details of this is step 1, step 2, step 3, and so on and so forth. This complication as you go through, as you generate more requirements, especially if you're working in an agile way, you can find that requirements change over time. You have to adapt and be fairly robust in terms of your working processes.
Play video starting at :6:28 and follow transcript6:28
When you want to do radius so try to break requirements down into the smallest possible steps and think about them as iterative steps. Think about them as processes, as dynamic processes that may change over time. Recognize the fact that users may have different ways of achieving things and build a system that encapsulates that. Dependencies may create problems down the line, so be careful about building too many dependencies into your work. I now exclude practice

and things like object-oriented programming too. Lots of dependencies have lots of interrelated things. Generally speaking, is good practice from a technical perspective but from a requirements perspective, if you depending on lots of other things, it can create complexity when those things change.
Play video starting at :7:6 and follow transcript7:06
When we talk about requirements, they often dynamic. We have to respect the fact that these things will change over time and that we are adapting and changing over time ourselves, and so we want to make sure that these are as granular and as smallest possible. Some requirements might be unexpected but still need to be captured, so you have to take into account lots of different types of stakeholder, lots of different types of users and use cases and try to understand that not everybody is going to interact with your system in the same way. What you generally want to do is be reflective. It's easy to focus on the technical aspects and ignore the context of use, but we have to validate the context of looking at this in some further examples as we go through, we'll look at some things like personas and scenarios to try to understand how context can help us to derive interesting and useful requirements, and due processes like verification and validation in a meaningful way.

We've had our general introduction to design and we've started to understand some of the terminologies used in design. Now we're going to look at some specific techniques and we're going to look at why they're useful and how they can help us to think about the way that we design systems. The first technique that we're going to look at is one of my favorites is the Eight Golden Rules of Interface Design by Ben Shneiderman. Now, this is all about understanding how to build interfaces. Most of us see this as representatives, DUIs or GUIs. But we see interface design in most of the systems that we use nowadays. Most systems that we engage with have an interface of some sort.
Play video starting at ::44 and follow transcript0:44
We're really thinking about how can we encapsulate the right design practices in the systems that we build. These eight golden rules will help us to do that. Let's start with first of all, consistency. Now this is obviously very, very important. We want to have consistent interfaces. We want them to look and feel the same way to our users. But we also want to encapsulate all that existing knowledge that they have.
Play video starting at :1:7 and follow transcript1:07
It's very unlikely that a user is coming to your system for the first time having not used any other type of system before. If you're building a website, for instance, you want to make sure that you're encapsulating good practice around consistency from previous experiences that the users have had. If they're used to seeing the navbar on the top, you probably want to leave it there. If they're used to clicking on buttons and things like affordances that we'd expect to see in a system, then we don't want to change the way those subtle components work. We don't want to create confusion for the user and in the same sense we want to standardize across our own system. We want to make sure that icons and language and the flow of information are all standardized and consistent between pages of our interfaces, between different elements of our interfaces. You don't want to be drastically changing different things as you go.
Play video starting at :1:51 and follow transcript1:51

If suddenly the button size increases or a button moves along the navigational bar or things are in different places on different pages, that's not a good thing and that's going to create some confusion. People will be looking for things, they'll be expecting certain things in certain places. Consistency is something that you want to build in from the offset. Now, the second practice is usability for all. Now this is all encapsulating and it's thinking about two extremes in this case to try to encapsulate that rich bits of information. We're thinking about the novice users who were using the system for the first time, are very new to the system. But we're also thinking about those expert users who may have used this 1,000 times or 10,000 times or even 100,000 times.
Play video starting at :2:32 and follow transcript2:32
We want to ensure that both types of users have a case for usability, have a case that the system allows him to do the things that they want to do in a meaningful way. In this sense, we want to give more familiar users shortcuts to the key functionalities of the system. This might be things like Control + C and Control + V to copy and paste. Most of us are probably pretty familiar with this modality of working. Having shortcuts or what we call accelerators in systems can really help those expert users to move quickly through the system without having to click through lots of menus and do lots of complex things. Over time those expert users can learn their shortcuts and they can utilize their shortcuts to save themselves a lot of time in repeat processes. We equally want to make it easy for novices to perform basic tasks.
Play video starting at :3:15 and follow transcript3:15
We don't want to get rid of those menu structures for the case of efficiency, because there are some people who might come to the system. We've never seen this way of working before, we don't know the shortcuts, and in that case, those menus, that way to click through things might be more effective for that type of user. We're thinking about all different types of users and encapsulating lots of different types of use cases in an interface that is capable of handling both extremes. We also want to provide feedback to users when they're using our system, of course, from the perspective of interaction. Interaction is not a one way thing. I don't click a button and then the system just does some stuff and doesn't tell me what it's doing. Generally speaking I'd expect to get some information back.
Play video starting at :3:54 and follow transcript3:54
That feedback that I get from the system should be useful. It should be informative and it should be relevant. It's giving me the right information at the right time to allow me to make certain decisions to say, well, I'm going to click off this, or the page is frozen, I want to do something to deal with that, I want to go to another page or I want to close my browser. When I click something, I expect something to change on the page or on the screen. I want things to happen, I want to know that that call to action when I've clicked on it has done something, it's taken me to the next page of the form for example. We're thinking about this type of feedback and how to encapsulate this within the system to make sure that the feedback is meaningful and useful and that the user is getting the right information at the right time so that they can act upon that. Now, one way of dealing with feedback is of course responsive buttons.
Play video starting at :4:40 and follow transcript4:40
You can have things like hover states and things on buttons, you can have onclick events that deal with these things; give the user feedback, change things on the page, and maybe even tell them that something's changed on the page depending on how much information you feel is relevant to give to them. Of course if things do go wrong, give the user a chance to deal with

that, to handle that situation. Say, well, actually, this isn't really what I wanted or I've got some unexpected behavior, but I know how to deal with that. I know how to subvert that as a processes and go around it, or go through the system in a different way to achieve the thing that I wanted to achieve. You don't want users to get lost or to get stuck and the feedback can help them in that process. You can help them in terms of their user journey through the system. Now we also want to have a dialogue going, and we want to make sure that we're talking to them and we don't want to keep them guessing, so we can use things like progress bars to ensure that the user knows that something's happening.

Play video starting at :5:29 and follow transcript5:29

If it's a long laborious process, then we might want to have a situation where a progress bar just tells the user look, this is wherever we are working on this process but it takes some time to copy these 10 terabyte files across to your hard drive or your backup drive, or your network address storage. I'm telling you where we are in this state of things. I'm telling you that were here and in a few seconds will be there. It's about keeping the user informed all the time, and this dialogue is really important to help the user understand that something hasn't gone wrong, that the system hasn't crashed, that it's not stuck. We are giving users the right amount of information at the right time. We also want to be able to handle errors in different types of waste and there is such a thing as elegant failure, so there's no such thing as a system that doesn't have error on, but you can deal with errors in different ways. You can help the user to overcome those problems.

Play video starting at :6:17 and follow transcript6:17

If there is an error, if something's gone wrong in your system, you probably want to explain what has gone wrong and how to get back to a safe state. The user can say, well, I can use this era in a meaningful way. If I have Error Code 9265B, as a novice user, that's probably not going to be very useful to me. I don't really understand what's going wrong, and I don't really understand how to get back to where I was. You want to tell the user what's caused the error? Why you're not accepting certain types of input on forms, for example, might be a good example of this. But you want to give the user the right amount of information at the right time to ensure that they can make good quality decisions, and they can say, well, I understand why this era's happening now.

Play video starting at :6:53 and follow transcript6:53

I'm not going to do that again, or I'm going to do things a different way to ensure that that error doesn't happen again. I understand what's caused it and I can work around it. Now, we also want to give people the opportunity to reverse their actions. Now, we have erasers and pencils for a reason. People make mistakes. People cross things out and writing that's normal. That's a normal feature of being a human being.

Play video starting at :7:15 and follow transcript7:15

It's okay to make mistakes and our system should encapsulate these good practices, so we can look at undo and redo as good examples of this control and zed is your friend. But we can also look at other examples of situations where people might get stuck in a system or might want to go back one step or one stage and generally speaking, it's good practice to design into your system the opportunity for users to get back to what they call a safe state or a safe space. You want to ensure that uses answering all the way back to the first form on a 100 page form every time they make a mistake, that's going to be really frustrating experience for your users. They're

going to lose a lot of information, is going to create problems in interaction. It's going to cause them to make more mistakes and more errors. It's a compounding effect is what we call it and so what we want to do essentially is to give users the opportunity to move back a step or move back a state, so we can delete characters in a form which can move back to a previous page and a form we can ensure that if there's something going wrong in the state of your app, we can move back one step and fix that and then move forward. We want to ensure that the user's flow is not broken, that they can move through the system in a fairly linear way.

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Now we also want to think about the internal locus of control, so what this really means is give people options, give them the control over the system. You don't want a user to feel helpless in a certain state. You want to give the user the opportunities to be able to interact with the system in the way that they want to, to get the right information back but also to be able to deal with certain types of states. A good example of this is when a system locks up without options for input, the user loses that internal locus of control they feel helpless. They don't know why the systems locked up and the classic case for this is that there might open a task manager or open some way of handling tasks and kill a particular process. But it's important to give people those options and give them the control over the system and say, well, if we're reaching an unintended state or if there are some unintended consequences, we can handle that and we can give the control back to the user to say, I don't want this state, I want to change. I was to move from this particular set of features to somewhere else in the system.

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Generally speaking, it's good practice to ensure that the user is in control of the interactions the whole time. Then finally, we want to reduce short-term memory load, so we want to make sure that we're not given the users too much to remember and think about at any one time. There's already a cognitive challenge in dealing with the system. In finding elements on a page, clicking on things, typing. There's a whole manner of things that we talked about in modeling that we want to be concerned about and so giving uses lots of information on top of that can be really problematic. We can do this through a number of different techniques and tools. We can think about the idea of giving iterative steps and stages in a set of processes to ensure that we don't have a form with a 100 different input types.

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We can just have one different input type, and then they can click through and deal with the question at a time. If we were to hand somebody a questionnaire, we probably wouldn't have all the pages laid out in front of them. A 100 page questionnaire, one page at a time we probably give them the questions individually, wouldn't we? We want to do the same thing with interface design, we want to make sure that we are reducing that short-term memory load. We are making sure that the user can navigate through these tasks without getting confused or stressed or anxious, or creating a situation where they're not sure about what's going on because they're having to remember something that you said to them five or six pages ago. Make sure that you're thinking about this iterative approach to processing different bits of information and ensuring that the user doesn't have to remember lots of different things at different states in the system.

So in our previous example, we looked at some golden rules that we can use for interface design. Now, let's look more broadly at things like design standards and try to understand why they are also useful in helping us to understand how to design good quality systems. So when we think about standards and standardization of design, there are lots of different options and they exist both in the hardware and software domains. You might see lots of examples of standardization in the hardware domain, but we're thinking about engineering tasks. When we're thinking about building things. It's very important to have standards. It's very important to understand that those standards have to be conformed to ensure things like safety and to make sure that people are kept safe, that we're not creating things that could be dangerous.
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And that we're ensuring that certain types of hardware are accessible and reliable and consistent so that as people use them, they can ensure their own safety. They can ensure that they are using something that has a certain quality metric attached to it where I can say when I drive a car, I expect a certain standard of safety, I expect it to be safe on the road. I expect it to be safe in terms of performing maneuvers and different types of mechanisms. This is also true of software. It's not just about hardware. We also have software standards and standards around how we do things as well as the sorts of things that we build and design. So one of the most common ones in this field is the ISO standard 13407/ISO 9241-210 and this is a standard around user centered design.
Play video starting at :1:34 and follow transcript1:34
So there's a more common version of this are more recent contemporary version of this in the 9241-210 standard, which talks about how to design systems from a kind of usability perspective. And so this is a very high level set of construct some steps involved in designing usable systems. But these standards are there for that very purpose. They're there to give us some opportunity to reflect on our practices and think about how we're doing things and make sure that we're doing things in the right way. So why do we use standards? Well, first of all, standards are published by well established industry. So we know that they respected institutes, they know a lot about these things and they can tell us a lot about how to do things based on their own understanding and knowledge.
Play video starting at :2:17 and follow transcript2:17
They've generated lots of expert opinions and they've been able to encapsulate a set of standards that allows us to understand how to design good quality systems. So, places like the IEEE and ISO, provides standards for doing things, provide ways in which we are supposed to do things to ensure that we get the best out of our products and services and it keeps everybody working in the same way. So most of us are already familiar with lots of standards. Http is a standard. HTML is a standard most of which use the web on a regular basis. So we know the value of that, we know the value of having standardization. We use a standard language this courses taught in English, which is helpful in that everybody can understand what I'm saying.
Play video starting at :2:56 and follow transcript2:56
If I started talking to you in French, that might be problematic. I be converging away from the standards and it will create confusion. We have different ways of working then that would require other people to go away and understand other components of that. And so what we want really is standardization. We want protocols and ways of doing things that ensures that we

are all working in the same manner. Now, there are some challenges with conforming to standards and the standards don't always provide the best quality information, every different vector of interaction. So, first of all, standards can be rigid, which is both good and bad.
Play video starting at :3:29 and follow transcript3:29
The rigid standards mean that everybody's working in the same way, but it also means there's very little opportunity to diverge away from standards and say, well actually, I think I've got a better way of doing this. If you're following a standard set of procedures, you're not able to do that, you're not able to move away in part different parts of knowledge in how you do things. Standards also don't tell us a great deal about the application space. So they'll tell us general high level features about how to do things, but they won't tell you what to do under certain circumstances. So they don't give you very specific types of information that you might need when you're designing smaller elements of the system or smaller components of interaction. So the standards are good at a very high level. A lot of the features of standards are generally visible, but they don't give us very, very good properties at that low level of granular detail, where we might want to know more about how to do a particular thing.
Play video starting at :4:17 and follow transcript4:17
What do you do under certain circumstances? Now there are alternatives to standards and we can use standards alongside these things as well. So we don't have to use one or the other. We can follow certain standards and we can also take principles and guidelines. So principles and guidelines tend to be more abstract. They're quite generalizable, we can look at these things and say they're kind of high level features to their very similar to standards in that way. But we can also look at things like design patterns which are very, very important in understanding how to design a system in a contextually relevant way.
Play video starting at :4:48 and follow transcript4:48
So this is a sort of ready made solution to a problem. It's like saying, well we've had this problem before and we've got a general design pattern for this, we know how to design for these sets of circumstances. So if I'm designing a playground, I know that I would have certain things in it would have certain features. They have grassy elements. There might be rubbery elements underneath all the things that the children playing with to ensure safety you generally speaking, have bars or some kind of perimeter around. So we've got to design pattern for that. And we've got design patterns for most things in the context of interaction design.
Play video starting at :5:18 and follow transcript5:18
We can look at previous examples. There aren't very many examples of things that are completely novel or completely contemporary. So it's good to learn from the past. It's good to learn from what other people have done and learn from those standards, learn from those principles and guidelines and also learn from these design patterns. And we can take all of those elements together and we can use them in different ways to get at different bits of information that we want access to. So design patterns again are borrowed from another field. So we're very good at this in HCI were very good at taking other people's ideas and in part of them on what we do.
Play video starting at :5:50 and follow transcript5:50
And this time we've borrowed from the architecture field. So in the past we've seen kind of lean manufacturing techniques as part of things like agile. But here we're looking at architecture as a means to understand how we can follow certain design patterns. So understandably a building

is a very constructive set, a way of doing things. We're thinking about low level constructs around setting foundations right the way up to how we're going to decorate a house. What's it going to look like? What color are we going to paint it, how bigger the window is going to be?
Play video starting at :6:20 and follow transcript6:20
So we've got these high and low level considerations and we borrowed this from the field of architecture to help us to understand how we can design systems for human beings. Now the application space in the architecture field is very similar to HCI. So what we look at is other people have done these things these ways we want to impart the same amount of knowledge that they've got from those processes in designing a new type of system. So there may be some subtle changes, but we follow those design patterns in a kind of wrote based way we say, well they've done these things and they've done these things because of this. So we're going to follow that pattern, we're not going to diverge away from that pattern, we're going to do things the same or a very similar way. Now design patterns are focused on the what rather than the why? So they're focused on what kind of task you're trying to do.
Play video starting at :7:4 and follow transcript7:04
What's the context of this type of activity? And then you're thinking about, well, in relation to that context, we have an existing set of design patterns that we can use. It's a bit like template, Germany speaking design patterns encapsulate good properties of design. So we take those good properties from other examples and we impart them in our own designs and generally speaking design patterns are easy to follow to achieve a think. So, if we want to design a specific type of system that say it's a payment system or some type of system that's dealt with other types of constructs, maybe it's about purchasing things, maybe it's about identifying particular types of products. We can look at design patterns from popular websites, from commercial places and then we can borrow those tools and techniques and we can use them in our own design to ensure that we are following good practice.

So let's have a little think about how we might organizing structure information in a meaningful way. So information can come to us in a number of different formats. We don't necessarily get to dictate or decide what information looks like. So we could have information looks like this, we've all seen this before. Hopefully this is an unfamiliar to any people here doing this course. And this is of course binary information. We could have text based information that might be kind of scribbles of what people have written.
Play video starting at ::36 and follow transcript0:36
So it could be handwritten information. We can get information from a range of sources. So it might be things like pictures with kind of visual information. And we could have things like charts and graphs describe certain relationships between X Y. So we can think about X being related to Y in some way. Now a lot of these pieces of information have certain descriptive value in them, don't they? You can say right, I can look at this and see that it's got some meaning.
Play video starting at :1: and follow transcript1:00
I can see some meaning from a picture. It's a picture of some object with some things inside. If it's a chart or a graph, we can see that there's some clearly identifiable information we are comparing or contrasting things against one another. We're making sense of some bits of data and adding an extra layer of understanding about what the data means. Now in this format, in the binary format, it probably doesn't have any meaning until you actually load it into an

application. But what we're getting at here is trying to understand how we can decide on this information structure based on a number of key factors. So when we get information, we don't automatically know what we're going to do with it or how we're going to store it, how we're going to present that information in a meaningful way.
Play video starting at :1:41 and follow transcript1:41
And we actually have a number of options. So what will typically happen in normal scenarios, we get something called data. It's not quite information yet. It's just some data. We don't really know what it does. And what we have to do is we have to impart our knowledge on this to convert it into something meaningful. And this is where information comes from.
Play video starting at :1:58 and follow transcript1:58
So we take some data, some data about a subject. It could be pretty much anything and we'll do things to it to ensure that it has some meaning. So we can capture it and store it in particular way. So we could store it in a particular format. And that would be one way that we'd handle the representation of that information to give it some meaning and understanding and say there's some context here. So the information has some meaning. Now, a binary file might be a number of different things that could be audio, it could be video, it could be images, pictures of things.
Play video starting at :2:29 and follow transcript2:29
It could be code, it could be compiled code. It could be some language that we're using to describe things in. So there's a whole host of types of information and they're all categorized in particular ways. So we can look at information actually is an example of information. So in language we know that this has some meaning. It has some construct components, so individual letters for instance, so the letter end appears in their multiple times so that some data, but we know that this has some information attached to it as well. So it has some meaning.
Play video starting at :3: and follow transcript3:00
We understand that the word information means something to us. And if we had a series of words, we could have things like word clouds and we could have representations of text using natural language processing techniques to understand that. So what we're getting at here is trying to understand the structure of information in relation to other things. So everything we do in interaction design is really about understanding where these things fit in a broader context. So let's think about how we might store information that people want to access. So the classic cases that we're building some system and we've got some information in there, we've got information that people want to get at, and we've got to think about how to structure that in a meaningful way that allows people to access it. So one way we can do this is we can take the smaller elements and we can break them down so we can say actually information is lots of things.
Play video starting at :3:47 and follow transcript3:47
So if we took the example of the text that I gave you before, we could break that down into smaller components and we could break it down into letters. Or we could break sentences down into words or phrases. There's a lot of different ways that we could do this and we could represent this data in different ways. But this meaning is really, really important. And if we structure this in the wrong way, it will make it very, very difficult for people to access it. So what we have to think about is how are we going to manage this process? And this is a very involved task.
Play video starting at :4:15 and follow transcript4:15

So if we were to take the example of building a website, for example, we have a number of pages won't be. So we'll probably have a main page or an introduction page. We'll probably have some about me page or about us, depending on whether it's a company or an individual. So we'll just call it an about page because we're not sure who it's about yet. We might have some other things in there as well. So we might have bits and pieces of data, so we might have a gallery, for example, we have lots of pictures of what we do. So maybe we are car sales company and we've got lots of pictures of lovely vehicles, or maybe we are selling really nice cakes and we've got pictures of cakes on our website.
Play video starting at :4:49 and follow transcript4:49
Now, that's the main content, isn't it? But we can't just think about this in isolation and say, well, we'll just build some pages and hope that people arrive at the content. We've got to think about destruction. We've got to think about, well, how are people going to navigate through this system? How are they going to get to our beautiful gallery? And ultimately, in many cases, if we've got commercial perspectives, we want to think about how can we make a sale? How can we ensure that people are going to buy things from us?
Play video starting at :5:13 and follow transcript5:13
And we can do that through having this very, very descriptive structure where elements are organized in a meaningful way. So, if we had a larger case, for example, where we wanted to build a system that was selling lots of things, So maybe groceries, for example, we've got some interesting challenges there, haven't we? We've got to think about what's the makeup of a person shopping, What does it look like to shop? What those key things that we have to think about and how do we categorize that information? So that's one strategy we can take. We can categorize information according to some predefined structures. So we could categorize things according to alphabetical order.
Play video starting at :5:47 and follow transcript5:47
So we could have eggs come before some other element, let's say milk, milk comes next in the alphabet, doesn't it's further down the line, maybe we'll have some bread. So bread comes before eggs. So that would be up there and we could categorize according to this alphabetic structure, but that probably doesn't make a lot of sense to most people, does it? It's not really a sensible way to categorize things. So straight away, we're starting to understand where information can help us to, or where structure can help us to understand information and how it can be useful. So most of us would expect to shop according to maybe where the items are stored. So you might have tinned items in a cupboard, You might think about that as a kind of physical relationship.
Play video starting at :6:25 and follow transcript6:25
So we've got some tins here, we got some examples of some tin products that we want stored in a particular cupboard. So these are generally speaking Long life products and lost a while. So we might put them in a particular place in our store, whether that's a digital or physical store. You can walk around the shop and see this can't, you can walk around and see how the items are stored in particular ways. And then we might think about other kind of staple products. So maybe milk and bread are in there. So I just use the letters for those milk and bread things that most people tend to need and we can think about now compartmentalizing these things, but there's also an inherent structure in them as well.
Play video starting at :7: and follow transcript7:00

So we have to think about how people going to navigate from here to here. How are people going to get from one section to Another? So let's say I have a recipe that includes something from box number 1 and something from box number 2, I have to follow this structure. Don't I have to make sure that I can get to all these different things. And so we have to think about compartmentalizing these things in meaningful structures. Now, it might be the case that somebody has to follow a very, very particular route to get somewhere and that's fine. That's understandable if we have lots of information and we have to really, really carefully think about breaking that down into a meaningful structure.
Play video starting at :7:34 and follow transcript7:34
So it might not be a simple process of where we have an XY decision. It might be the case that we have to go across long trajectories to allow us to get access to the information we need. Now, what we essentially don't want to do is we don't want to make this journey too long. We want to see a success scenario from the minimal number of steps possible and to do that. We think about how we compartmentalize bits of information together, and we think about adding structure, an imposing structure and things in a meaningful way that allows us to achieve our intended goals and allows us to map to the user's cognitive model their understanding of the way the world works and being able to access these things in the quickest, most effective, most efficient way possible.

Now I'm going to talk to you about information and how to structure information in meaningful ways. When we think of information is this broad set of bits of data and things that we know about the world. We want to impart some meaning on that. We want to put that in a place in which people can get access to it. Now, this might seem fairly intuitive, but actually is quite an involved process and there's a lot of thought that has to go into this. We're going to look at a specific technique that allows us to do this later on, we're going to look at a technique called card sorting. But for now let's just focus on how we store and organize information in a meaningful way.
Play video starting at ::43 and follow transcript0:43
First of all, in terms of organization, there were lots of techniques that we can use to organize data or organize information in meaningful ways. Now we cannot possibly encapsulate other people's worldviews in our minds. There's lots of different perspectives about how we should organize things, how people interact and understand the world, their own conceptual understanding of where things fit together. Thinking about things like proximity of objects to one another. How might you store things in a kitchen cupboard, for example, is a very specific thing to you. It's very individualistic. We can't really impart all those bits of understanding that we might want to.
Play video starting at :1:19 and follow transcript1:19
But we can encapsulate some techniques that will allow us to think in this holistic way and to think about different ways to organize and structure information. There's some meaning there so that some people can get access to that or that most people can get access to that and understand how that information is structured. Now we're going to look at the latch technique. So the latch technique stands for location, alphabet, time, category, and hierarchy. We're thinking about a number of things that are in a number of different perspectives about how we can organize our information according to these different types of structures. We essentially

have a strategy for organizing the main bits that we want to focus on. We can focus on these different elements depending on what type of system we are using.
Play video starting at :2:1 and follow transcript2:01
You do have to use some of your own knowledge here. You can't just apply these things verbatim and hope that they work. You have to really think about which technique is the right one for the sort of information you have and the type of interaction that you're trying to create. First of all, we'll look at location. Now, locations are fairly similar construct to a lot of the things that we've looked at in the past. It's very basic. Most of us have a general understanding about location-based data.
Play video starting at :2:25 and follow transcript2:25
We can think about a map as being location-based data. Things are related to one another according to how far apart they are. We can think about, towns and cities as being generally speaking, in the same sort of proximity. Then we might think as a country level or is a state-level, there's a larger set of constructs, more elements encapsulated within. Then as we think about continents, we've got even more elements encapsulated within. So you've got this kind of outward structure where we build things up over time. You've got smaller related components that fit into a larger component.
Play video starting at :2:55 and follow transcript2:55
Now this is a good way to think about information. This is a useful way to think about information. We can organize information into geographic regions. So we could say, If you're buying some items on a website, you can say, well, we're going to organize all the products that are sourced locally into one specific category. We can actually organize information in that way. We could say, we're going to give you all the restaurants within a three-mile proximity of where you are now. That's another example of how to organize according to location and it's meaningful.
Play video starting at :3:22 and follow transcript3:22
It's relevant because I want food that is going to arrive warm. If I'm ordering some takeaway food, I want to make sure that it's warm, that somebody's not having to travel a long distance, that they're not wasting lots of fuel or causing harm to the environment. For me, there's lots of considerations I take into account to ensure that that is built into the system and that the information I'm being given is the right sort of information at the right time. From a design perspective, it's important to think about location as one of the ways in which you can sort your information. Now, quite a common way to organize information is through alphabetic characters. We think of a to z essentially as a way to organize information. There's lots of examples of this in the real world.
Play video starting at :4:1 and follow transcript4:01
We can think about an index, we can think about dictionaries as having constructs around. We go from a to z. So if I want to find a word beginning with the letter p, I never get to p. Then I navigate through to find whichever word I'm trying to find within there. That structure is easy to understand, most of us have used it before. It's generally speaking, quite a common way to organize information or data. Now, this can also be a challenging way to store information, there's one case in which I want to talk about which is problematic from my perspective.
Play video starting at :4:29 and follow transcript4:29

Whenever I click on a form or I'm filling in some registration details and people want to know my location. I'm never quite sure where to look if the data is stored alphabetically. I'm never quite sure if I'm from the United Kingdom, from Britain, from England or Great Britain. In each one of those cases, I have to look at a different character and have to scroll down to figure out, is it U, is it B, is it an E, or is it a G? This is problematic. There's different ways to think about London and England as being stored as data. There's different ways to categorize that.
Play video starting at :5: and follow transcript5:00
Unless people standardize, it becomes a bit problematic and it's problematic for me to find the information I need to ensure that I am selecting the right category. Make sure that you're using these things in meaningful ways. We can also categorize according to time. Now this, again is a fairly similar construct. Most of us are used to time as being this linear thing where we think about things in relation to when stuff happens. Time might be a good category for things like calendars, for scheduling events, meetings, etc. We can think about, "I can't have that meeting at that time because something else is already happening at the same time."
Play video starting at :5:33 and follow transcript5:33
But we can also identify things like useful email. We could say, "Well, the emails I've got today are probably the ones I'm going to respond to. It's very unlikely that I've got an email from six months ago that I haven't quite got round to responding to yet, and if I haven't, there's probably a good reason." How we sort information. We can think about time as being another way to do that. We can sort information in terms of how long it takes to do something, or in terms of these types of different bits of information around when events happen. Is there a clash between two events?
Play video starting at :6:2 and follow transcript6:02
Does something start at 2:30 and run to 4:00, and then something starts at 3:30, in the middle of that event? That's going to create clash. We can think about time as being an important way to categorize information. Now, we could also think of organic categories. We could think about generating our own categories of information. This might be things like weight classes. If you're into boxing or physical sports, a lot of physical sports have weight classes.
Play video starting at :6:27 and follow transcript6:27
You have heavyweight, light heavyweight, cruiser weight, and so on and so forth. That's one good way to categorize information. To think about a group of people organized into some specific categories. If you're categorizing houses, for example, you might look at the number of bedrooms they have. That's a good way to categorize the size of a house in the UK at least. We look at houses in terms of how many bedrooms they have, not in terms of floor space. Most people don't search houses by floor space.
Play video starting at :6:52 and follow transcript6:52
Again, that's another construct that is specific to a group of people. Now, categories can be a good way to organize information, for instance, around things like movies or shows, because we have genre based classification. We might have a horror, we might have a comedy. There are certain types of shows that people like watching, and so navigating by that structure is meaningful to lots of different types of people. This is generally quite a good way of organizing large data sets as well. We can inform categories as we go. As we navigate through big data sets, we can categorize data into smaller categories, and then we can do things like comparison

and analyses to understand a bit more about those organic categories, and how they relate to one another.
Play video starting at :7:32 and follow transcript7:32
If we're doing statistical analysis of things, we might want to use category based representations of things like gender. Now, hierarchy is another way to organize information. This is a derivative technique in that you are taking often categorical data, and then sorting it or organizing it according to a specific structure. You might be showing people the products that have been most purchased or have been most purchased by similar customers. You might be showing similar items, you might be sorting items by price or popularity, again, you could sort by distance in this manner. You could think about hierarchy as a set of distance-based constructs. But one of the most common ways of thinking about hierarchy, and one of the easiest ways to interpret this as a structure, is to think about a family tree.
Play video starting at :8:18 and follow transcript8:18
A family tree is a good example of the hierarchical structure where you have ancestors, and then you might have grandparents, and parents, and children, and so on and so forth. You can derive those relationships and start to understand how the different elements are related to one another by going up or down that hierarchical tree, that family tree to see those representations in different ways, and understand how people are related to one another. Now, hierarchies are a really useful way of storing data too, but again, this process of deriving different bits of data from other places is generally how hierarchical data is formed. You might want to do this where you have things like analytics, or lots of rich data around categories that you're then using to further refine things according to your own set of structures. Now, there are some challenges with storing data, and this isn't an exhaustive list. This isn't the only group of ways that you can sort and organize information and data. We're still relying on some expert knowledge here as well.
Play video starting at :9:12 and follow transcript9:12
We want to categorize data according to our own understanding, our own lenses. This isn't just a case of, we can categorize age according to specific ages. It will vary from person to person, from study to study, from activity to activity. Again, it's not a case of when you hit 19 or 20 or 21 or 25, you fall into a specific category. That's not as easy to understand, and so you might have to have some knowledge about your users, and some knowledge about your data, and your analytics before you can actually start to do this work. Again, data might not organically fall into the right categories, so you require some predisposed knowledge to enable you to form these categories, and understand which of the techniques in the LATCH techniques you'll use to impart the most amount of information, and to ensure that that information is stored in a reasonably sensible way. Now there are advantages. There are lots of options and opportunities.
Play video starting at :10:6 and follow transcript10:06
Human beings like organization structure. LATCH at least provides a good basis for structuring and organizing information.

Let's have a think about how we can get our users involved in describing and defining information architectures. Now this doesn't all have to be about the designers, it can be about our users too. We can get our users involved in the process and this is a good thing. You want as much user input as possible if we're doing user-centered design, we have to do it in a

meaningful way, don't we? We want to ensure that our users have inputs so that they have ownership of the system so that they can understand how the system works, and that the system is designed to meet their wants and needs. We're going to have a look at a technique here called card sorting. Now this is a really important and useful technique, and it allows you to impose structure at very different levels.

Play video starting at ::46 and follow transcript0:46

We can start at a very high level and we could say, well we have some pages on our website and we think that they'll go 1, 2, 3, 4, and 5, that's how we think that the system should be designed but we don't know. What we're going to do is we're going to give some cards to our users, some little cards, that describe each one of these pages, and each page is an individual card, and we can say, we've got a car for all five steps. Only seems to think that's a circle, that's strange, and what we want you to do as a user is to describe the interactions that people will have with this system. There's a number of different things you can do with card sorting, it doesn't have to be a prescriptive set of exercises. There's a few ways that you can approach this in order to get the most out of it. You can take these and you can simply ask the users to describe them as a set of actions or as an orientation, you can say, well we want some numbers here. The user might say, well, I do that second, I'd do that third, do that first, I do that fifth, and I do that fourth, and that's fine.

Play video starting at :1:46 and follow transcript1:46

We can then look at restructuring our data, restructuring our pages according to some given structure that's been provided to us by our users. If we have for example, 90 percent of our users say that this is a sensible structure it might make sense to actually design and build it this way. But we can do this at a more granular level as well, we can actually take smaller components of the page and do this. We can say, well, if we have this page that's about x, what individual elements would you put on this page, and where would you put them? We can have a structured that's imposed by our users that says actually, we're going to take this and we're going to do 3, 2, 1, and then maybe they'll group up some other activity. They'll say, well actually y is a different activity, so we're going to stick with y, other page that we've got now is we're going to have five and four. We've broken the structure down into two separate structures here, haven't we?

Play video starting at :2:34 and follow transcript2:34

or our users have, they've described a set of ways of working with the system. Card sorting can be really useful for these sorts of activities. We can get an understanding about how things should be designed, and should be built in terms of information architectures based on user feedback, and this is really helpful. This is really good at helping us to understand the mental models that people have about how information should be organized, and make sure that we have the right mapping, we have that good one-to-one relationship between what the user expects and what they see on the screen, and the more user feedback we get, the more we can use this to inform better design decisions. Information architecture is not just about what you think you should do with information, and how you should categorize, and compartmentalize information, we want to get as much user feedback as possible, and card sorting is just one of the many techniques that we can use to do that.

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Hello and welcome. In this video, we'll explore how to run online card sorting using Optimal Sort by Optimal Workshop. Card sorting is a simple but powerful technique for understanding how users group and label content. Card sorting helps you uncover how people mentally organize information. It's commonly used in information architecture design like structuring websites, apps, or menus. You can do card sorting activity in person, but Optimal Sort allows you to run

these studies remotely, saving time and giving you access to participants globally. So we'll go through a quick demo of setting up card sorting activity using sample data.
Play video starting at ::51 and follow transcript0:51
Okay, so now we will just type Optimal Sorting, and we'll head to the Optimal Workshop website. Here you can see that Optimal Workshop offers a lot of different research methods that you can use for your studies, including surveys, qualitative insights, but for this specific demo, we're just going to focus on card sorting. Before we begin the setup, I will tell you a little bit about their LearnHub. So if you go to the community and click on LearnHub tab, you can see that they have a lot of resources that you can use to learn about the research methods and the services they provide. I will recommend you to go through their 101 guides. Here you can see that they have a guide on card sorting as well. If you click that, you can see that they tell you everything about card sorting, introduction to card sorting.
Play video starting at :1:55 and follow transcript1:55
They even tell you about different types of card sorting, open, closed, and hybrid. What does that mean? If you go to one of these, you can also read about different types of card sorting. For this specific demo, we are going to focus on open card sort, which is when your participants can group all the cards and all the labels according to the categories they want. They can come up with their own categories. They also tell you about the setup, how to create your first card sort, and how to recruit participants. And they tell you everything about analysis, how you can analyze the data, what type of different analysis methods they offer.
Play video starting at :2:41 and follow transcript2:41
So it's a complete guide that you can use. I would really recommend that you go through this thoroughly before you begin your own study. So let's just start with setting up our own first card sorting study. So the website, it offers a free trial, which is a seven-day trial. So if you already don't have the account set up, you can sign up using the form here, or you can sign up with your Google account as well. I already have an account created for this demo. So I'll just go ahead and log in.
Play video starting at :3:23 and follow transcript3:23
So this brings us to our dashboard. This is where you can set up your own studies. It gives you options for a lot of different research methods. But since we are focusing on card sorting, I'm just going to explain that. So you can see that you have two options here, which is you can use a template and you can create a study from scratch. If you click on use a template, it will take you to the template library. And here you have ready-made templates for different types of studies.
Play video starting at :3:58 and follow transcript3:58
You can do A-B testing. You can do user flow. We will click here, which is organize and label your content, which is related to card sorting. So if you click here, then it will take you to the template. It tells you everything about the template, why it is useful, what you will need. You will need a set of cards that represent a concept, an item or piece of information that people could find on your product. What you can learn about through this study and some of the best practices.
Play video starting at :4:29 and follow transcript4:29
It tells you like you should define your goals, how many cards you should have, how many participants you should aim for. And it also tells you like this type of studies are run in

combination with like website navigation and task surveys. So you can preview this template to have a look at like what does that already have. If you click on the preview, it will take you on another tab where you can actually run through the whole study. It starts with the welcome page. We are not going to use the template in this demo. We are going to create a study from scratch.

Play video starting at :5:7 and follow transcript5:07

But if you want to use the template, you just click on use and then you have a ready-made template that you can edit. You can edit the welcome page, you can edit all the template cards they have. It will be a quicker way to set up your study. But since we are going to learn how to create the card sorting from scratch, so we'll go back to the dashboard and we're going to click create from scratch. So this opens up a page where you can create your study. For this demo, I'm going to give it a name. Let's go with card sorting demo.

Play video starting at :5:48 and follow transcript5:48

Then in study type, you're going to select which method, which research method you're going to use. We're going to use card sort. For project, you can keep it as default project and you can select no folder. Since we are using the trial version, so it gives us limited features. For example, you can only use two participants per study. But if you are going to run your own study, then I would recommend that you upgrade and then you can have access to all the features that Optimal Workshop provides to run the complete study. You can have access to more than two participants and you can analyze the data better.

Play video starting at :6:35 and follow transcript6:35

So once you have given it a name and you have selected the research method that you want to use, you just click the create study button. And now it's going to create this very first study for you. You can see that the study page has different tabs. The first one is setup, which has different sections. We'll go through each one of them one by one. So first is study details. Here you give what is the purpose of your study and your participant recruitment.

Play video starting at :7:11 and follow transcript7:11

It could be like any number of participants that you want to come up with. For this specific demo, I'm just going to use some dummy data. So the purpose of the study can be very simple for this demo. We just want to test how people group the sample data. I'm just going to put 30 as my required number of participants. You can select the number of participants based on your study goals and how many participants you would need to have a better analysis. Then we'll move to the cards.

Play video starting at :7:55 and follow transcript7:55

This is the main area where you can give what labels or what items you actually want the users or participants to sort. So I'm just going to enter a few details. For example, the first one can be apples. The second one can be maybe eggs. So I'm just going to have a list of all the grocery food items. You can also import the data if you already have. And if you want to use like a large set of data, then it's better to kind of like just import from an Excel sheet instead of typing everything yourself.

Play video starting at :8:36 and follow transcript8:36

So I'm just quickly going to copy stuff from my Excel sheet and I'm going to paste it here. So once you press the import button, you can see that it will give you all the labels from your Excel sheet and you don't really have to type them yourself. You can delete any of the labels if you

want. You can also add new cards. So this is the main items that you are going to have for the activity. So these could be like your website navigation bar. It could be like menus.
Play video starting at :9:13 and follow transcript9:13
It could be an existing website that you want to see like if the information architecture is good enough or if you are designing a new website, you can ask users to categorize or label or group all of the categories in your new website to see if it makes sense. So after you have entered all the labels that you want, you can see that on the right-hand side, there are a few options. For this demo exercise, we don't really need all those options. So I'm just going to leave it as default. You can see that you can also show participants either all of the cards or you can select the number of cards that you want to show them. For this demo, I'm just going to show them all the cards. And if we then move to the next category, it asks you the card sort type.
Play video starting at :10:8 and follow transcript10:08
This is the type of card sorting that you will use in your study. For example, you can select between open card sort, closed card sort, hybrid card sort. Open card sort is when participants sort cards into categories they create, which is the one we are going to use. Closed card sorting is when participants sort categories that you provide them. And hybrid card sort is when participants sort cards into categories you have created and can also create their own categories. You can also see that the website provides some information in terms of the analysis tools that are available for each type of card sort. So for the open card sort, you can see that you can analyze the card grouping, category grouping, standardization grid, similarity metrics, and so on.
Play video starting at :11:5 and follow transcript11:05
So on the right-hand side, you can see an option that you can choose the participants and you can ask them to give all the categories a name or you can deselect that, then they can group the items, but it's not necessary for them to give each group a name. I would suggest that we keep it checked because it is useful to see like what type of names do participants come up with. And it's always useful to see how they are grouping and what type of names they are suggesting to understand their thinking process as well. So I will keep it checked. So the next one is study flow. This is basically how your whole study will look like. So it starts with the welcome page.
Play video starting at :11:55 and follow transcript11:55
For each of the sections, you can either keep them on or you can keep them off if you don't want them. For the welcome page, I'll keep it on and you have some pre-written text here, but you can edit that in terms of like if you want any specific details for your own study, so you can type them here and customize the welcome message. I'm just going to leave it as default. Then the next one is participant agreement. So you can select that for your actual study, but for this demo study, I'm not going to use that. But this is a very useful page that they provide because this is the consent form that you ask the participants to fill and you can ask them to sign before they start the study, which we require as an ethical requirement to perform any study with the users or participants which are human. So here you will be asked to provide certain details.
Play video starting at :13:6 and follow transcript13:06
For example, specify who you are. So you will specify if you are a researcher or a master student or a bachelor student conducting the study, you will define the purpose of your study and then you'll explain what the participants can expect from the study, the content, how much

the duration would be if you are providing them with any incentives. Then lastly, you will be asked to let the participants know how will you ensure their privacy. This is in terms of if you will collect the data anonymously or if you'll collect something that is personal like their email ID or their names by which they can be identified. You will also provide like why you are collecting that data. You can also provide and ensure that you are not going to share their information with any third party. So this is all the participant agreement.
Play video starting at :14:5 and follow transcript14:05
You can edit it based on your requirement, your study details and the way you will be handling the data. So after that, I'm just going to turn this off because we are not going to use that. After that, you have a screening question. So the screening question helps ensure that you only receive responses from participants that meet your criteria. For instance, like if I want only female participants to respond to my study, then I can maybe select this drop down select and then I can ask them about what is your gender and I can give a few options here. For example, male, female or transgender. prefer not to say and for each option you can see that you have further actions that you can take for example for male then I can select reject for female I can select go to study and then prefer not to say I can again select reject so it will give me only the participants that I need for my study so if the participants don't meet the criteria then it will reject them and send them a rejection message which is the next section so if we see the rejection message it says thank you and then it gives them the reason why we are not selecting them because they don't meet the requirement again this is like a template message but you can edit it and modify it based on your requirement if you want to give some additional details or if you want to even make it a bit shorter so I'm just going to leave it as default so the next one is participant identifier this is how your study participants will be identified in your results so you can select if they will be anonymous you can ask for their email or any other format for example you can ask for their student ID so it really depends on how you want your participants to be identified in your results normally I just keep it anonymous and that's the normal way you do the research studies the next question is pre-study questions here if you want to ask any demographic questions any questions related to their experience or any other type of data that you feel will help you analyze the data better or categorize the data better in terms of different participants and their and their like characteristics so you can add questions here it gives you options to select different type of questions for this one I'll just select a radio button question and maybe I just can ask them on average how many times per month they do grocery shopping because we gave them a list of all the labels that were linked to grocery shopping which is very useful if you are designing an online grocery shopping website so you can check like how people normally group all the food items so then I'm going to give them some options here for example one it could be once a month twice thrice maybe four times so they are doing it once a week and we can say more than four so I think that's enough for this demo you can add as many like pre-study questions as you want for example you could be asking them about their age about any other data that you want to collect after that we move to the actual study which is card sorting so the first thing is instructions instruction instructions are how actually participants are going to conduct or perform this card sorting it has two parts you can see that part one message is displayed before each participant has sorted any cards and then there is part two which is displayed after each participant has sorted their first card so they have some preset instructions but you can edit them if you want and you can provide some detailed version of this

or add or delete any details that are not relevant but for me this looks quite good for the demo so I'm just going to leave it as it is the next one is card sort it has cards and categories which we have already defined so if you want to change anything you can click on cards and categories and go back and edit anything in terms of the labels that you are giving or in terms of the card sorting technique or the type that you are going to use for example if you want to switch from open card sorting to close card sorting you can do it now and then you have post study questions which is kind of like the last thing so after they have completed the card sorting they have done the grouping you can ask them certain questions this is very useful for getting insights into their experience with card sorting activity or the labels or the categories that you were asking them to define so again like it gives you options to select different types of questions for this one I'll just go with multi-line text and we can ask them a simple question like maybe describe any challenges that you faced while grouping the cards you can also ask them to give any suggestions of how these categories can be improved or what they thought was particularly challenging or any other thing that you want to ask them for example maybe you want to ask them in terms of rating rate from 1 to 5 how challenging this was to group into categories so it really depends on your study purpose and what you want to achieve from that and what type of data you want to collect either it is like qualitative data or if it is quantitative data okay so I'm going to leave it as this as it is the the only one question okay so the last one is the thank you message here you just thank all your participants for participating in your study and that their feedback was really helpful they again have a template message but if you want to edit anything or if you want to make any changes and sometimes you want to ask them for a follow-up interview or anything so you can add a link here where you can ask them if they would like to be contacted again for a follow-up interview you can do all that here once you are satisfied with that you can then move to the next part I'll just keep it as default and the last one is closed message this message is displayed when the study is no longer available for example if you have reached the required number of participants then you can end the study or you have just manually ended the study yourself then this message will be displayed for all the participants that try to access the study afterwards so that's the study flow it has like pretty much everything and after that you can move to the settings here you have a few options like URL slug where you can add something at the end of your URL the next one is really important which is session replay here you can turn it on and then you can create a session recording of participants completing your study this is very useful sometimes if you really want to understand how they were grouping the data you're not only interested in the final grouping but you are also interested in terms of which card they select first what is their thought process how they were doing everything so it will give you the recording of the complete session next you can select the language I'll just keep it as English UK and then you can password protect your study as well you can add a password to control who can access your study so for instance if you password protected then only the people who has the password can access and give response to the study the last bit is the closing rule which determine how your study will finish for example by default if your study exceeds 20,000 participants it will be automatically closed but normally you would not have that many participants normally a small or medium level study would have like 30 to 50 participants so you can select either if you want to manually close the study whenever you want you can set a date for example at 30th of August the study should close so it will close at that date you can select limit for the number of participants for example if my study collect

responses from 30 participants it will close you can also set a date and participant limit both and that's your choice if you want to end it on a specific date and if it has like a participant limit those are the settings and then the last bit is the branding this is just like making your study a bit more personalized here you can add a logo for your study you can add a social media image you can customize the color and text for your buttons for example maybe I can select a blue color okay and then instead of continue maybe I can use next as the text for continuing the study and instead of finish I can use submit so when you change the text you can see that it asks you to make sure your study instruction references your new customized button text so whatever you're putting in your study instructions in your study flow you have to make sure that you use the same text of which you are putting here so it is consistent and users can understand that so once you have set up everything it's good to press the save button it has autosave so it saves automatically after a certain time but just to be sure it's good to press save after that you can preview your study it will take you to another tab where you can go through your study so you can see here is a welcome page you can also see some of the customizations that I made it is showing me the button in blue color instead of the actual it is showing the text which is next on the button instead of continue so it means like it's working so if we move to next it is asking me about the gender so if I select male let's see what happens if I click next so it is giving me a rejection message and saying thanks that I cannot participate because I don't meet the study qualification so it means like it is doing everything that I set up we're just going to close the tab and go back I'm not going to go through the whole study but just to give you an idea you can preview your study and once you are satisfied with everything then you can press the launch button which means that you are now going to launch your study so it is giving you a pop-up message where it's asking you to check everything because when you press the launch except for messages and design settings everything will be locked and you cannot edit or make any changes anymore this is to make sure that your data is consistent so make sure you have everything that you want and once you are happy then you can click the launch button and it is going to launch your study it will automatically take me to the recute tab but you always have that tab in the in the menu so you can click on the recute tab and here you can see that it gives you different options to recute your participants so the first one is invite your own which is free so you can get a link which you can share with your participants and they can join the study and give you responses the second one is standard recruitment which is through the optimal workshop so it gives you an option to get a quote so if you press get a quote you can actually tell how many participants do you want for your study for example if I put 20 you can choose the age range for example if I want between like 20 and 30 years old and then you can select the country it also gives you all other options like what education level occupation gender you want and then it will give you an estimate of how much it will cost to recruit that number of participants and how much it will be for per response we're not going to use it we're just going to go with the free version which is you can get a link if you click that you have an option to either copy the URL or you can get a QR code as well that you can download and then you can share with your participants so after that once your study starts getting responses you can go to the results and there you can see an overview or the analysis of the data this is empty now because we don't have any participants responses yet but if I use the link and if I complete one study very quickly we can then see how it will look so if I start categorizing all of these very quickly I'll just group them in terms of how I think it makes sense and then I can just quickly name this one meat I can

quickly name this one fruits and vegetables and this one maybe dairy and I can then submit and I just for the sake of the demo I'm just writing no and I'll submit so once we have one response now let's go back and see if I refresh this you can see you can you have one response now so you can see the completion time you can see how many participants have completed the study how much time did it take for these 49 seconds and the location of the participants then you can also see the categories that were created you also have option for standardization but we haven't said that. You can also go to the participants tab to see like the individual participants data. You can also see the questionnaire results which is like the pre and post questionnaire.
Play video starting at :30:15 and follow transcript30:15
So in pre post questionnaire what type of percentage or frequency do they have and then you have a lot more options for analysis in terms of like cards, categories, standardization, grade, similarity index but you will have more details here when you have like more number of responses from participants. You also have the option to download your results. You can download your questionnaires, raw data, participants data, similarity index and the last one is that you can share your results. You can click and choose which results do you want to share and then you can create a URL and you can copy that and share with anyone. Maybe if you are working together on a study and you have different team members you can share the results with them and that's it. Then you can end the study when you have collected all the responses. Okay so that's pretty much it. It's a quick demo for using OptimalSort to set up and analyze a remote card sorting study. Use it whenever you are designing menus, categories or navigation especially during early stage design. Remember the best structure isn't what you think may make sense it's what the users expect.

I thought I'd save the best for last and I'll talk about my favorite evaluation technique for both early and late stage evaluation. I'll talk about why this technique is so useful, why it's so helpful, why it's so valuable, and why I use it so much in my own work. Heuristic evaluation is essentially a guiding set of principles for design that is evaluated against an actual design. What we're effectively doing here is we're creating some general rules that we're going to use. We set some standards, we set some ideas of things that we want to achieve, and then we're looking at evaluating them against the design we have. I've got 10 or five, or seven or nine things I'm evaluating the contextual implications of our design against. I'm thinking about are we meeting these general criteria?
Play video starting at ::54 and follow transcript0:54
Some people would describe these as themes, some described them as rules, some described them as guidelines. There's a number of different terms you might hear to refer to heuristic evaluation techniques. But what they are is effectively a way to evaluate our systems at high level and try to understand where there might be problems and challenges. A quick caveat for this one. The rules can seem somewhat obvious if you have practice designing or building types of system. All of you have some kind of technical expertise, and some experience, and some knowledge about how technical processes work. You're probably going to look at some of these design was and think, well, that's really intuitive, that's fairly obvious.
Play video starting at :1:30 and follow transcript1:30
It's common knowledge that we want to design with these things in mind. Now, it might be common knowledge, it might be obvious, but in many cases, people don't use these. They don't

use them in their design, they don't think about these as a set of processes for designing and evaluating, and then we get horrible systems that break experiences and cause chaos. What are the advantages of heuristic evaluation? Well, it's a quite lightweight technique, it's relatively easy to run, it's relatively cost-effective, it doesn't require a huge amount of investment in resources or time. We can run these quite easily in fact. They can usually be performed by different experts from a range of domain practices.

Play video starting at :2:7 and follow transcript2:07

We can think of two types of experts here when we talk about expertise. We can talk about experts in terms of usability evaluation, people like me who have PhDs in usability engineering, and then you can think about experts as people who have expertise in different areas. That could be a programmer, a designer, a developer, somebody working in particular capacities around things like information architecture, mathematical areas, working in physics domains, for example, for game engines. There's a lot of different types of experts in our system. In fact, we can take experts from other areas and train them up quite easily to do these types of heuristic evaluations. When you talk about experts, you might be talking about usability evaluation experts, which are obviously the sorts of users we want to do usability evaluations. But if we don't have access to those people, we can use people from a range of backgrounds, and we can train them up quite quickly to do these things.

Play video starting at :2:59 and follow transcript2:59

What are the concerns with heuristic evaluation? Well, if you're predisposed to domain knowledge, you might do a better job than other people, and this is true of anything in life. If you have some knowledge, you're probably going to do best than somebody who doesn't have any knowledge. This can be difficult to account for. You might not have access to people who have that specific domain knowledge. You probably don't have readily accessible usability engineers at your fingertips where you can just call them and ask them to evaluate things. It might be the case that you have to settle for other types of users who might not have that expertise.

Play video starting at :3:30 and follow transcript3:30

But we can bring in domain experts in some capacity and they can help us to facilitate this set of processes. Now, the lightweight nature of this also means that it might not uncover much, if anything. If you're working with non-domain experts, you're working with people who have less expertise, they might have problems with heuristic evaluation, both in terms of applying these techniques and in picking up issues from your system. Of course, the whole point of evaluation is to find issues and problems. If we're not doing that, it's a waste of time and resources and money. Depending on the level of expertise of your users or your evaluators, in this case, you might find that you get different value depending on who's actually evaluating the system. There are other concerns with heuristic evaluation.

Play video starting at :4:14 and follow transcript4:14

It doesn't have the systematic rigor of some of the experimental design processes we talked about earlier or techniques like A/B testing and multivariate testing. It's not as rigorous, it's not as systematic in a set of processes. Of course, it's not as comprehensive as think-aloud studies. We might find specific issues around general features of the system, but we can't point to very, very specific things and say that over there is causing the issue that's causing users to have this type of problem. We're thinking in much more general terms about general issues in terms of

the user experience. How do we overcome these challenges? With every one of these techniques, we've had challenges and we had to try to overcome them.
Play video starting at :4:52 and follow transcript4:52
How did we do it here? How do we do it with heuristic evaluation? Well, the first way we can do it I've already talked about is multiple evaluators. Having different people who can tease out different types of issues, drawing from domain expertise to bring in programmers who can look at specific types of issues that they're used to seeing, and other types of domain experts as well as our usability evaluators are experts in the domain of usability. Now, you'll probably still miss out on some context here. You're not going to pick up the rich context of user experience design. There'll be things that you miss out on in terms of this not being a contextual experience, you're looking at high-level themes, high-level rules for design.
Play video starting at :5:30 and follow transcript5:30
It will give you some general ideas, but won't give you specific context. Of course, there'll be some degree of bias here, so anybody who sits down and looks at a system will have some degree of intrinsic bias. This is less so for people who've looked at more systems. As I say, the usability evaluators, the experts in this domain will probably pick up a lot more issues. That again, if you have a certain bias, if you come from a technical background, you might be looking for issues that are more technical, and so on and so forth. You'll see different issues depending on your background, and your knowledge, and your experience. What about flexibility? Now, this is an interesting one.
Play video starting at :6:5 and follow transcript6:05
Now using heuristics at any stage in development can actually have some positive influence. We can think about this at a very, very early process of generating and eliciting requirements. We can look at these techniques to evaluate requirements in context of a system. We could actually have heuristics as requirements in many ways they are requirements do have things like clear visibility of system status. In looking at these things in those early stages, they can impart value, but we can also use these later on and we can use them in a way to validate things and validate our assumptions. We can look at early designs, we can look at designs that are medium fidelity, and we can even look at high-fidelity designs and evaluate them under this lens and get value in all of these processes. Heuristics can be used at any point in your interaction design process.

Let's finish up by talking about the practicalities around heuristic evaluation and what it means to do a heuristic evaluation. Now, there are a few heuristic techniques out there, and my own techniques are also out there, so you can look at my own heuristic evaluation techniques, and we'll actually look at one of my papers where I design and develop some heuristics from scratch. But there are also other types of heuristics out there. Lots of people have designed needs for both generalized cases and specific cases in areas like medicine, looking at designing systems for specialists purposes. So the one you'll often see referred to when people talk about heuristic evaluation techniques are Jakob Nielsen's. They are one of the first ones and they are one of the most robust and most generalizable set of criteria we can use to evaluate a system. They're not the only ones out there and lots of other good ones as well.
Play video starting at ::57 and follow transcript0:57

People like Donald Norman and Ben Shneiderman have their own ideas about design goals and design techniques and how we can heuristically evaluate systems. What is heuristic evaluation? Well, it's essentially a set of rules for both design and evaluation. Now, a lot of people look at design and evaluation and see them at opposite ends of the software development lifecycle spectrum. Actually, I see them as pretty close together and in fact they're interrelated. Evaluation feeds back into design. We make changes in design, and then that's reflected in how we evaluate things.

Play video starting at :1:29 and follow transcript1:29

Those changes should be positive changes and should reflect in positive evaluation criteria. Hopefully, these two sets together. If you're working through an iterative user-centered design sets of processes, that will be happening, that feedback loop will be happening. Now, with heuristic evaluation, some are context-specific. There's domain experts who write heuristic evaluation tools for specific types of systems. People like me who talk about designing music systems for purpose. There's lots of other contextual design cases where you have a very specific set of heuristic guidelines for designing and building and evaluating those systems.

Play video starting at :2:4 and follow transcript2:04

Some are more generalized and the more generalizable ones are probably the ones that you're going to use the most. Talking about Jakob Nielsen's ones or Ben Shneiderman or Donald Norman. Those are the three main ones. Now, when we start heuristic evaluation, we usually begin by selecting the elements to evaluate. What are we trying to evaluate from the system? Usually, we don't evaluate the whole system because it would just take too long. If you've got a complete version of a system, we might want to focus on specific evaluation criteria.

Play video starting at :2:32 and follow transcript2:32

You might be looking at small post type of a system. We might be looking at a single page. We could be looking at a form, series of group-together elements. We probably don't want to evaluate the whole system unless we've got lots of time and resources available to do that. What we also want to do before we start, is settle on some rules, some heuristics for evaluation. What are we evaluating against? What we're going to set as a design, an evaluation criteria we're going to measure against?

Play video starting at :2:57 and follow transcript2:57

Have we achieved x, y, and z? Those x, y, and z attributes are what we have to decide on before we start evaluating. We can look at the pre-made ones. People like Jakob Nielsen have loads. Or we could look at specific ones that are out there for our domain, or we could generate our owns. There's nothing stopping us from generating our own set of heuristic evaluation criteria based on a thing that we're trying to achieve. Now, we start out by recruiting evaluators and in some cases providing necessary training.

Play video starting at :3:25 and follow transcript3:25

If we're bringing people in to evaluate, we might want to prime them and brief them on what they need to be evaluating for, what the different criteria are all about, and what they should be looking for in this process of evaluation. You also probably want to identify to them a route through the system, some things for the evaluator to explore. Again, this is all about scope. We've talked about scope in other courses you've done with me. You define the scope for exploration. You tell them what you want them to achieve through the system. They're exploring

a particular page, or they're exploring a particular prototype, or a version of the system, or a variant of the system.
Play video starting at :3:58 and follow transcript3:58
This heuristic evaluation could happen at any point in your design process, right up through late-stage evaluation, but it needs to be clear what people are evaluating and what they're evaluating things against. Typically this will be, I'm evaluating a prototype against the set of rules or predefined criteria. What you want to do here is consider the misuse cases where a designed feature breaks a rule. We're essentially looking for, which of these heuristic evaluation criteria are not being met or where in the system are they not being met? What parts of the system are causing problems, are causing challenges? What parts of the system are creating issues where we're not able to recover from an error very easily, or we've not got meaningful error so we don't know how to act? We've got Error 564237B.
Play video starting at :4:44 and follow transcript4:44
That doesn't mean anything to me. I'm suddenly in a situation now where I've got this big error code that's coming in front of me that is 64 characters long, and I've got no idea what to do. This is a problem. This is how we'd characterize these sorts of issues that we'd have in this type of evaluation criteria, and we'd be thinking about, this particular design feature is causing this problem. This is where this rule, or this heuristic criteria is being broken. Now we want to consider the implications of each of the features in here. Is a change somewhere breaking something else?
Play video starting at :5:15 and follow transcript5:15
Have I changed something in the system that's resulting in some problematic case, and problematic situation. What are the impacts of these different elements together? We're looking at the system here in a bit more context than we have done in some of the other examples. We're thinking about where is an issue being caused in the system. If I click on a button or I click on a menu, or a toolbar, or somewhere in the system, that changes things in the page, I'm now confused, I don't know where I am, I want to get back to the homepage, I don't know how. These are the sorts of issues that you might want to experience and try to evaluate using heuristic evaluation criteria, that things are being broken by other things, that there are certain features of the system that we need to fix to characterize the right process and to ensure that we're able to work through the system in a meaningful and useful way. What do we want to do when we've collected all this feedback, when we've identified a number of challenges with our system, and we talked about where those challenges live, and the ramifications of those challenges.
Play video starting at :6:11 and follow transcript6:11
We want to collect and collate or document the feedback from the expert reviews. All the people who've done this heuristic evaluation, we want to take their feedback and we want to take a good look at it and really understand where those problems occurred in the system and what the manifestation of the activities that caused those problems were. Where are the system elements that are causing issues for people, essentially, is what we're looking at here. We want to conceptualize all these different characteristic features in some way. One of the ways I like to do is, is I want to look at each of the issues that has been experienced by each evaluator and characterize them according to some criteria. I like to use what's called a traffic light system, so red, amber, and green, thinking about a red issue is something that's really critical, that's really

drastic, that's cause the system to fall down, that cause absolute chaos for the user. You might have amber issues, which are really annoying and frustrating but aren't completely system breaking, and then you might have green issues where it's a bit of a problem and it's a bit annoying, but you can skip past it or ignore it.
Play video starting at :7:10 and follow transcript7:10
It's not a huge issue. This idea of categorizing these issues is really helpful for you to think about A, what you're going to prioritize in terms of fixing, and B, what the manifestation of the activities in relation to these issues are. You can think about this is a big problem. We might need to think about whole redesign, or this is a small issue, we can change this just by changing some text. We're thinking about that idea of actionable feedback again. Can we do something, can we take an actionable response based on some feedback that we've got from our evaluation criteria? Again, it's all about iteration, formative experience design.
Play video starting at :7:45 and follow transcript7:45
Thinking about how we can take this formative criteria, evaluate and make a change and improve the system. Of course, with the more severe issues, those changes might be more problematic. To actually solve that problem you might have to spend a lot of time redesigning an entire page or redesigning a workflow or user experience, whereas the smaller issues, the green issues I talked about, they might be simple fixes. In terms of your expert review, this is where the challenge comes in with this. It can be tricky and expensive to recruit usability experts. Not everybody has a PhD in usability. It can be a real challenge to find people who have experience of doing these types of reviews and picking up on these specific types of issues that you tend to see across different types of systems.
Play video starting at :8:28 and follow transcript8:28
Now, these sorts of experts are really good at picking out lots and lots of issues but again, that comes from experience, from having lots of experience of seeing the same sorts of design challenge problems that you experienced elsewhere, and then reflecting on that feedback and having a schema of where to look for things and knowing where things will cause problems. You can train domain experts up to do this, and they often perform exceptionally well. You can take somebody who's a programmer or somebody with some experience of coding or design, and you can bring them in and you can train them up in terms of this process. It's usually pretty quick to train them up to give them an idea of what they're evaluating against, give them some criteria, and they can use that domain knowledge in interesting ways to figure out what the user will be doing with the system and where problems would occur for specific types of users. Because in a sense they are a user, they have a specific experience. It might be a different type of experience. They might be much more technical people working with the system.
Play video starting at :9:21 and follow transcript9:21
It's all about customer-centric focus. But a technical person can understand how the system works and how the components are interrelated, and they can see those problems from afar. Again, you can train people up quite quickly, quite easily. It doesn't cost a huge amount of money either, so it's a very nice way to bring people up to speed, and to use this technique. This is why I think this technique is used so much across different platforms and different industries because it's very, very easy to train people up, and you can have lots of people who are expert reviewers once you've trained them up to a certain level. In summary, heuristics can be helpful

at almost every stage of design. We can use them whenever we want to, and they'll almost always impart some value and understanding if we do them right, and we have good experts.
Play video starting at :10:3 and follow transcript10:03
You might want to select heuristics that are appropriate or specialized for certain types of systems depending on what you're doing. Of course, this is contextually relevant, so if you're building a website, you can probably use generic ones by people like Jakob Nielsen. But if you're building a very very specific type of system, you might want to look at the literature. You might want to look at heuristic things that are out there that are for specific types of systems.

Let's talk about prototyping. Now prototyping is an interesting thing. It's something that we often have lots of experience of as developers, designers, as engineers. We tend to prototype ideas and concepts in different types of ways depending on how confident we are about the thing that we're building. I describe prototypes here as a manifestation of a system. It's some version of a system. That might be a low fidelity version where we have basic components on a page but don't have huge amounts of descriptive quality.
Play video starting at ::39 and follow transcript0:39
It could be a medium fidelity prototype where we have some descriptive quality, some value, we know most of the elements on the page and what most of the interactions would look like, or maybe even we have a high fidelity version of this where we've got all of the functionality built in and some of the aesthetic components with some slight changes and tweaks that we want to make before we release this thing. Prototypes can exist at all of those levels. In fact, you should be moving through fidelity as you go. You should start out with low fidelity prototypes, work through them, get some feedback, make some changes, and then move on to the next state. Make changes as you go based on your user feedback. Make sure that you're engaging in this process of evidence-based reasoning. Every time you make a change to a system, there should be a justifiable reason for doing so.
Play video starting at :1:24 and follow transcript1:24
This is so important and this is why we prototype. This is how we get the most value out of this prototyping process. Now a prototype could be a very basic sketch of a system. It could be just a paper prototype where you sketch out all the ideas of the designs that you have on paper or it could be at the other end of the spectrum where you have a really complex working version of a system. Both of these are fine as types of prototypes. In fact, as we go through, we'll be moving from this idea of low fidelity, a very basic sketch through to high fidelity examples i.e. close to the version of the thing that you are building. Now, a prototype may also be a physical manifestation of a system.
Play video starting at :2:3 and follow transcript2:03
It could be a piece of plastic with no functional aspects of the system. But you would feel the shape of it. You could hold it in your hand. You could put it in your pocket, you could stand up, you could sit down, you could move around with it and you get a sense of how this device feels to use. You wouldn't get that utility proposition in this sense, but you get the aesthetic value, you get the look and feel of it. How is it in terms of that experience of actually having this on your person from day to day. What does it mean when you move around?
Play video starting at :2:31 and follow transcript2:31

How is it affecting you? How are you effecting it? That interaction with prototypes can be really be invaluable. That's how we get the most value out of this process by understanding that these prototypes have value in some given context. The look and feel, the design of this, the physical proximity of objects on the device, the shape, the weight all play a role in terms of how you experience that device and how it works for your personal circumstances. When we think of prototyping as a step that's imparting the knowledge of our user evaluations to improve upon a set design processes, we can clearly see the value in this context. Now we've seen examples of prototypes of processes on this course.

Play video starting at :3:12 and follow transcript3:12

We thought about models as being a manifestation of a prototype essentially. We've looked at different examples of ways that we might validate our prototype too. We can think about prototypes as a system that we are evaluating based on a set of given criteria. Depending on where we are in our process, it's important to look at this through the right lens. We've looked at some of these techniques before that we might use to validate our prototypes. We've looked at focus groups and interviews. We could use questionnaires.

Play video starting at :3:40 and follow transcript3:40

We could do some interactive type prototyping activities with participatory design, there's a lot of different ways that we can actually look at our prototypes and understand their value and understand how we can improve upon these things. But we've already got a sense of what prototyping looks like from a modeling processes and we already understand how we can capture user data in a meaningful way. What we essentially want to do is to impart these new processes and these new set of learning objectives in building interactive prototypes. By now we should all have the skills to be able to prototype. But this is also a reflective process where we have to think about how and why we do these things and what the value proposition of this is. The main thing to understand here is that there's a reason for prototyping, that we want to get some interesting information back based on the feedback that we can get from these prototypes. Prototypes are really helpful to give us a sense of the tangible elements of the system.

Play video starting at :4:35 and follow transcript4:35

We can think about, there's a button here, there's some text there, there's an image here. What does this look like when all of these elements come together as a user experience. You might want to prototype at different levels or you might want to prototype the whole thing. But generally speaking, prototypes will encompass some of these things at the very least. We could be prototyping for functionality and thinking about how the system works. We could prototype for aesthetics and think about, is this a pleasant experience? Then we can all prototype based on our design criteria we talked about before and think about, why this is a reasonable design decision for some assumptions that we're making.

Play video starting at :5:13 and follow transcript5:13

You can test and evaluate these things as you go. What's the purpose behind prototyping? Well, we are scientists to engage in the practice of evidence-based reasoning so we want some evidence to allow us to make interesting and useful decisions. We design, adapt, and use feedback to reflect upon things. What we really want to do is, we want to take all this feedback and decide on what the right decision going forward is. Should we remove a button, should be changed with text on it, should we make it bigger or smaller, should it belong to another group of

things, and so on and so forth. We get a sense of how things fit within the world of designing and developing systems.
Play video starting at :5:52 and follow transcript5:52
We start with a concept. We start with some very low fidelity prototypes, and as we validate our design decisions, as we go through, we get much more feedback, and much more rich feedback to help us to make choices about what we do next. We can think of this as a continual process of test, evaluate test, evaluate, and so on and so forth. This process of refinement, this cyclic process, eventually leads us to what we call a release candidate, a version of our software. You may have also seen the term alpha use before and beta, these are other versions of working software that maybe missing some basic functionality or maybe haven't been tested to the right degree, but they are a much more high fidelity version of a prototype. They're much closer to the end version of the system. Even when we finish building systems, we might find that we're still in that prototyping phase, that we are still engaging in those processes of trying to understand how to improve our software, how to improve our hardware, how to improve the user experience.
Play video starting at :6:49 and follow transcript6:49
Now, I'd like to move warning here that prototyping is a process of continual improvement and it's not a chance to practice as an artist. What we should be doing is thinking about the feedback, the feedback we're getting based on our evaluations and trying to improve upon the things that we're building. This is not a design exercise, is not a chance for you to go away and build the most beautiful interface that you've ever build, there are other factors at play here. You can't look at this in an isolated way and say, I'm going to prototype entirely around functionality, or I'm going to prototype entirely around design or entirely around aesthetics, it has to be a good mix of all these processes and taking into account that rich user feedback. Now it's quite challenging to think in a high-level way, especially when you're thinking about those low-level implications too. You're often thinking about, well, I want to change something very, very small, like some text or a button, but what's the impact on the entire system if I do that. Then might be changing the way in which we interact with the system, it might be changing the user's workflow, the flow of information through the system.
Play video starting at :7:53 and follow transcript7:53
You may be disrupting processes by making small changes. We have to think of both the high and low level here. We have to understand that low level decisions will have high level impacts and vice versa. If we're thinking about high-level design decisions, we're thinking about the impact it will have on low-level processes, can people achieve the things that they want to do? Are they likely to get lost or confused in the interface? Or what kind of problems would we create by doing these things? We're looking both at improving upon the process, but also reflecting upon the fact that as we make changes that might change things elsewhere and change the way that people will interact with these systems.
Play video starting at :8:30 and follow transcript8:30
Let's look at an example of a prototyping process. A car has a lot of components that make it work, it has an engine has various components that connect the parts together. We're not just evaluating functionality in a car, we're not just thinking about what it can do mechanically, we're thinking about car in terms of those broad set of criteria too. We think, well a car need to be safe, if I'm going to drive it I want to make sure that I'm safe in the vehicle, I also want to make sure that the car is economic, so I don't want spend huge amounts of money on fueling the

vehicle and ensuring that I can run the vehicle. I also want to think about things that can be spacious enough to accommodate my needs, so I want to have space for passengers, I want to have space for luggage, and so on and so forth. In many cases this can create problem. If we think about larger vehicles, they have more storage space, but there may be less economical.
Play video starting at :9:21 and follow transcript9:21
This is where it gets tricky. We have some choices to make, we have to decide and weigh these things up in our mind based on the feedback we have. We might have a calculation of the most predominant that a car can do, we might also understand the storage space in terms of capacity, and then we're thinking about these things and weighing them up. Is it more important for me to have more space in my vehicle, at the cost of it requiring much more fuel to be able to move the vehicle from one place to another. When you're prototyping, you're often engaging in these rich processes of weighing something up against some other things and trying to understand what the value proposition of all this is.

Let's think about another example of prototyping and let's try to conceptualize this a bit more before we move forward with some of the more concrete aspects of prototyping. Let's think about prototyping is a process. What I would like you to do is to think about a time that you have engaged in this prototyping process. Think about something that you've done to prototype something, to create a basic version of a system or some functional aspect of a system or some version of a system that's not the complete version. Take a moment to think about this and think about how you've engaged in that process and the considerations that you've taken into account. Can you list all of those considerations and can you list how you've weighed these things up in your mind against one another and make some intelligent decision based on the understanding and reasoning process? Now an example of this might be something like moving furniture around your house or planning how furniture might fit in a new room.
Play video starting at :1:4 and follow transcript1:04
You're thinking about individual objects, aren't you? You're thinking about the sofa, the television, tables, chairs, and so on and so forth. Lots of different elements that you might have within a room. You're thinking about moving those objects into a place in which they're going to fit. The first consideration is thinking about the fit, the shape of the objects in relation to the shape of the room. If you've got a corner sofa for example, that might be problematic if you don't have a good corner or a long space that you can put it in. There's lots of different design considerations around those individual elements that we have to then think about where they're placed within the room and why that's relevant and why that's important for a set of interactions.
Play video starting at :1:42 and follow transcript1:42
Of course, in a house you have to live there, you have to interact, you might have with other people, and so you're taking their considerations into account as well. We can go much further than this then think about these objects in relation to how they interact with one another. Can I see the television when I'm sat on the sofa? Might be a very simple, basic interaction that you might want to make sure that you can engage in as a set of processes. Again, it's about thinking about orientation of objects in relation to one another. Now the value proposition of this is ultimately that we can see things together. That we can start to group elements up and we can think about the interactive processes that people can impart on our system.
Play video starting at :2:22 and follow transcript2:22

If we think about the example of the living room, we might find that the position of the sofa obscures the light from the window. We don't get very good natural light in our homes and that could be a problem for us. We have to make changes based on all those different elements of feedback, of those things that we can understand. Maybe if you had the idea to place the sofa in the biggest part of the room and the longest part of the room, that might just so happen to be near the window. In that case, if it's obscuring the light, that becomes problematic to us and we have to think about what if we're moving this object, then we're moving other objects around in relation to that to ensure that there's a positive user experience, to ensure that people can interact with all of the elements that they need to without there being any problematic types of interactions that might take place. Again, this idea of prototyping is all about understanding things in relation to other things; objects, spaces, places, interactions, design, aesthetics. All of these considerations have to be taken into account and we can't work in isolation in his process, at least not when we think about more broad spectrum type activities.
Play video starting at :3:24 and follow transcript3:24
When we think about high fidelity prototypes and even medium fidelity prototypes, we have to consider the implications for design. What does this mean amongst a broader set of activities of things that people want to do or achieve with our system? There's very few examples of things that live in isolation. When we first start prototyping, we might have very limited interfaces, we might have very limited propositions. But as we go through our processes, we start to get more and more information and more guidance about how to design. We want to ensure that we are prototyping as we go, that we are validating that our expectations are meeting the user's needs, that we are meeting the user's requirements, their wants, their desires, and that we're doing things in a meaningful way to impart value on our system, to ensure that we're building the right thing for a certain group of people.

So let's have a think about what a prototype could be and how we could prototype certain types of things. So, a very basic level, we might think about a prototype as involving some basic set of interactions. A sample of our application might look something like this. So we've got some buttons, maybe you've got some text and then we might have a check box at the bottom. And then maybe we've got another button there. So that's valid as a prototype. It's got some basic components on the page.
Play video starting at ::37 and follow transcript0:37
We can think about what each of these buttons contains. We can consider the nature of the text. We can consider things like do people understand what the checkbox is for, do they tick it, do they cross it, and so on and so forth. And then we can also validate the buttons or the call to actions in terms of whether or not it's feasible, given a set of tasks that we've assumed that the user is going to complete. Now that seems pretty reasonable as a prototype. But we may also want to prototype in different ways. And I've spoken about this already at length in terms of understanding prototyping, being at different levels and different conceptual levels.
Play video starting at :1:9 and follow transcript1:09
So we might want to have high fidelity prototypes that show really complex working versions of systems. But when we start out, we're probably going to have some very basic ideas. So we're probably going to focus on things like structure, navigation. The elements in terms of the content itself that's going to go on the page, where those elements are going to fit together. So maybe

I'm designing a web page. And I want to start out by thinking about navigation. So I've done some research with users and I've looked at some existing websites and now I understand that navigation works a certain way.
Play video starting at :1:39 and follow transcript1:39
So most people expect the nav bar to be along the top. So we assume that we'll nav from the top and we'll have some elements. Of course there's different ways to do this. So we could have some divided elements like this that describes certain functionality. If they're links, they're probably going to be underlined. And then we can think about what's going to go on the page. So maybe I'll have a big cover image at the top of the page somewhere here.
Play video starting at :2:2 and follow transcript2:02
And we can actually describe things in prototyping ways as well. So we can think about images as being this idea of a description of a cross usually. Which could describe that there's going to be an image that goes there at the top of the page. Might be a nice cover image, which is a big picture of the thing that we're talking about. And then maybe we'll have some text, and we'll have it in in two columns like this. Now there's a basic example of a prototype of a website. I haven't given much thought to it.
Play video starting at :2:28 and follow transcript2:28
I haven't spent much time thinking about what's involved in the actual elements of the page. I'm more focused on where the content is going to go and what it's going to look like. So at some point in time, I might decide to change my mind and I might say actually, I'm not happy with having a cover image there. I don't like that cover image. I want to change it to something else. So I can very quickly and easily do this, and I could rub it out if I was working with pencil, but I'm not. So I'm working with a digital display, which gives me the option to rub things out.
Play video starting at :2:54 and follow transcript2:54
And I might decide now, okay, we've done some research and we find that people don't really like big images at the top of the page. They'd much prefer to see a video. So I could just put a play button there, couldn't I? And say, well, a video could be played, or it can be paused or stopped or skipped, I guess. We could define this and describe this at a high level way in a number of different techniques and tools. But ultimately, what we're really getting at here, at the low fidelity end of the prototype, is trying to understand what's going to go on the page and where. Now we might want to do more complex things here.
Play video starting at :3:23 and follow transcript3:23
We might want to describe certain functions of the prototype. So there may be elements of the page that I really want to spend a lot of time on. So maybe I'll think about the text elements. And I'll say actually, the text elements are really important because we're describing something relevant here, or that's the main content of the page. So I might want to spend some time thinking about things like character width, the number of characters that people can read in terms of the left to right orientation of text. How many characters is too much? Should I center my content, or should I align it in another way?
Play video starting at :3:53 and follow transcript3:53
And so I can start to think about these things and tease these things out as I go so I can change them, can't I? I can change them quite quickly. I can say, well actually, I don't want two columns of text. I would just like one column of text. And I'd like a picture to go alongside it to describe that thing, because people find that easier to understand. So my prototype suddenly changed

now and I've got a different view of things. I'm actually going to have a picture here alongside some text which describes some things.
Play video starting at :4:18 and follow transcript4:18
So at a very high level here, we can see that there are different elements on the page. We've described navigation, we've described where things go, where things go in relation to one another, and we've talked about how things can affect one another. So how having one thing on one side of the page might affect how people interact with the other side. And we can make these changes quite quickly and easily. Now, this is what we should be doing at the early stages of prototype. And we should be thinking about what changes can we make to improve things? Where should things go, where do they live?
Play video starting at :4:45 and follow transcript4:45
Where do they belong? What makes sense to our users? And how can we build the best possible experience for human beings who are going to use a system?

So let's look at some examples and where they might fit on our spectrum of activities that we're engaging in. So we can think about some of the low fidelity examples that we've produced for, and we can think about them in relation to a low fidelity exercise taking less time. So something down here, for example, in the bottom of the left hand corner, here would be a very low fidelity prototype and it would take us a small amount of time to produce something like this. It would be fairly straightforward to produce the example that we just saw the low fidelity prototype of the website that I did. It didn't take me much time, did it. And you got a sense of you understood where some of the components might live, but you didn't really get a good description of what the system might look like. So that's an example of a low fidelity, but also a low cost and a low time production process.
Play video starting at ::54 and follow transcript0:54
You can clearly see the advantages there are that we can produce things in a very quick way. We can work through ideas, we can throw things away very quickly and we can converge on better ideas as we go. So as we start out in a process will typically start out with low fidelity and low time production processes. So what we often call throwaway prototypes, things that we can produce and throw away our ideas and concepts will change a lot in that early stage process. So what we want to think about is, well actually, we will be moving in this direction at some point in time when we will be moving up here. So we'll be increasing fidelity, but we'll also be spending more time building things as we go. So if we imagine the stuff that's at the top end of the spectrum here, it'd be nice to have some high fidelity stuff, but it'll take us a long time to produce things because we have to think about every individual granular element in relation to one another and make sure that we're building things that are meaningful.
Play video starting at :1:45 and follow transcript1:45
So this is close to building what would be considered the final project or the final product thing that we're building would sit right at the top of our chart. And we'll be thinking about what actually, we want some meaning here. We want all the elements to come together in a meaningful way. And so that takes some time to build that. What we don't want to do is to try to start out with high fidelity prototypes. Because then you can see the problem here as we go in this direction, we're thinking about time and the cost of time to us, we might not have a huge amount of time to develop these things. And so it might be tempting to jump in at this top level

and say, well, we just spend loads of time developing the the finished version of this, and then we'll know that we have a working version, but in fact, we're working with the same problems here, we're working with the same challenges.
Play video starting at :2:28 and follow transcript2:28
So we want to make sure that we're working through those ideas that we started out with those problems, things that we're talking about, those assumptions that we're making, we want to validate them, and we don't want to jump from here to here without first ensuring that we can validate our design decisions. Because if we get to hear if we get to this top point and we think we've made a mistake, we've done something wrong, something is fundamentally wrong in our designs. And we effectively have to go back here to the beginning and we have to say, well, we've got to throw out this idea and start from scratch anyway, and we've wasted all this time. So if you imagine moving along this trajectory, we're wasting lots and lots of time. That's a real problem for us. And we don't want to do that, that's not a good thing to do. We want to make sure that we're maximizing our time and resources.
Play video starting at :3:11 and follow transcript3:11
And so we do want to spend quite a bit of time in those early processes ensuring that we are getting this right from the offset. We don't want to make design assumptions and then reach a point where actually there is a real problem for us and we have to go back and do all these things again, it's costing us more time, but we're not getting any value out of that proposition. So it's very important that we think very, very early on about this. So what about those stages in the middle if we've got that line. Imagine we're building something as we go. You've got a line that looks vaguely like that process might be different. We might be spending more time on certain elements unless there might be a curve in there.
Play video starting at :3:44 and follow transcript3:44
But we start out with this kind of early process of producing something like a paper prototype or a sketch of things. In the example we saw we saw a nice example of that and as we go through we're actually increasing the fidelity army. But we're also making sure that as we grow up this line that our design assumptions are correct, that we're using that user feedback to say actually we should be doing things this way. We're doing things the right way we're confirming with validating and we're verifying our assumptions based on our findings. And as we go through, then the fidelity increases and we move into things like wire frames or more functional prototypes up here, perhaps in the kind of close to finished version and then eventually we have some kind of application or service or product that goes out at the top where hopefully we've spent a reasonable amount of time on this. But we've actually got some value out of this process and we've understood that this is where we want to be and it's absolutely fine to follow this trajectory. Of course, what we don't want to do is go along and not increase the fidelity and spend loads of time on things.
Play video starting at :4:42 and follow transcript4:42
That's bad. We also don't want to go through a process where we do things too quickly because again, we've talked about those assumptions that we're making and the potential for them to be quite dangerous. So if we're increasing infidelity quite quickly, we might be missing something. We might be causing problems with this trajectory and saying actually, somewhere here, we made an assumption that wasn't correct. And so we have to go back. But of course, we can't go back in time. We can only go back in fidelity.

Play video starting at :5:6 and follow transcript5:06
So we have to start from that time point here. But we still have low fidelity. So what we're thinking about is actually, it's really, really problematic to make assumptions. We want to make sure we validate as we go, and we want to make sure that we're following a relatively linear path. That as we increase our fidelity, that we're spending a reasonable amount of time on building things and that we are using that time in a meaningful way.

So let's have a think about what this actually means in terms of manifestations of jumping into high fidelity prototypes too quickly and why this can be problematic and why we have to make decisions about everything on our page. So let's first start out by looking at our navigation. So we've got some navigational structure over here which basically describes all the elements on the page where the links are. So I've decided to put my navigation on the left hand side. Now, if I was to change that decision, you can automatically see how it's going to affect the things on the screen. So the other elements are going to be negatively impacted if I choose to do that. If I choose to move that navigational part at the top, it will move all the elements down it'll shift what we can see visibly.
Play video starting at ::48 and follow transcript0:48
So again, we're seeing some unintended consequence of making the wrong design decisions. So in this case I have to be certain that I'm going to navigate the page in this way. And so this is one of my early considerations. I'm going to think about this navigation first of all honor. I'm going to make sure that I'm navigating in the right way and that each one of these is a meaningful category and I can say yes, I agree with all of these things. They seemed like sensible choices for things to have as navigational structures. And if I change those things, I have to understand that there's some implicit agreement when the user that's the right way to do things and it's improving the experience.
Play video starting at :1:22 and follow transcript1:22
So what about the page itself? Well, maybe I've got some design ideas from my users and they've told me that they like this layout of the page. So one of my users have given me feedback and they've said 1st element on the page is the heading and I really like this. I can see what you're talking about from here and know where I am on the page. That makes sense. So, again from a high fidelity prototyping perspective. This is all coming from lots of feedback from my users and it makes sense to design things in certain ways based on that feedback.
Play video starting at :1:49 and follow transcript1:49
I've also got a gallery example here where I've got some images and these images are reasonably pleasant. They look right, I can see them, I can see what's going on in each of the image is unhappy with the alignment to shape the orientation. So, again, we're thinking about these things in terms of the box. So, I've boxed these in using some Divzs very, very simple basic HTML but this works reasonably well for me and I'm satisfied that this is a reasonable way to do things. And then finally at the bottom of my page I've got some text which describes what the images are saying. I've got some text up here as well, but I've got maybe an introduction here and then some more text that tells you a bit more about me and what I'm doing. So this is fine as a high fidelity prototype.
Play video starting at :2:28 and follow transcript2:28

And what's the challenge in designing in this way if I start out like this? Well, if I started out with this design, it might be right. It might work for me. It might work for my users, but it might not. And if it doesn't, I've got real problems here, haven't I? So if I'm going to change the way that the users navigate the page, we've already talked about that now bar having some impacts on the users. But there may be other things going on on the page that could cause problems.
Play video starting at :2:49 and follow transcript2:49
So if I've chosen the wrong text size for instance, or the wrong phone or typeface as you might describe it, then that's going to create problems in accessibility. And if I change the typeface, what kind of things do you think might happen? If I change the text that I'm using, what might happen? That will affect the orientation, the layout. So all these dependent components now are going to have a negative experience as a result of that, images might be pushed down. So these images might end up further down the page and then the text might not be readable. There might be other properties that I can't see.
Play video starting at :3:20 and follow transcript3:20
Images might get pushed across in their containers. There could be all kinds of challenges here if I make those changes at that kind of granular level. Thinking about changing a fun is something that's fairly specific, but it has wider implications on the whole page. So again, we have to think about these things and we have to decide on these things early. So instead of some text, I might just have a generic description of where I want the text to go here rather than having that prescriptive text. And if I'm early on in my prototyping process, it makes sense to do that. In the same sense, I might want some placeholders for my image to start with rather than having specific images and then having to deal with things.
Play video starting at :3:55 and follow transcript3:55
Like if I'm building this in CSS grid for example, I have to decide on which grid categories I have and so on and so forth. So I'm making quite prescriptive decisions that are going to affect everything else. So what might make sense is to just design with this process in mind and make sure that I'm thinking about where things fit and what the orientation and the user experience is like first and then thinking in the out over time and then eventually writing some code that will bring these things to fruition. So there's another challenge here that I haven't talked about and it might be in terms of the orientation of things in relation to one another in a more broad perspective. So this has given you one view of the web page. This is showing you the desktop view of my web page. But the problem here is that I'm assuming that users are going to be navigating my view based on a fairly large screen.
Play video starting at :4:39 and follow transcript4:39
Most desktops tend to have large screens. So I'm making the assumption that all of these elements are going to display really nicely on the page. And what might happen is if I'm thinking about this type of structure, if they're using a mobile phone with a small screen, they might get a different view. So on this page I've actually got some media queries which will put a question mark four and we'll say, well actually there's some dependent components that are feeding in and out here. So I'm making certain decisions based on things like my view, port width, my view, port height and other properties in the page. So again, it might be the case that somebody will get a different experience of this. And in fact, I might have three different experiences of this depending on which version of a system they're using.
Play video starting at :5:19 and follow transcript5:19

So there might be a small screen version, there might be a larger screen version for what we often refer to as phablets. And then there might even be a much larger screen version that I'm using right now as my desktop interpretation. So we could actually have three sets of processes going on here to describe different types of designs and again, each one and have two prototype and work through to make sure that I'm doing the right things in a meaningful way for those groups of users to ensure that they get the best possible experience. So, again, be careful with this and make sure that if you are moving on with your designs and you're moving through fidelity from low fidelity to medium to high, that that's based on some evidence and you've got some reasoning for doing that. And it's not just a design decision that you feel the designs perfect for a specific purpose, that you're actually validating that based on some kind of evidence.

One of the really important things to think about when you're prototyping is to consider the idea of a user journey and the experience involved in working through a set of systems to achieve some given scenario, a task, goal or orientation. When our users use our system, they want to do something with the system, they want to achieve things, but they do that through a certain level of interaction and interactivity and it happens at different stages. We're thinking about prototypes in relation to a number of things. Firstly, we want to think about the individual components of the person using the system and making sure that we are working within the constraints of that individual to ensure that they have the best possible experience. Every individual is different and we want to make sure that individual has an experience that's going to map with their expectations. Moving forward, we also want to think about the contextual implications of how this system is going to work for that person. There's certain context here that we have to think about.
Play video starting at :1:4 and follow transcript1:04
What are the interactions going to look like? Where are they? What are they doing at any given stage? When we prototype, we're not just thinking about a static system. We're thinking about a set of interactions, some steps and stages. These are what we describe in terms of phases of the user experience with thinking about a phasic modality of delivering. Interactions are ready, just input and output, and so we think about this trajectory of actions that allows a user to achieve something and ultimately we want them to achieve that goal.
Play video starting at :1:33 and follow transcript1:33
We can think of this as a very static thing as they want to achieve some things. They have some ideas in mind about some main success scenarios, some things that they want to do with the system and that makes sense. But ultimately, there's all these broader considerations that we have to encapsulate in our design too. What might this look like? Generally speaking, we want a very positive experience for people and we want them to follow along a specific trajectory that looks something like this, wherein they get the best possible experience. That at each stage they're doing something that's validating their expectations and reality. They're not confused, but nothing's too complicated that the experience maps with their version of the system.
Play video starting at :2:11 and follow transcript2:11
But in fact, most systems don't work this way. If we're prototyping, then we're probably not working with a system that looks even remotely like this. What we want to do is, we want to encapsulate those experiences and understand what's happening in relation to the things that

people are doing that are giving them positive or negative trajectories. Let's go back and think about how this might look. We might have a personal context and a goal working through the system and we might think about this as a set of stages. If we imagine we've got a stage here of activities, of phasic stages here, they'll be working through a set of processes. We could imagine them as step 1, 2, 3, 4, and 5.
Play video starting at :2:50 and follow transcript2:50
These are generic steps and you'll have more descriptive processes in the work that you do. But ultimately, we can think about these as generic things that people want to do at different points. At points during their experience, they're going to have highs and lows and those highs are really good. It shows that the system's working well. If we're up here in terms of the user achieving what they want to, then having a positive experience that the system's working in an effective and efficient way, that's great. But at times we might also have problems and we might be dipping down here at some points in time. This is where prototyping can also be useful.
Play video starting at :3:21 and follow transcript3:21
We can think about the highs and the lows of the experience and we can imagine this as a kind of up and down trajectory. We can think about this as we're moving through different levels. We might be having a really shaky experiences, very positive and very negative, then very negative for a while, then very positive again. What we want to think about in terms of our design is looking at these peaks and troughs and trying to understand what's happening at different stages. Through the prototyping process, what's happening at this particular stages, allowing a user to have such a positive experience. What's happening down here that's causing real problems for people? Why is this such an issue?
Play video starting at :3:55 and follow transcript3:55
We need to look at this specifically and we need to change things in the prototype that related this experience here to ensure that we can improve our designs and make sure that people have a much more positive experience. You can think of this like a journey, it's a journey through a system. We have highs and lows, much like in life. The experience is all about understanding how we can improve upon those lows and drive them in this direction but also understand what we're getting right in terms of these highs too.

So here we have a framing for our paper prototype. And we're going to talk about why we do this, why it's valuable. Why it's useful and helpful and how we can use paper prototypes for all manner of things. Now, I'm going to show you how quickly we can create a paper prototype from scratch. Now imagine I'm doing this on pen and paper. Obviously I'm using digital tools because I have to for this lecture. But I could just as easily do this on pen and paper if it was a talk class and I could hand these out to you and you could see them.
Play video starting at ::34 and follow transcript0:34
So, if I was in front of you, we could do this this way. So let's imagine I'm designing some kind of profile based website. Well, we've got some characters, some individuals. So straight away I've got some feedback. There's some things I can put in there. So I could have an image there of the person that we're talking about. A general description of who they are and what they do.
Play video starting at ::51 and follow transcript0:51
And maybe I'll have some other elements down the bottom of the page. So maybe I'll have a kind of divided section which has maybe some quotes from this person just to give a sense of

who they are. So straight away, we've already got almost a third of our page designed. Now, we've thought about where all the elements are going to go. We've thought about this in terms of a wireframe. We're not thinking at that high level descriptive level of detail just yet, but we will get there at some point in time, hopefully. So, the idea now is that we can design these things as we go.

Play video starting at :1:20 and follow transcript1:20

We can demonstrate the value of what this is going to look like in a tangible way. We can think about where all the elements live in relation to one another. We can make changes very quickly with paper prototypes. So I can rub things out. I could say actually, I don't like this idea of the bottom. I'd like to have a little bit of white space there because I think it's a better design. I can do that and get away with that.

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That's not cost me a huge amount of time or resources, has it? I've just made that change almost instantly and straight away now things look better. So I can make these improvements and I can get feedback on these things straight away. I can ask the user to use this system and I can change it and then I can ask them to use the next version of the system. So Straight away, I've gone from version 1 to version 2, which is really quick and really efficient. So this is the value of this. If you're running workshops or you're getting feedback in some kind of manifested way like that.

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You want to ensure that you can get that feedback and make changes as quickly as possible. Especially early on in your design process is we want to get the bad ideas out and get the good ideas in as quickly as possible. Now, the other advantage of this is that we can think about this in relation to the sets of activities that people are going to do. So we can think about what's this going to look like as a user experience. Now, if we facilitate this in the right way, we can do lots of interesting and quite clever things. So first of all, we can see the bits that are missing on the page, can't we? So if we've got some elements here and some elements on the page to do some tasks and we ask the users to engage in that process.

Play video starting at :2:45 and follow transcript2:45

And we actually hold up a second, something's not quite right here, they're not able to do that thing. We can bake that into the system. We can just say, okay, we'll change that, will adapt that, and we'll make sure that this structure works. Will make sure this works for a particular type of experience. And we can think about that user flow through the system. So you can look at where they go and what they do. Do they go top to bottom left to right?

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Did they go across the system? Do they go through the system? What do they click on? What do they not engage with? What do they engage with and so on and so forth. We can get a rich understanding about what the context of uses like. Even from these high level prototypes that don't give us a lot of description.

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Now, we could do lots of clever things with this in relation to the tools we have. So I could have a pair of scissors to cut bits out and move them around. I could mark things off, like I've just done with a marker. I could have some sticky notes, I could have some things in here to change some of the elements. I could use different color schemes to identify things to people. There's

lots of different things I can do with paper prototypes that are quite useful. And provide quite nice interesting descriptive value of things.
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And I can change over time, and I can actually move through fidelity just in terms of these individual versions of things that I'm producing. As I move from version 1 to 2, to 3 to4 and so on and so forth, these small changes yield some value each time. So I'm getting a very, very quick improvement on my system. Without having to worry too much about how I've got to change that huge code base. This is going to affect all these other things. I can make changes on the fly quite easily with paper prototypes and there's some value in that. Now of course, there are limitations as there are with any technique so we can't represent all things with paper prototypes.
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So if I wanted to represent a gesture for instance, I might use a visual structure for that, but it's not simple to do. So imagine kind of swiping left to right, for instance, or right to left. I might describe that that way, but that could be interpreted in a number of different ways. Gestures are often quite complicated things that could be described in cycling quays or in terms of movements or trajectories or actions. We could have multi gestures where people are doing multiple things at once. And so you can't really do that in paper prototypes. You can't really manifest those sorts of interactive things that people do.
Play video starting at :4:54 and follow transcript4:54
And of course, if I have things like animation in my interface, I can't do that. So I can't show animation. I could mimic it, so I could use something called Wizard of Oz prototyping to actually manipulate these on the fly while somebody is using them. And in this sense, I could move things around on the page. So I talked about the example of getting rid of things. I could take this out and then I could go and get a pre cut out piece of paper that has another thing on there and I could put that in. And I could pretend that I'm the computer changing things on the system.
Play video starting at :5:23 and follow transcript5:23
And then as people interact with things, I can continue to change them over time manifesting a set of interactions. Without having to build that interactivity in the system. So things can change continually as a set of processes that I engage in that activity. And that's called Wizard of Oz prototyping, it's a very interesting and useful technique that still use quite a lot today. Now, what else can we think about in terms of paper prototypes? Well, there are other challenges with these. We can't necessarily think about the high level manifestations of activities as that kind of granular set of components.
Play video starting at :5:55 and follow transcript5:55
So I can't see any problems with this image because it's just a high level description. I can't see problems with this text because they're just high level descriptions of where things live. So, if I wanted to get at that really descriptive value, I couldn't do it with paper prototypes, really. As we move through fidelity and we move through complexity. We find that there are more tangible elements of the system and there are more things that we can test. And of course, there are other challenges with this that eventually we probably will throw them away. Even if we have a nice manifestation of a system that has relatively good feedback.
Play video starting at :6:27 and follow transcript6:27

At some point in time, we're still probably going to have to sit down and write some code to make this all work. So again, this doesn't solve all of our problems, but it's a technique that we should be using often. And it's a technique we should be using often early on in our process to help us to understand things. And we can do lots of clever things with this, as I've mentioned. So if there's an error, if somebody's done something wrong in the system. I could throw up a sticky note that says, actually there's a problem here. There's an error, you've created an issue and I could show that to the user again.
Play video starting at :6:55 and follow transcript6:55
That level of interactivity could be manifested in other ways. Of course, there are challenges and the things that I can't do. But there are challenges with all the techniques that we use. And so it's important to understand the value of paper prototyping and how useful it can be. Particularly for collaborative activities where you've got multiple people working. That's either the user and yourself, manifestations of you changing the system based on their feedback. Or you could have multiple people working on an individual prototype.
Play video starting at :7:21 and follow transcript7:21
So I could have three or four people around me here we're sketching things out as I'm talking. Doing clever things on the interface and changing things, and we could adapt things according to feedback. And I could have lots and lots of people working on this one sketch, couldn't I? So I could have a team of people all working together to solve all these interesting design challenges. And make sure that we're building the right thing. And then if we decide that it doesn't work, we can always throw this away and start again. It's very simple to do.
Play video starting at :7:45 and follow transcript7:45
And in fact, even with my digital system, it's very easy to just go like that and start from scratch. And say actually we'd like to design this a completely different way. So hopefully you've started to understand the value of this process and moving between different levels of interactivity and interaction. And how that level of fidelity will change over time. Thinking about inherent structures and complexity and systems. But also understanding that techniques like paper prototyping are really useful at those early stages. And that as we prototype as we go through as our prototypes improve in terms of fidelity.
Play video starting at :8:15 and follow transcript8:15
What we're also doing is making sure that we are actually validating our assumptions. And saying that we're doing this for a very good reason. That we've made these changes based on some feedback or some research we've done. And that there's some value added by doing it in this way versus the previous the way that we chose to do it. So again, as your prototyping, make sure that your understanding what those interactions look like. Make sure you're taking on board as much feedback as possible and you're reflecting on that process as you go. And understand the tools like paper prototyping are really useful to help us to do meaningful things.
Play video starting at :8:45 and follow transcript8:45
And move through fidelity quickly and cheaply.